The working (6,7): a complete named derivation

121 graphical steps | Four explicit sphere moves | Portrait US Letter

VERY BRIEF SUMMARY. Remove the incident half twists, use a five-point disk, turn the remaining positive factors into FOUR exterior twists by PC lanterns, roses, daisies and Hurwitz transports, then cap their fourth power.

The previously missing geometric boundary closure is now supplied. The final equality is obtained from actual named substitutions and factor transports, rather than a whole-product action recognition. All four sphere moves retain their disk discrepancies, exact sphere certificates and separately checked genus-two lift.

What is still open. This detailed route has 121 retained graphical steps, shorter than the earlier 177-step archive. It does not yet meet the goal of fewer than twenty meaningful operations. Reusable named patterns and fixed-factor drags are identified, with all adjacent checks retained in the certificates.

How to read it. Read each factorization downward. Curves are paired with the braid representing that exact support; blue marked-point dots continue at braid separators. Only the changed interval is printed at a move, with the unchanged factors counted above and below. Every actual SPHERE MOVE has a yellow banner. Individual supports are not cumulative joint cut-system certificates.

Logical scope. Common frames and cyclic starts are explicit conjugations. The terminal four-exterior product is central, so reversing those conjugations recovers the source equality without assuming it was central. Marked points are not framed boundaries; signed framing balances are retained for PC substitutions. A single five-point cap has lift -I; the sixth- and fourth-power caps used here have +I.

Contents

Full diagrams, named sub-derivations and sphere discrepancies.

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121 graphical steps retained. Four sphere moves labeled; full discrepancies in Section IX.

What the steps will do

1. Reach the five-point chart. Split the required squares, match and absorb the incident half twists, change the exterior point explicitly, and retain then separately cap the six framing corrections.

2. Build positive named configurations. Use transported PC half-lantern, daisy and cluster-lantern configurations to eliminate every half twist. Two actual exterior twists are extracted and remain visible.

3. Rebuild a round four-point curve. An actual rose supplies T34; a lantern makes T234, which removes the precise winding of a 12 curve. A cluster lantern produces round T1234.

4. Close the inside-four rose. A common frame and short cyclic starts collect the inside-four factors. One PC lantern, one Hurwitz swap and one PC daisy give two round T1234 units.

5. Untwist the eight point-pushing petals. The saved route has 21 actual noncentral adjacent interchanges, grouped into 16 genuine fixed-factor drags. Every final petal is checked individually against two canonical PC daisy lists.

6. Apply two daisies and cap. Each T1234 and its four actual petals becomes an exterior twist. The four identical exteriors combine to the fourth power, followed by a separate labeled sphere cap with +I lift.

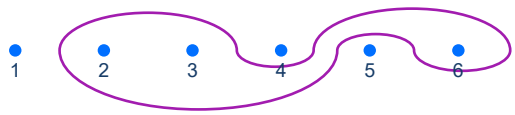
I. The fixed-point boundary chart

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before The fixed-point boundary chart

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. F1
T3^2



2. F2
H



3. F4
T3^2



4. F3
H



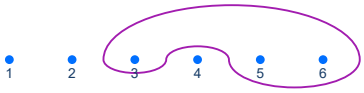
5. F5
T3^2



6. F6
H



7. F8
T3^2



8. F9
H



9. F7
H



10. F10
T3^2



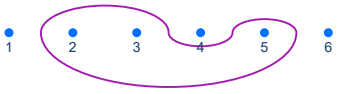
11. F11
H

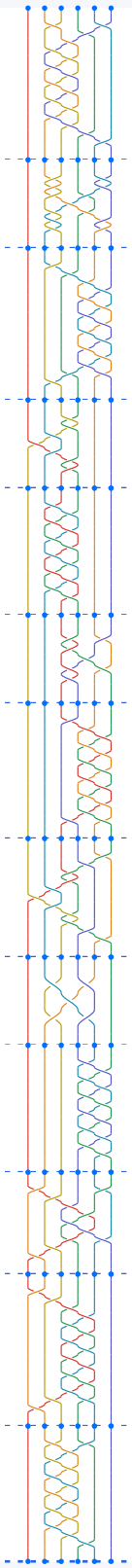


12. F12
T3^2



13. F13
T3^2





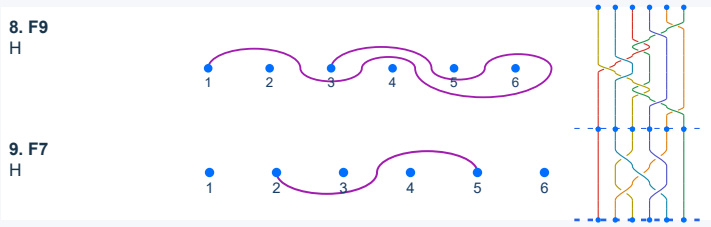
I - step 001

Change the working ordering by an inverse Hurwitz move

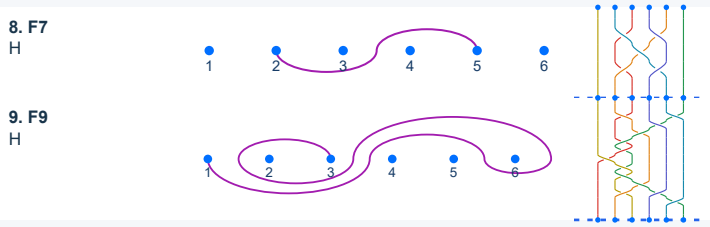
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



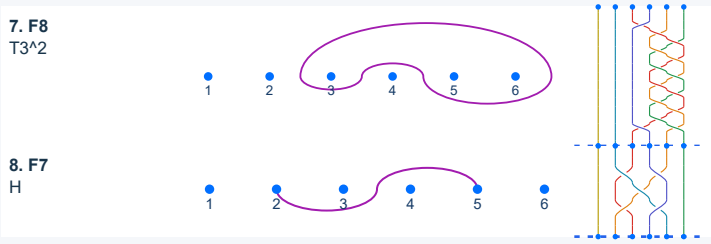
I - step 002

Change the working ordering by an inverse Hurwitz move

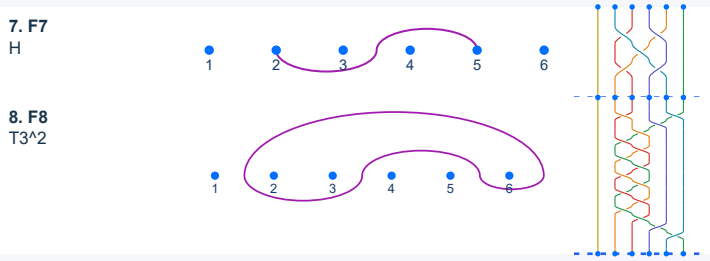
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



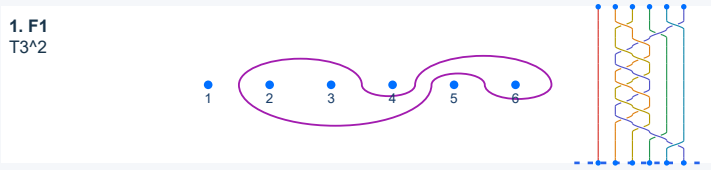
I - step 003

Split F1 into ordinary roots

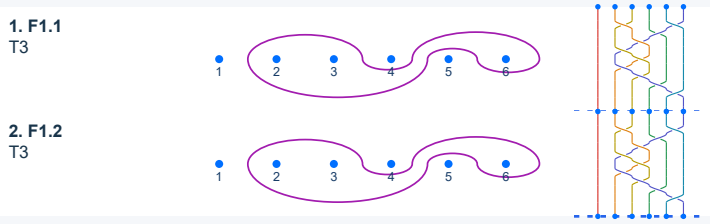
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

Before



After



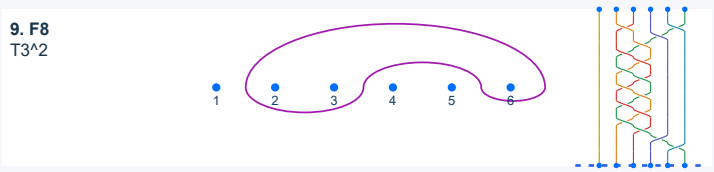
I - step 004

Split F8 into ordinary roots

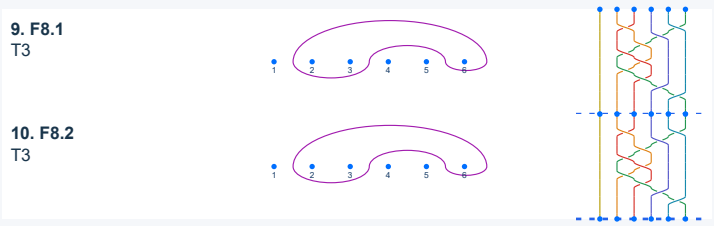
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



I - step 005

Split F10 into ordinary roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



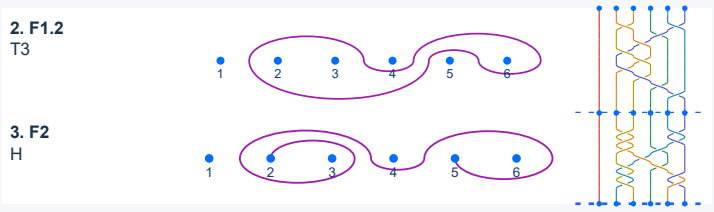
I - step 006

F1.2 down past F2

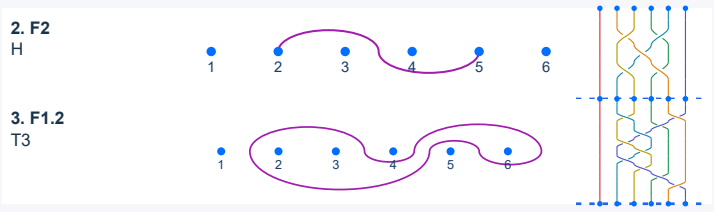
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



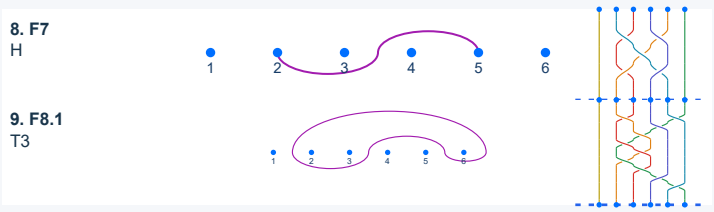
I - step 007

F8.1 up past F7

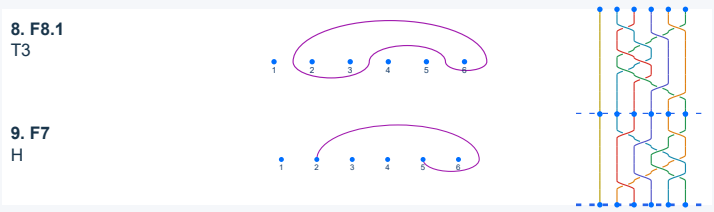
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



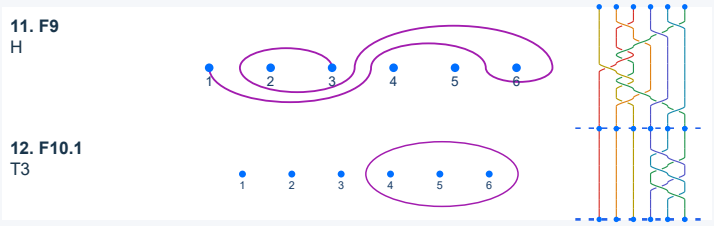
I - step 008

F10.1 up past F9

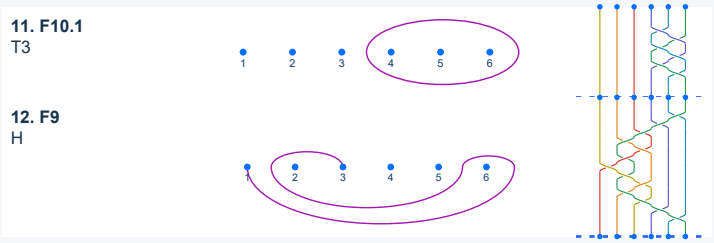
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



I - step 009

F10.1 up past F8.2

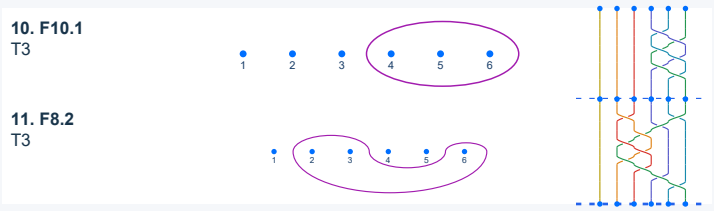
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



I - step 010

F10.1 up past F7

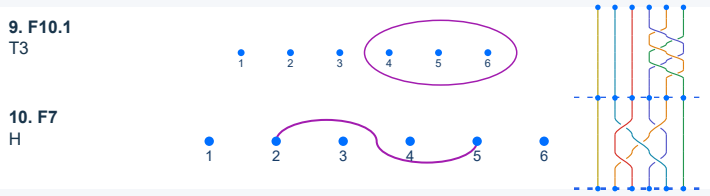
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



I - step 011

Use the exact disk equality of the matched arcs

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



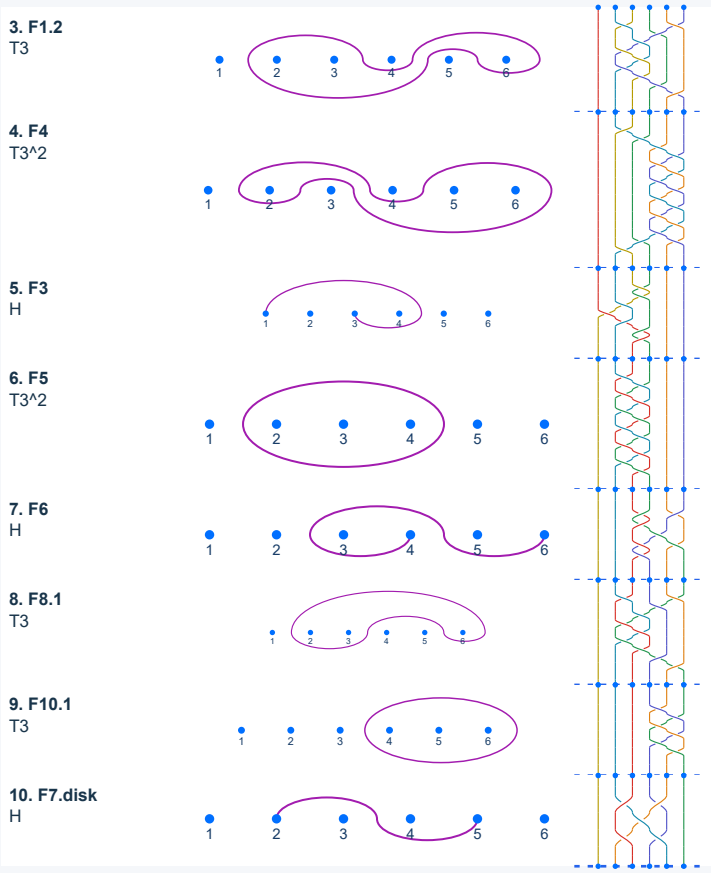
I - step 012

Drag the matched arc up across SEVEN factors (one Hurwitz drag)

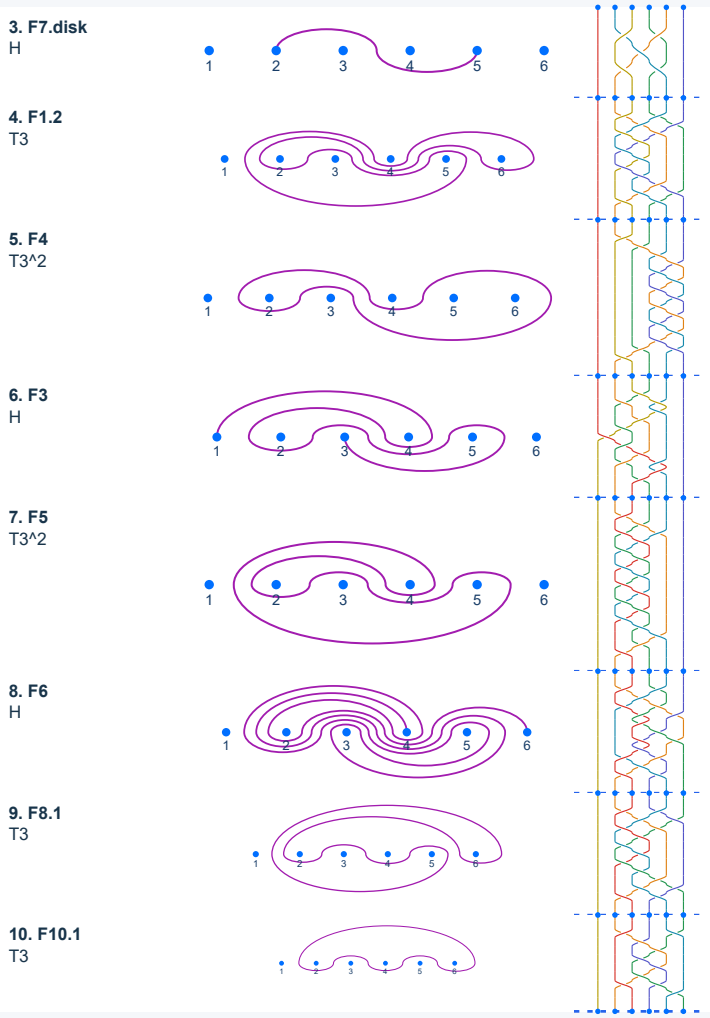
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 7.

Before



After



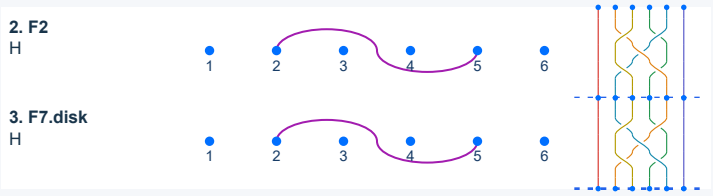
I - step 013

Combine the two matched half twists

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



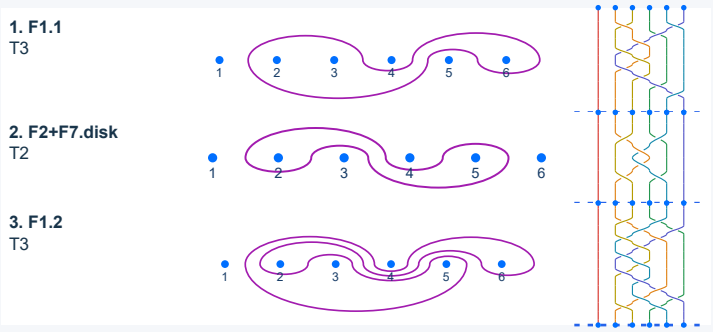
I - step 014

SPHERE LANTERN: absorb the matched pair and the two prefix roots

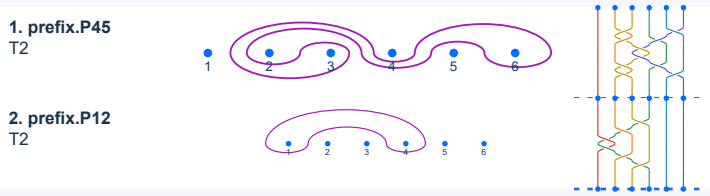
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



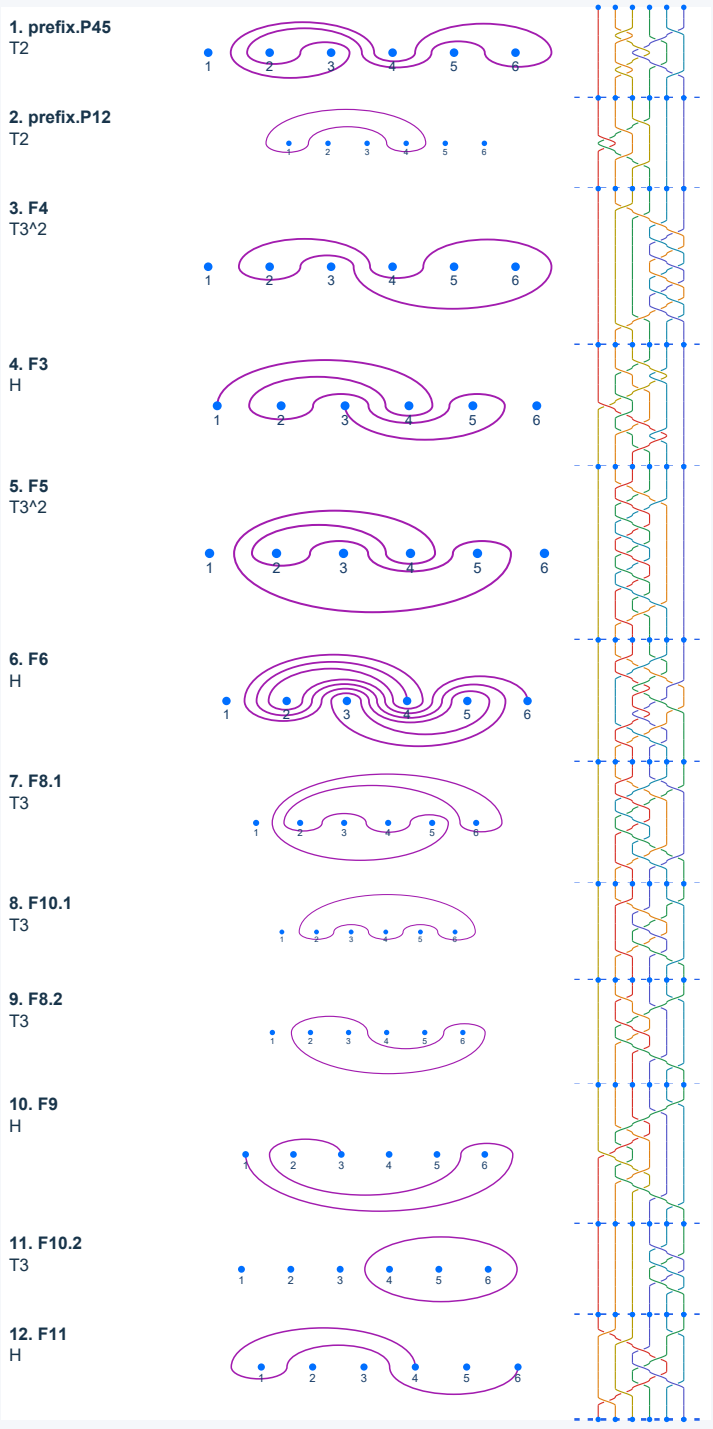
I - step 015

Global frame: fixed working point 2 becomes displayed point 6

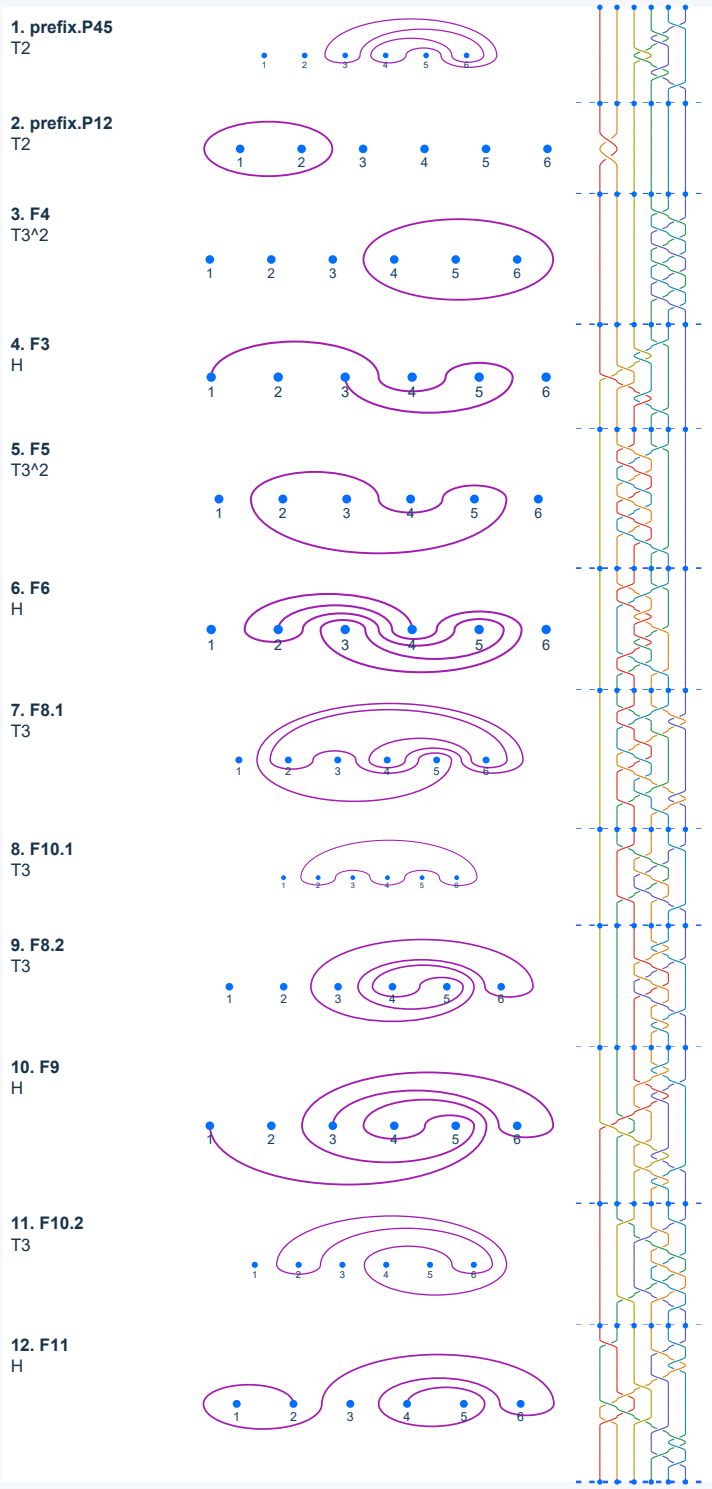
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



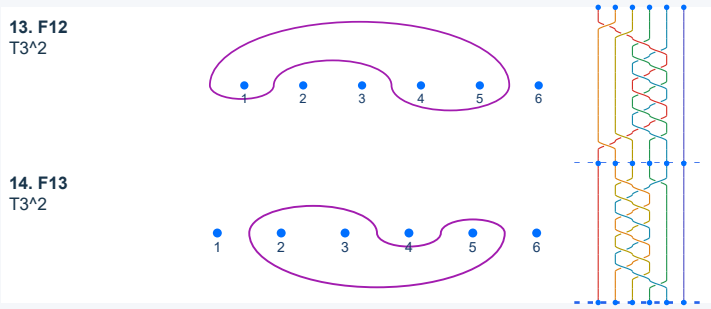
I - step 015 - continued

Global frame: fixed working point 2 becomes displayed point 6

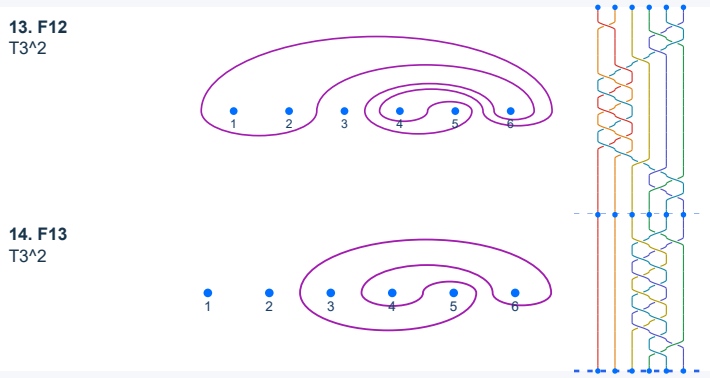
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



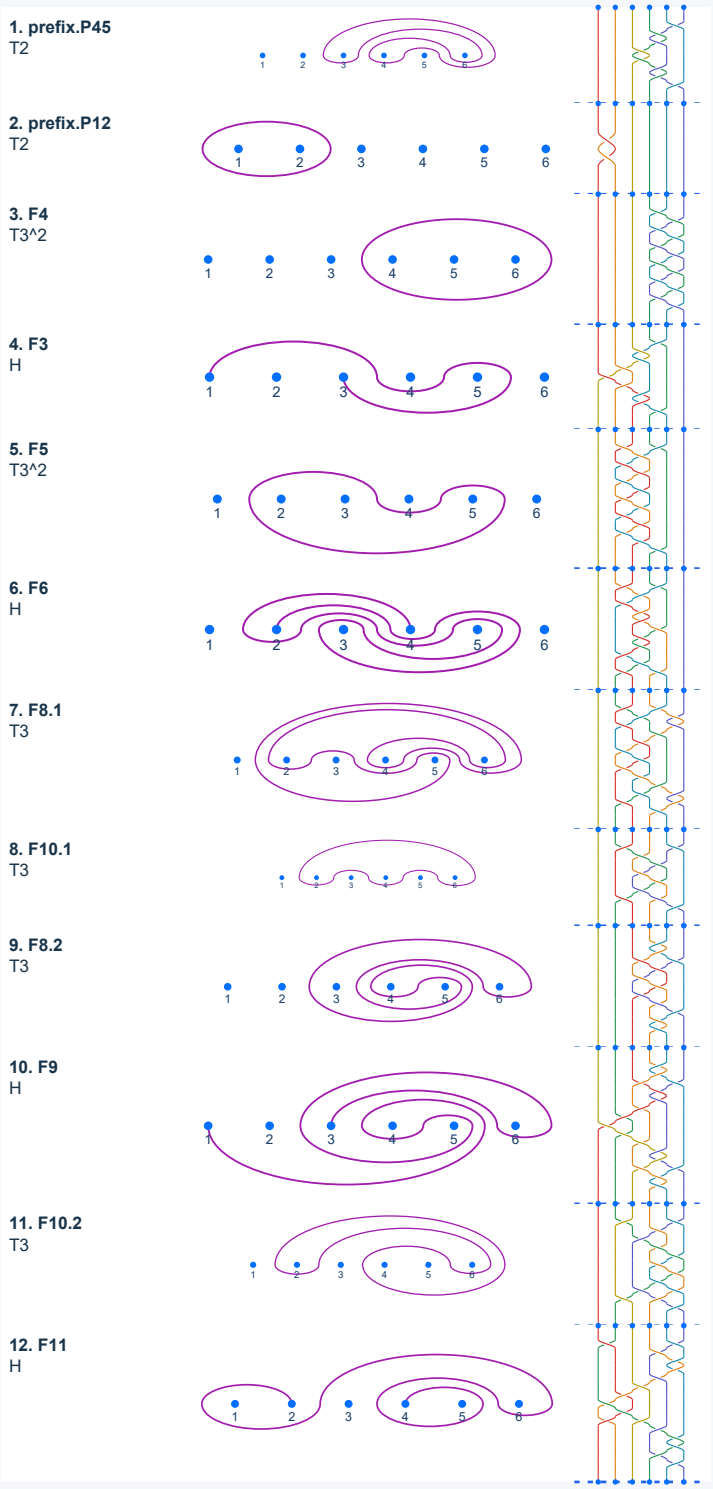
I - step 016

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

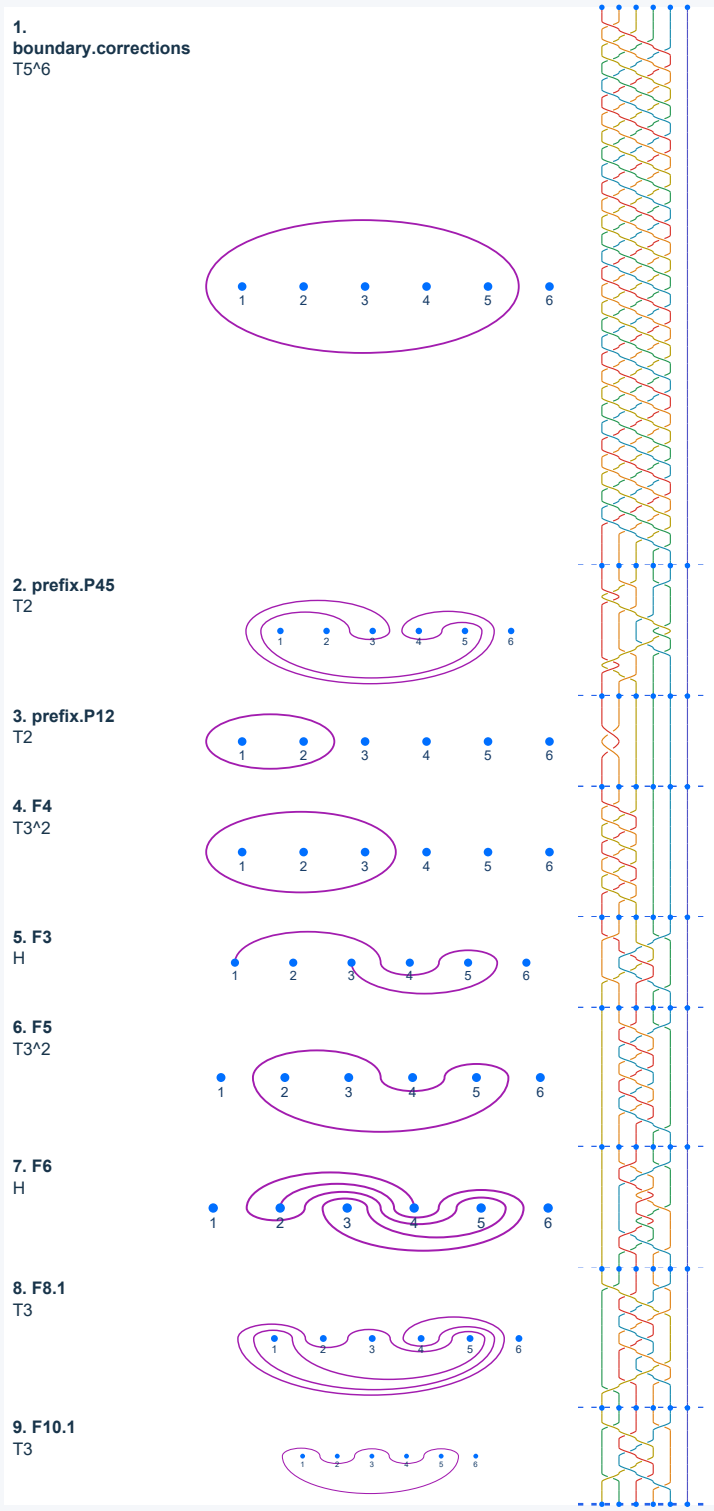
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



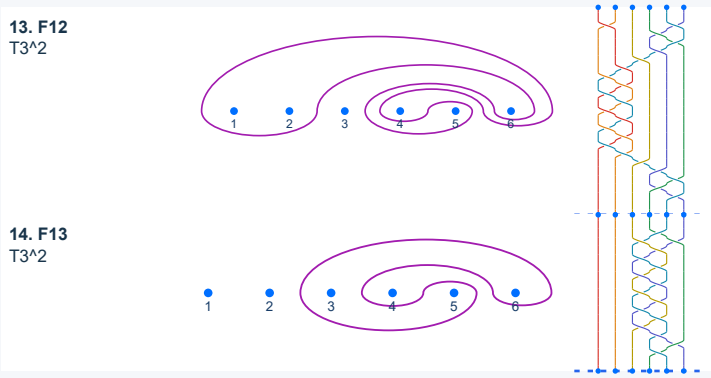
I - step 016 - continued

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

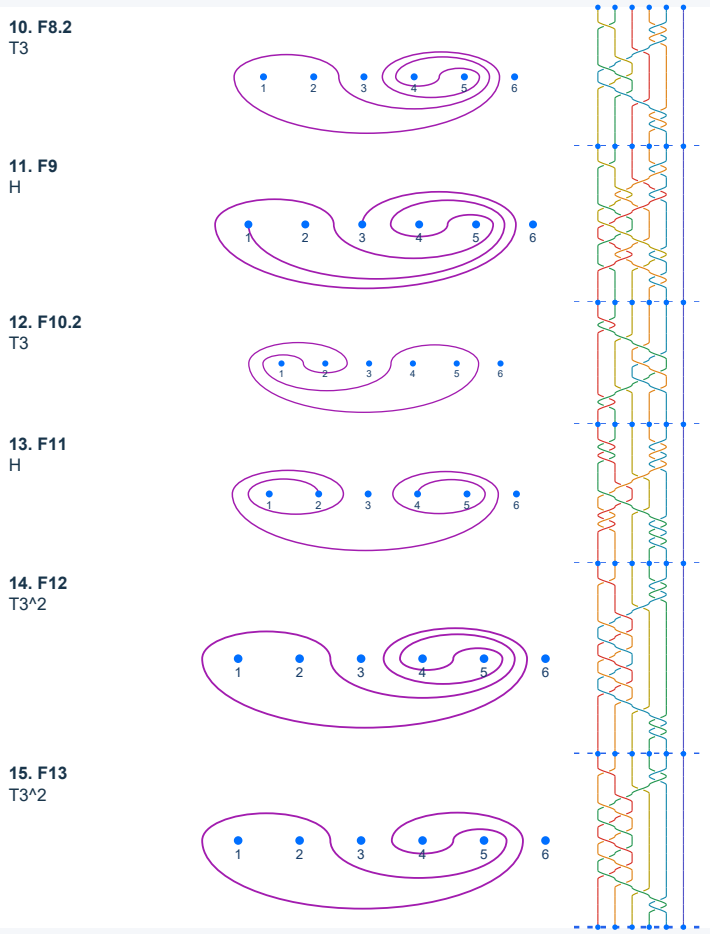
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After



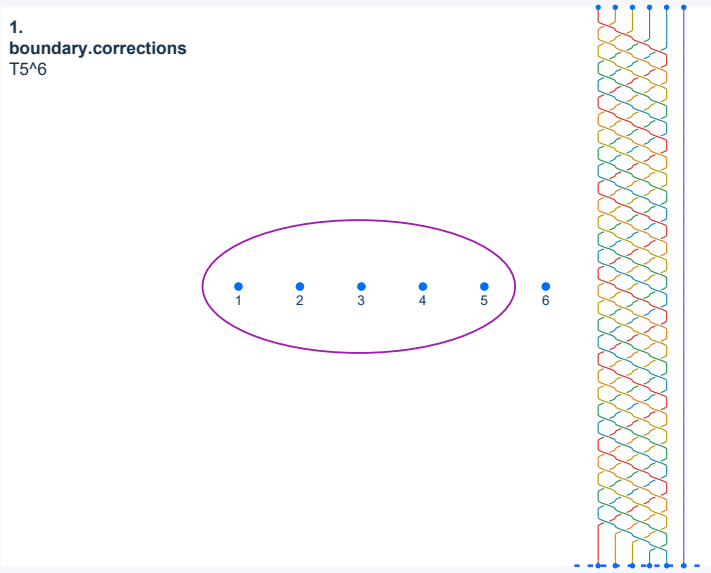
I - step 017

SPHERE MOVE: cap the SIX framing corrections

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After

Identity / no factors in this interval.

II. Positive clusters and removal of half twists

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

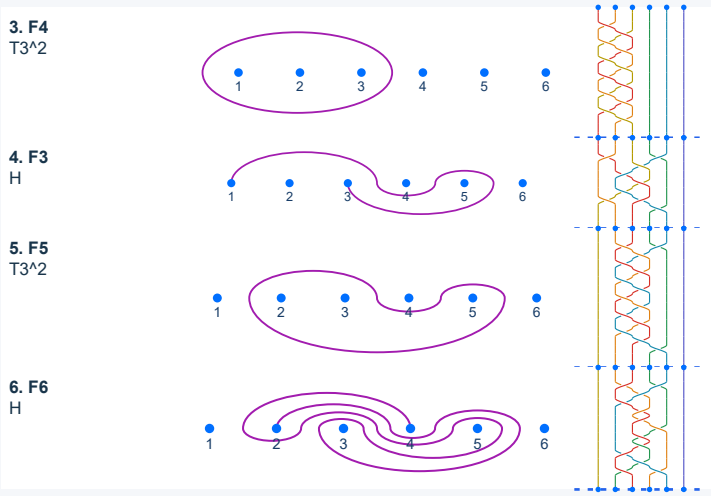
II - step 018

Expose K = sigma2 sigma1 from the round squared twist

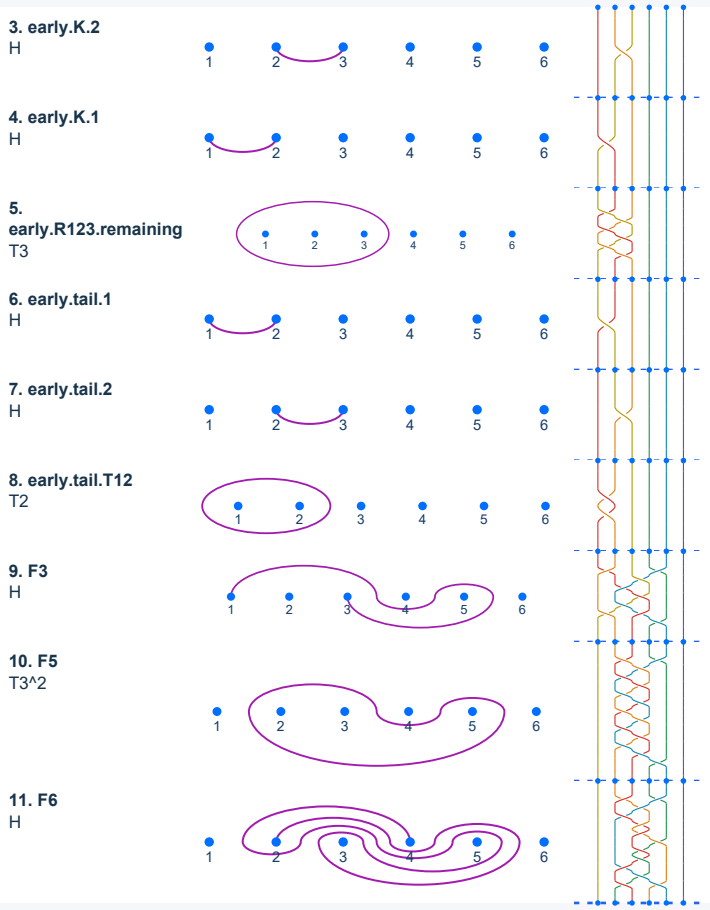
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 8 below. Only the changed interval is shown.

Before



After



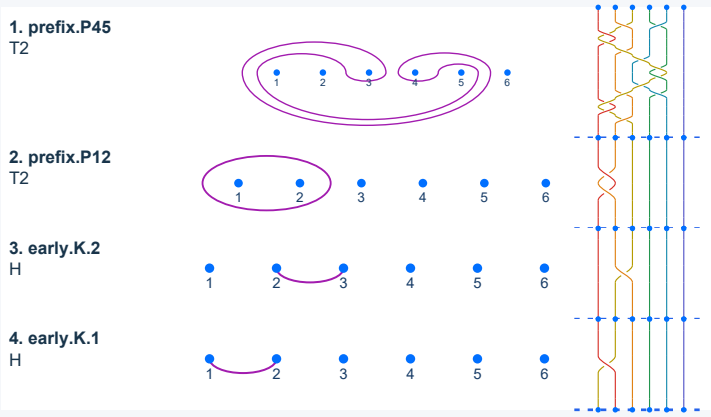
II - step 019

Carry K up across the first two pair twists

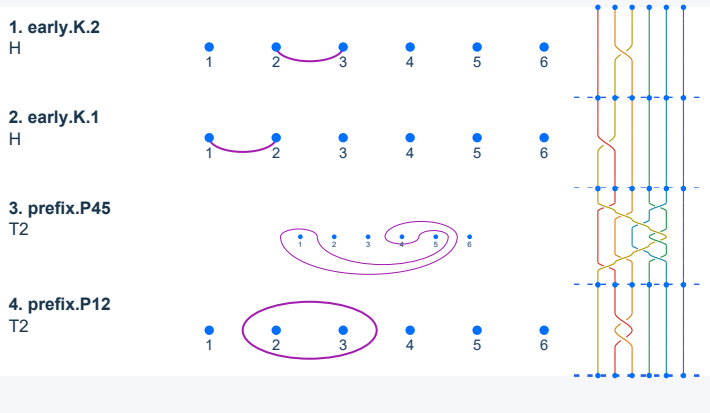
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



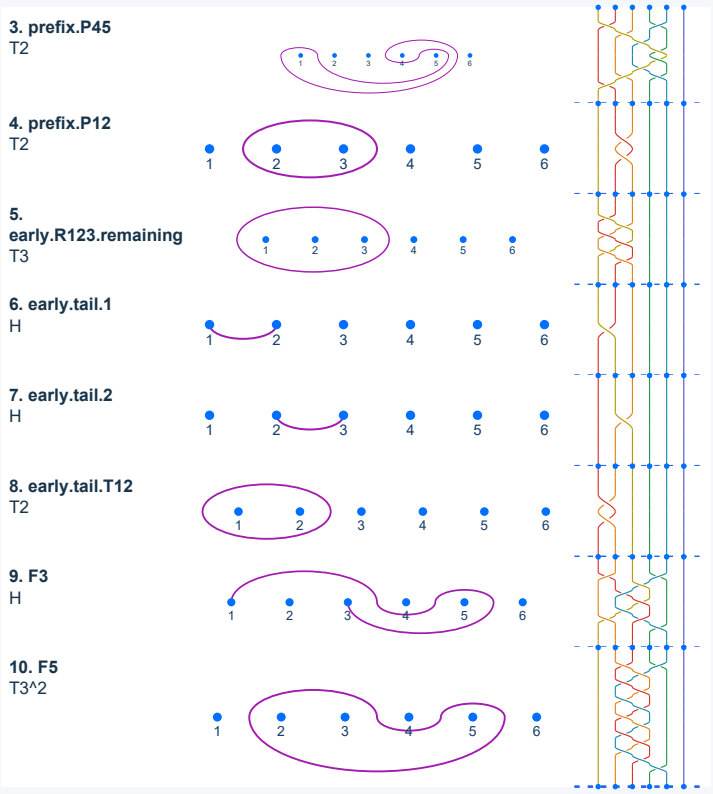
II - step 020

Drag the required Q14 down across seven neighbors

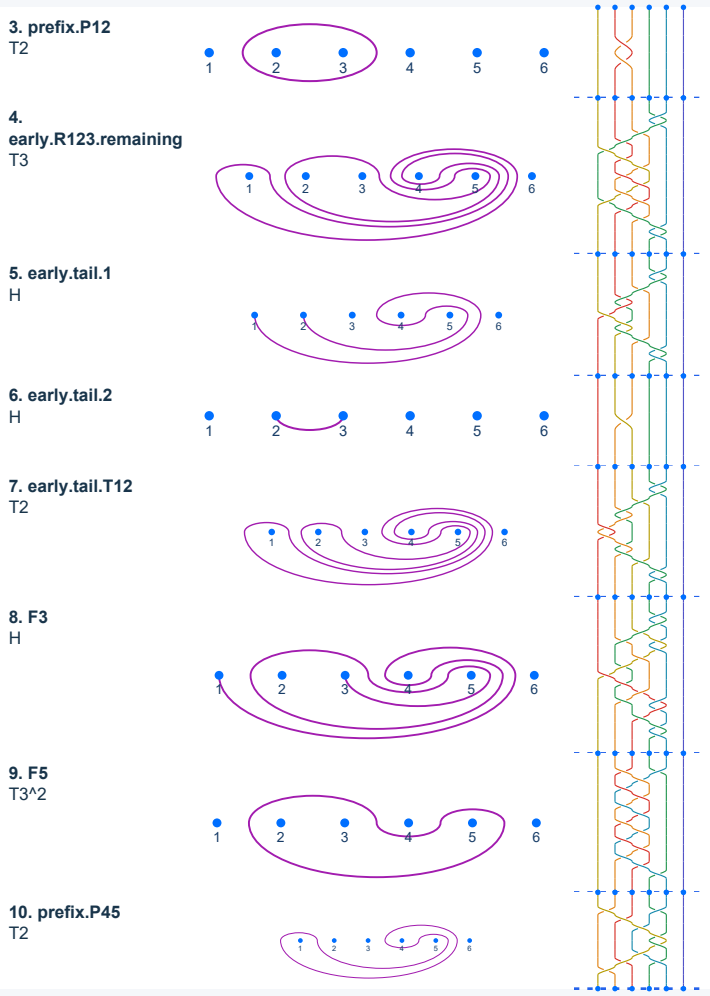
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 7.

Before



After



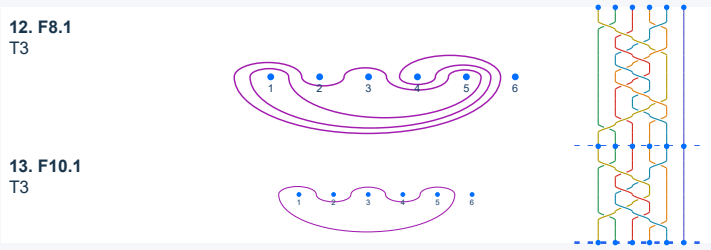
II - step 021

F10.1 up past F8.1

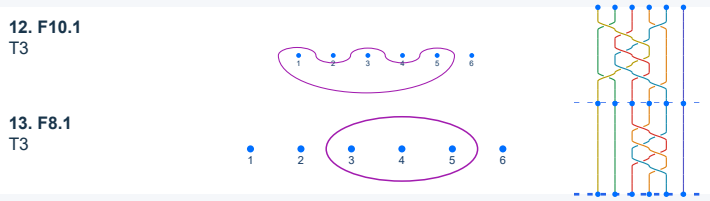
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



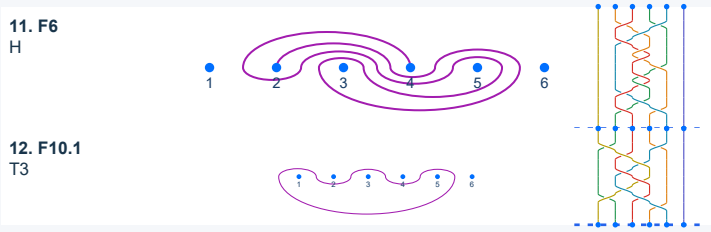
II - step 022

F10.1 up past F6

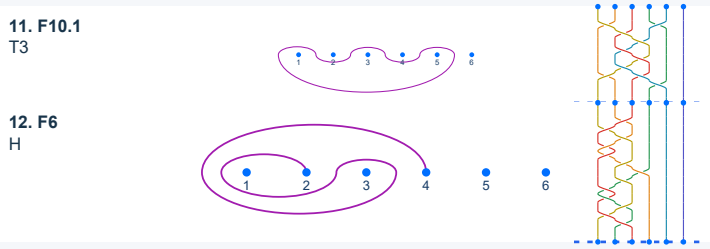
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



II - step 023

F8.1 up past F6

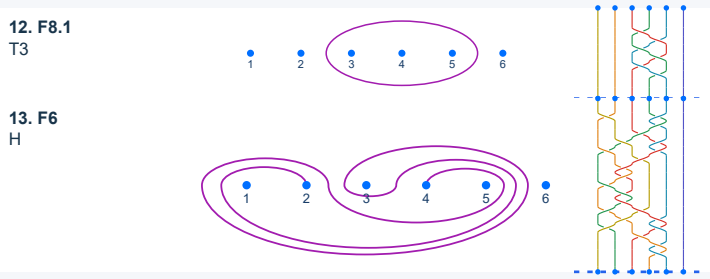
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



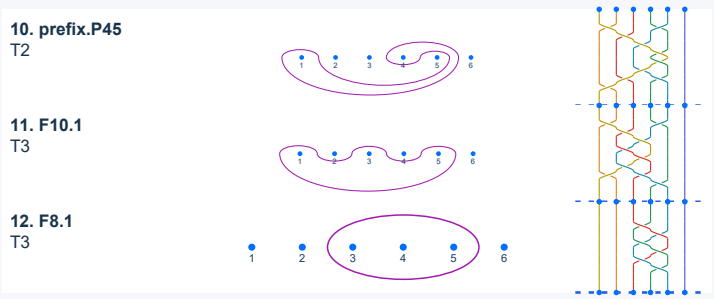
II - step 024

PC cluster lantern: Q14 and the two roots become O1345 and P35

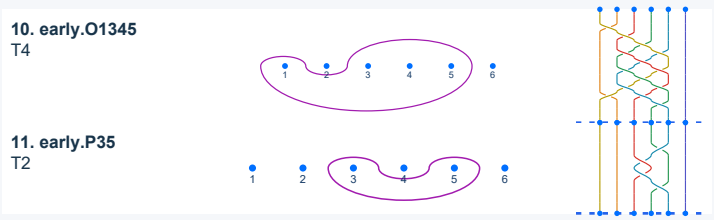
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 7 below. Only the changed interval is shown.

Before



After



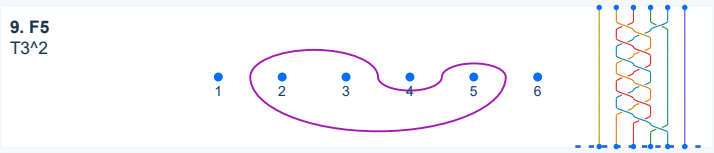
II - step 025

Split F5 into its two positive units

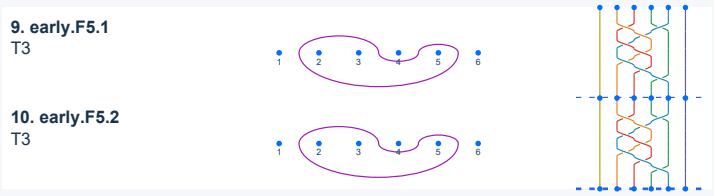
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before



After



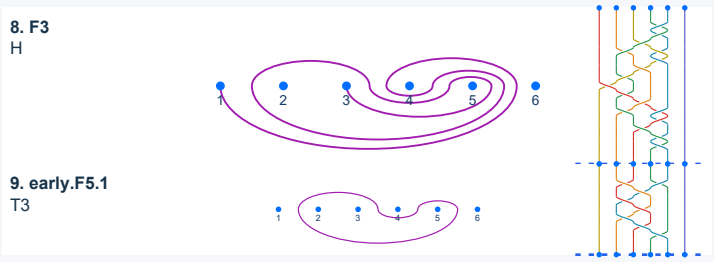
II - step 026

Carry one F5 root up across F3

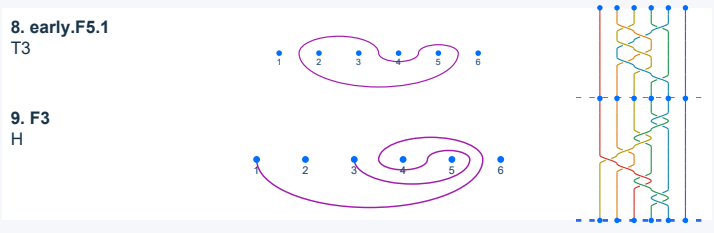
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



II - step 027

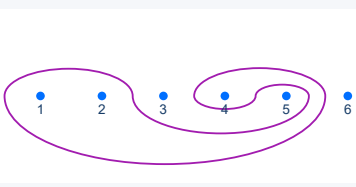
Split F13 into its two positive units

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

18 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

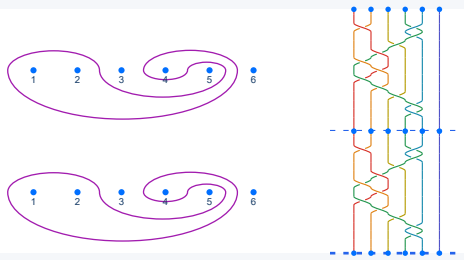
19. F13
T3^2



After

19. early.F13.1
T3

20. early.F13.2
T3



II - step 028

Carry one F13 root up across four neighbors

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before

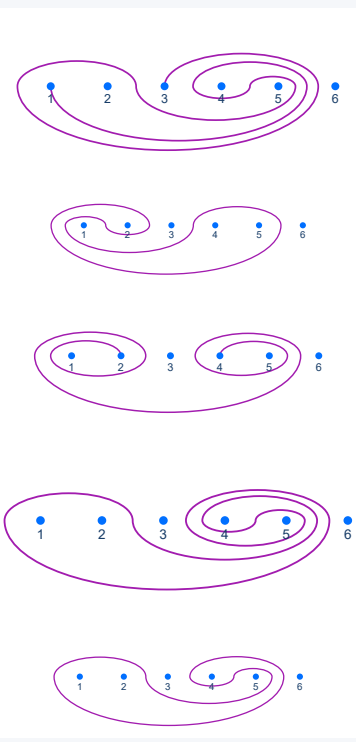
15. F9
H

16. F10.2
T3

17. F11
H

18. F12
T3^2

19. early.F13.1
T3



After

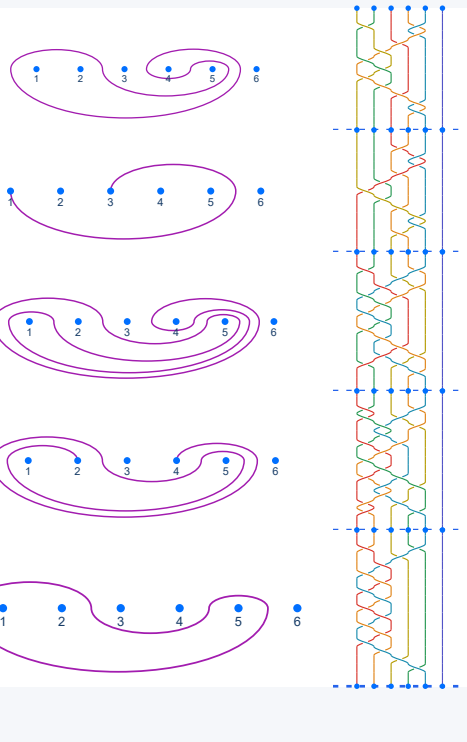
15. early.F13.1
T3

16. F9
H

17. F10.2
T3

18. F11
H

19. F12
T3^2



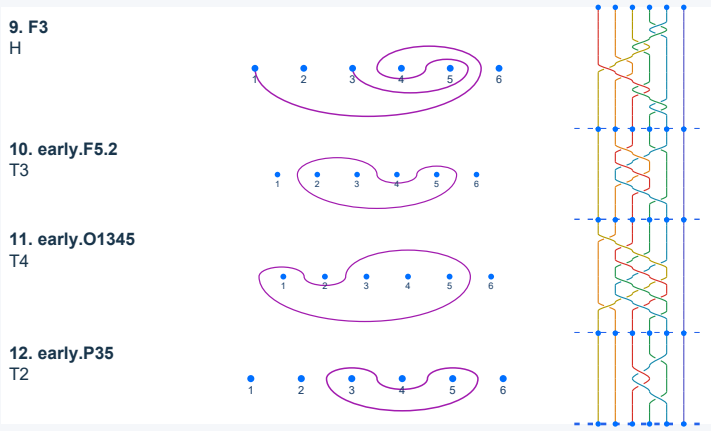
II - step 029

P35 transports F3 to exactly the current F9 arc

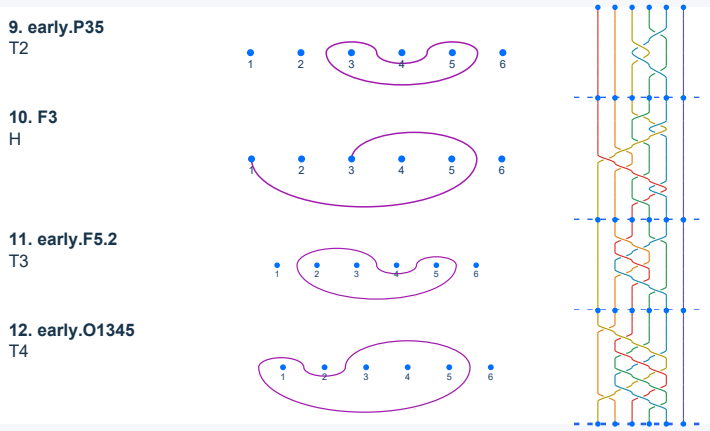
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



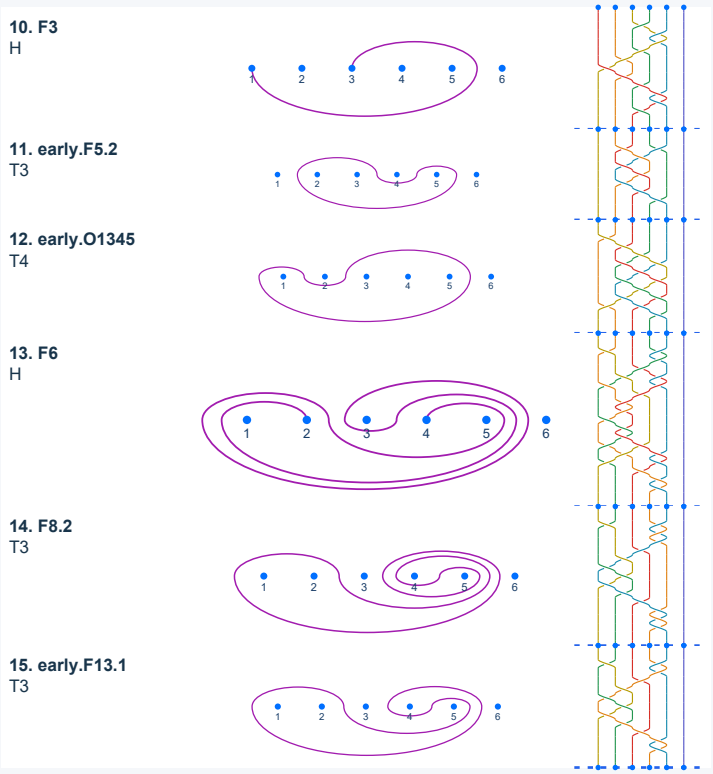
II - step 030

Drag the matched F3 down to F9; F6 becomes short

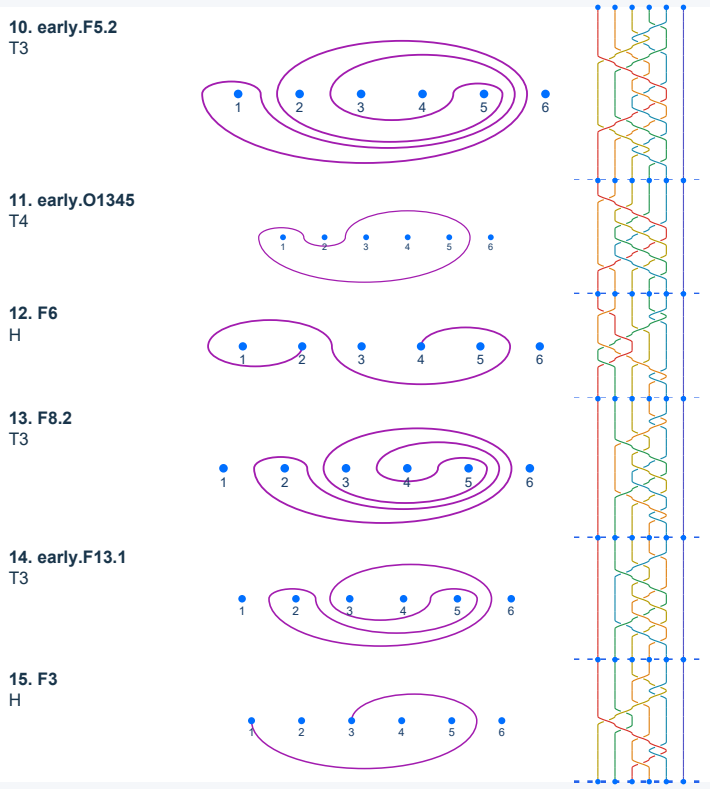
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



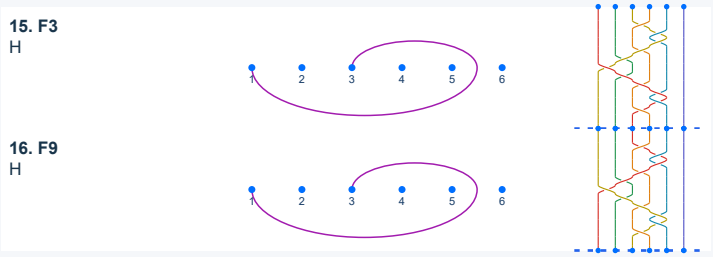
II - step 031

Combine the matching F3 and F9

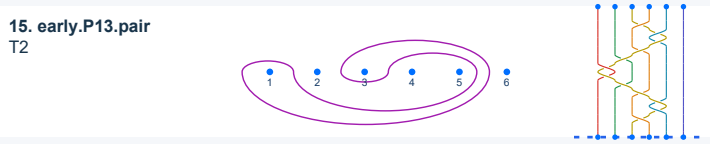
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



II - step 032

PC daisy of O1345, conjugated inside by sigma4 inverse

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 8 below. Only the changed interval is shown.

Before

11. early.O1345
T4



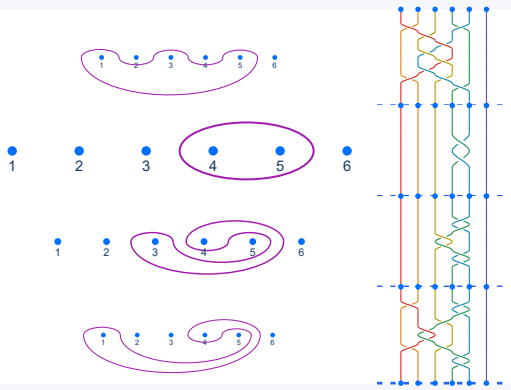
After

11. early.daisy.1
T3

12. early.daisy.2
T2

13. early.daisy.3
T2

14. early.daisy.4
T2



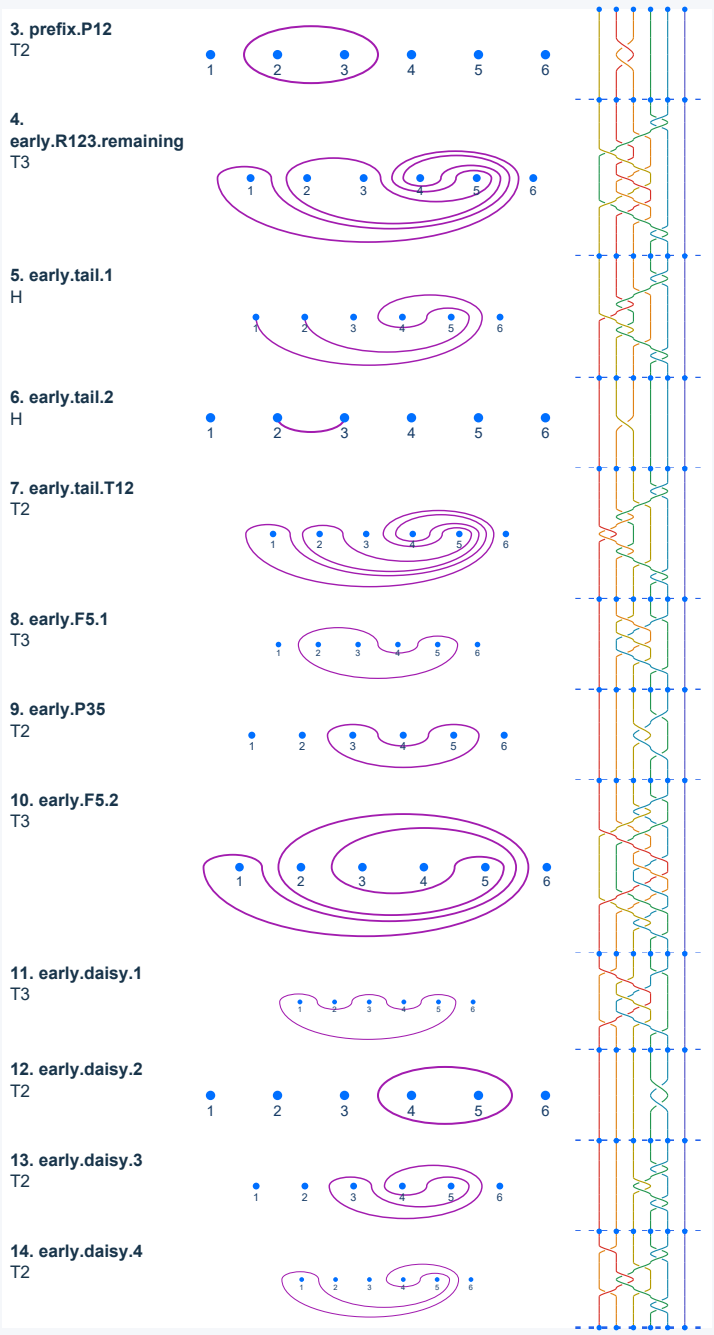
II - step 033

Carry the actual Q14 petal up; the two H12 arcs match

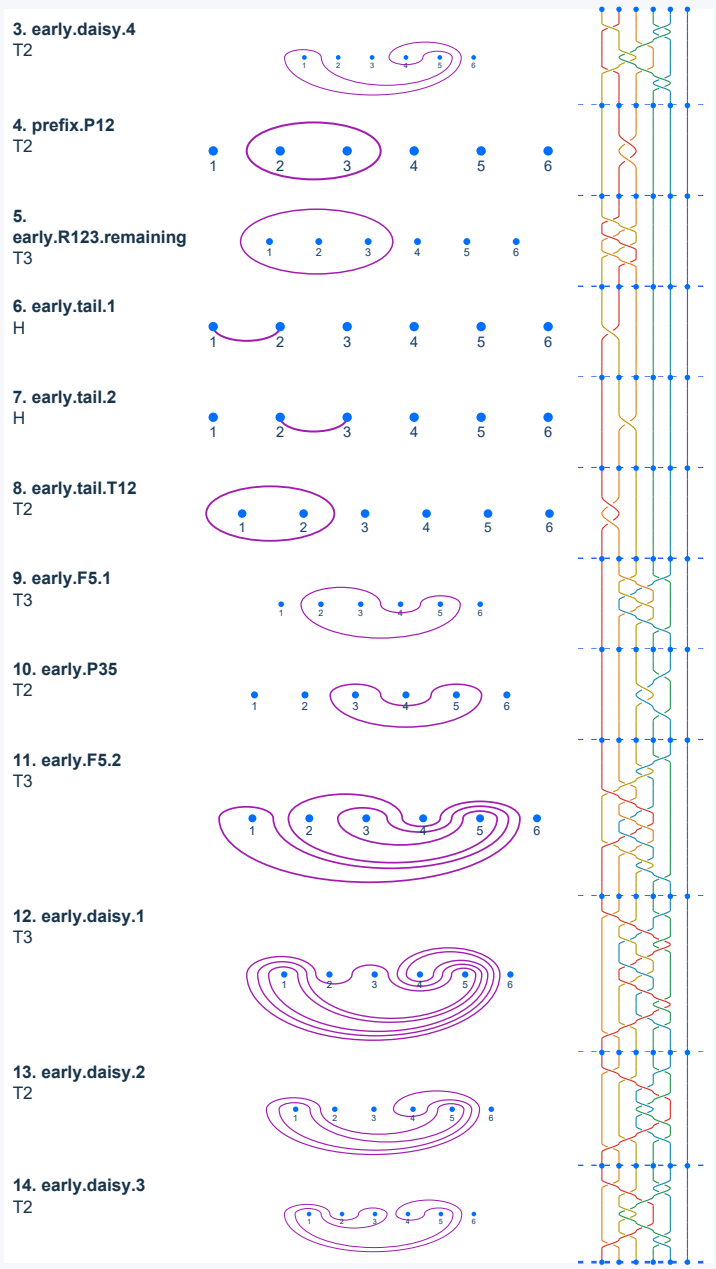
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 11.

Before



After



II - step 034

Gather the two exact H12 arcs

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 16 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



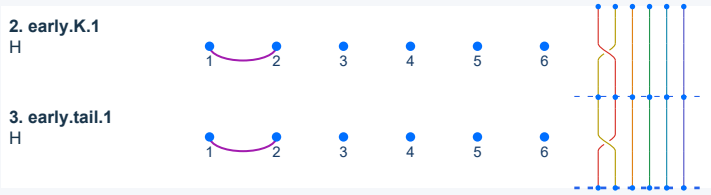
II - step 035

Combine the matching early.K.1 and early.tail.1

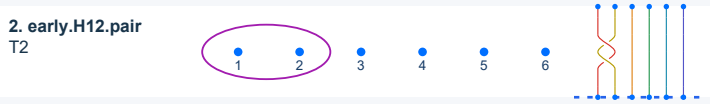
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 19 below. Only the changed interval is shown.

Before



After



II - step 036

Gather the two exact H23 arcs

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 15 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



II - step 037

Combine the matching early.K.2 and early.tail.2

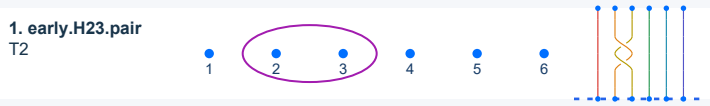
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 19 below. Only the changed interval is shown.

Before



After



II - step 038

Split F12 into its two positive units

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

18 unchanged displayed factors above; 1 below. Only the changed interval is shown.

Before

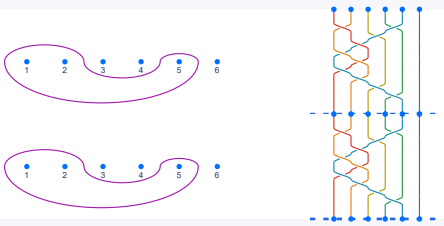
19. F12
T3^2



After

19. early.F12.1
T3

20. early.F12.2
T3



II - step 039

One C12 root shortens the final F11 arc

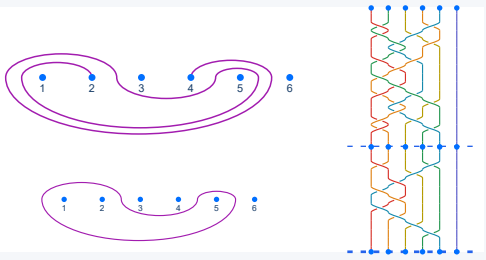
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

17 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before

18. F11
H

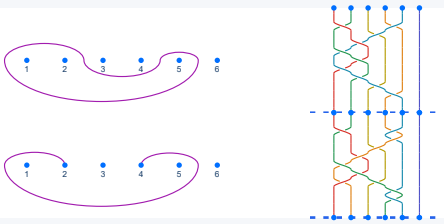
19. early.F12.1
T3



After

18. early.F12.1
T3

19. F11
H



II - step 040

Gather the two round T12 twists across R123

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 15 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before

5. early.R123.remaining
T3

6. early.tail.T12
T2



After

5. early.tail.T12
T2

6. early.R123.remaining
T3



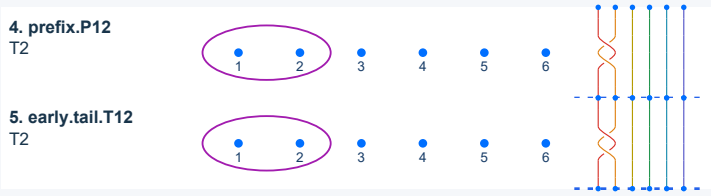
II - step 041

Combine the matching prefix.P12 and early.tail.T12

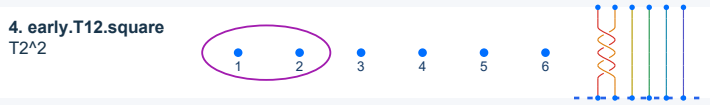
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 16 below. Only the changed interval is shown.

Before



After



II - step 042

Carry the round T12 square to the top

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 16 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



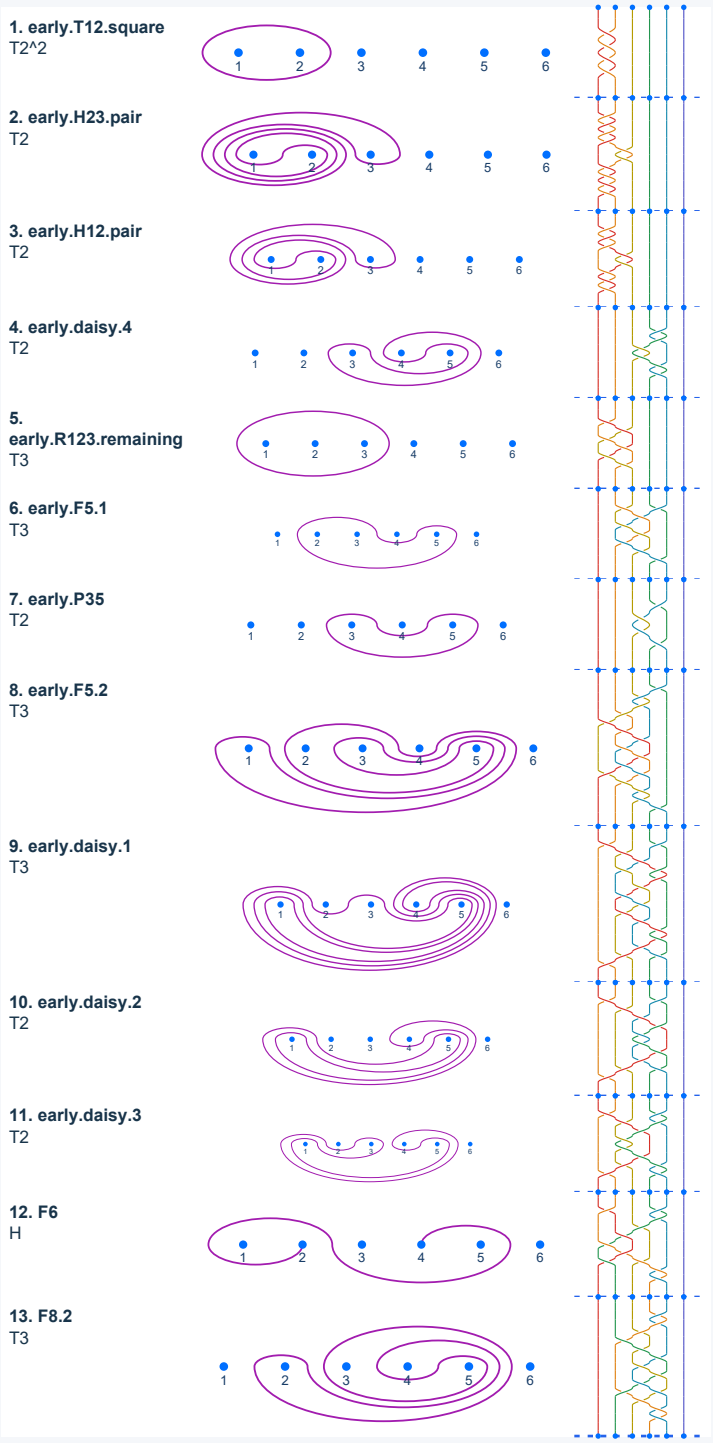
II - step 043

Rotate the first factor to the bottom of the central product

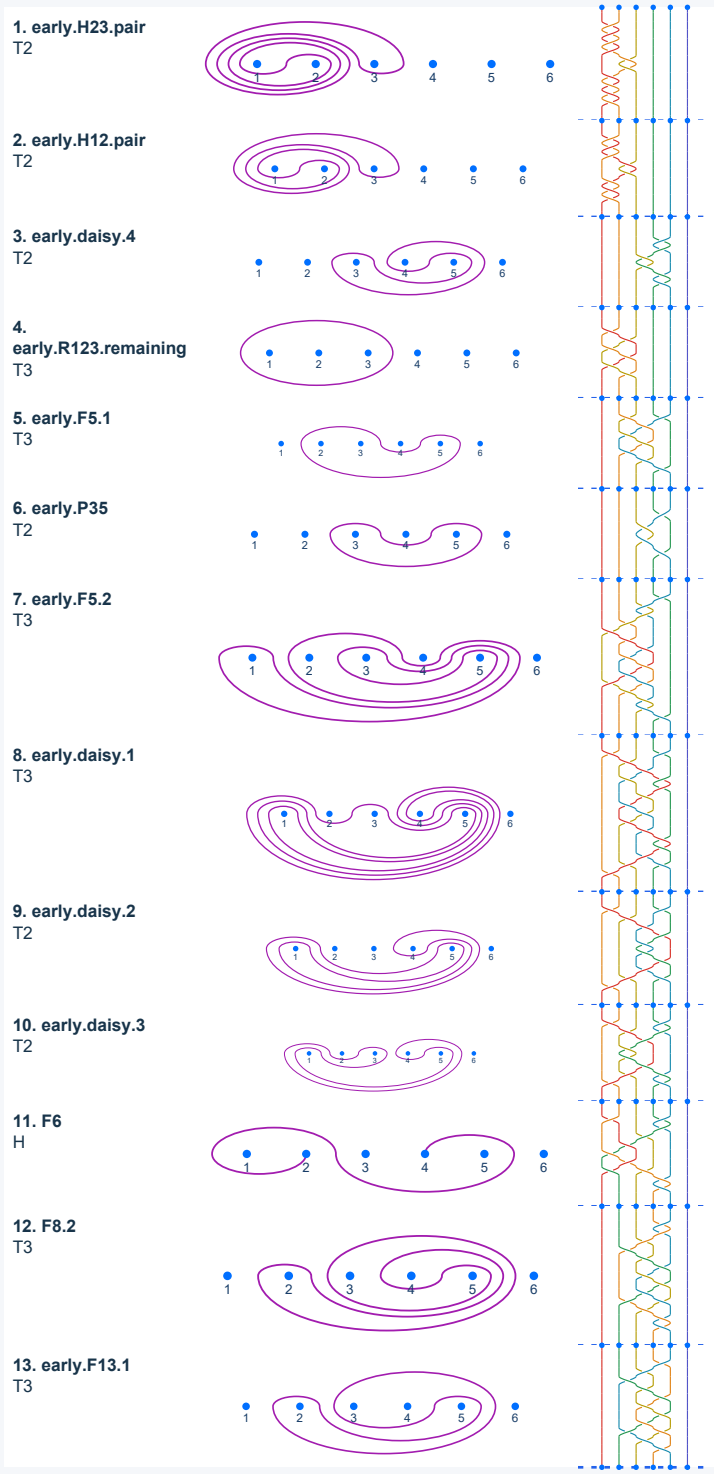
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



II - step 043 - continued

Rotate the first factor to the bottom of the central product

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before

14. early.F13.1
T3

15. early.P13.pair
T2

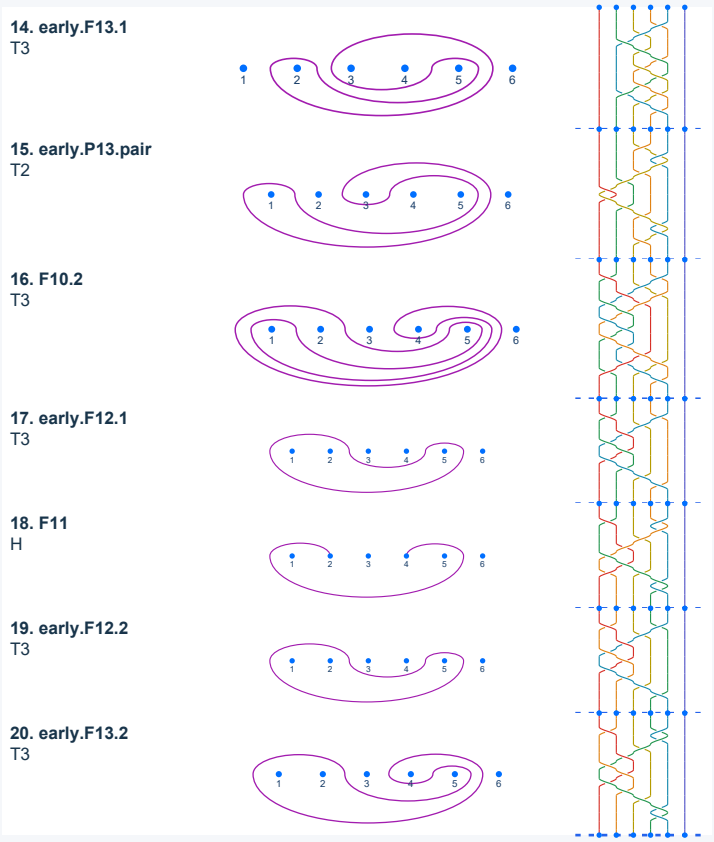
16. F10.2
T3

17. early.F12.1
T3

18. F11
H

19. early.F12.2
T3

20. early.F13.2
T3



After

14. early.P13.pair
T2

15. F10.2
T3

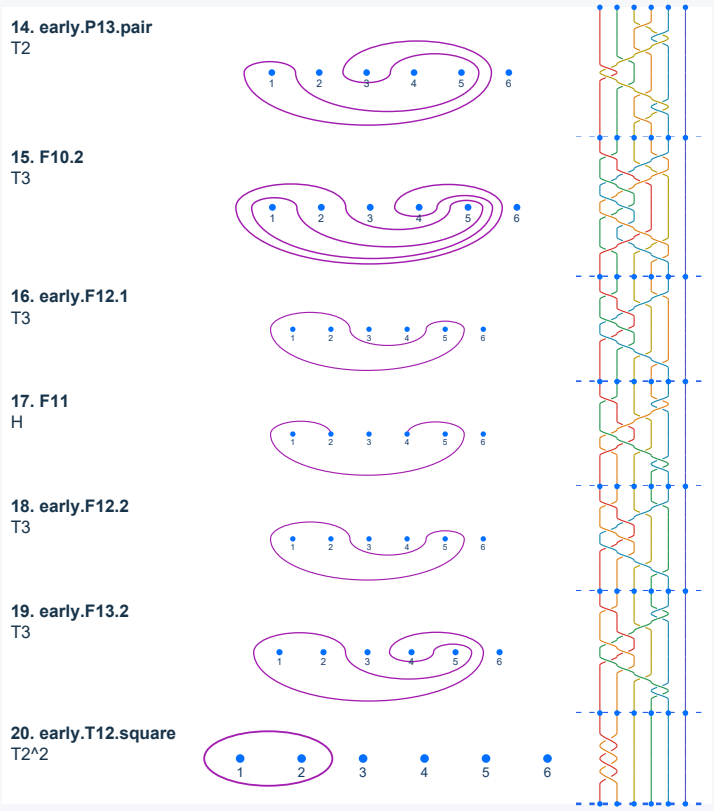
16. early.F12.1
T3

17. F11
H

18. early.F12.2
T3

19. early.F13.2
T3

20. early.T12.square
T2^2



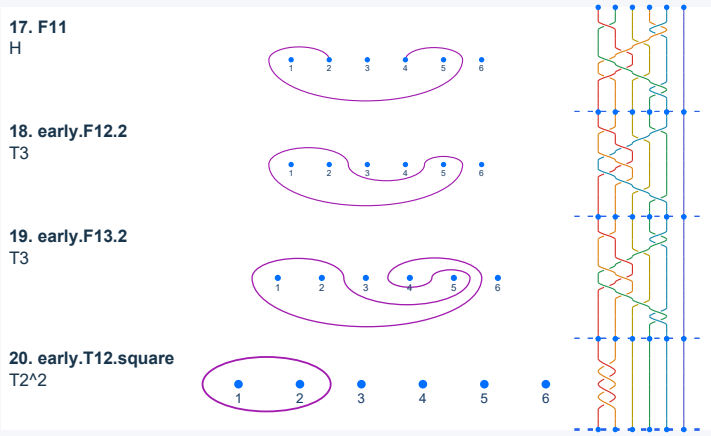
II - step 044

Carry T12 squared up the short side; F11 now equals F6

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



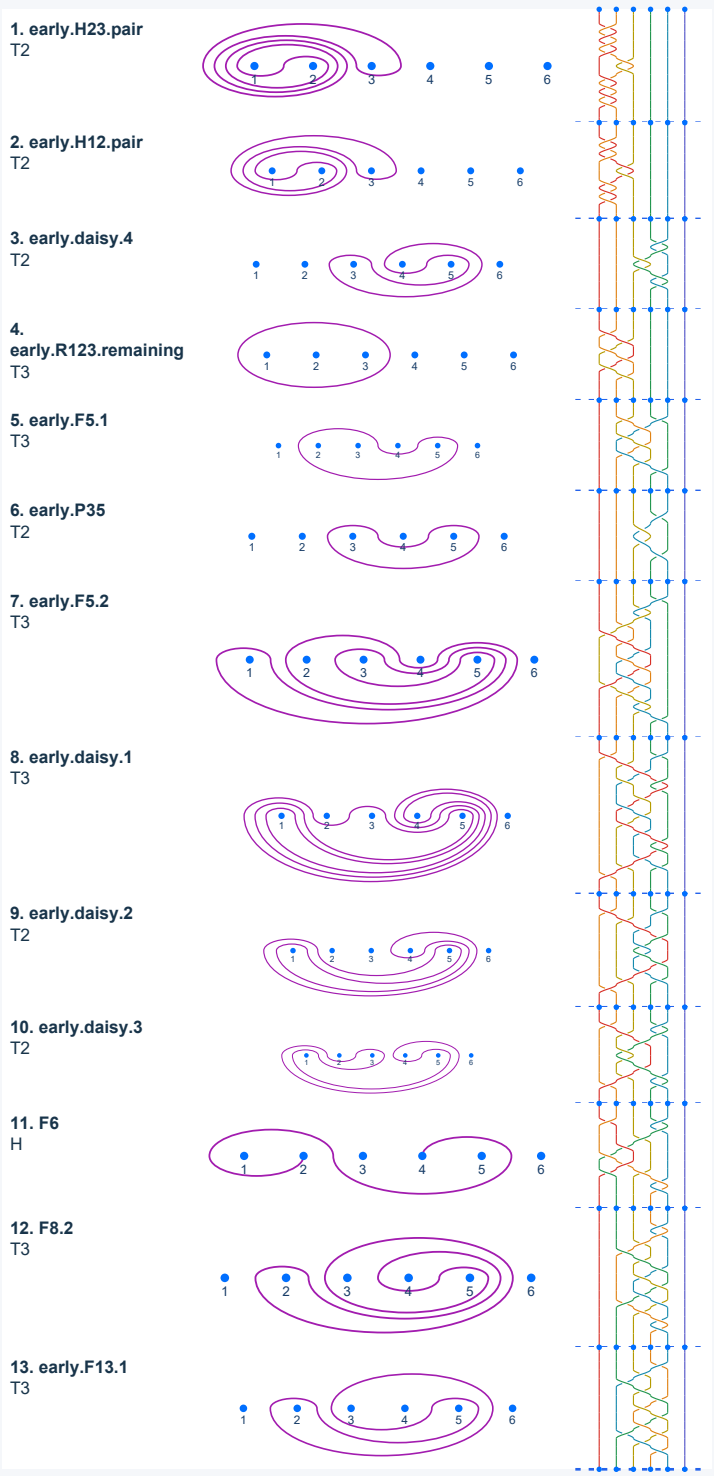
II - step 045

Common disk frame removes the shared winding before the last gather

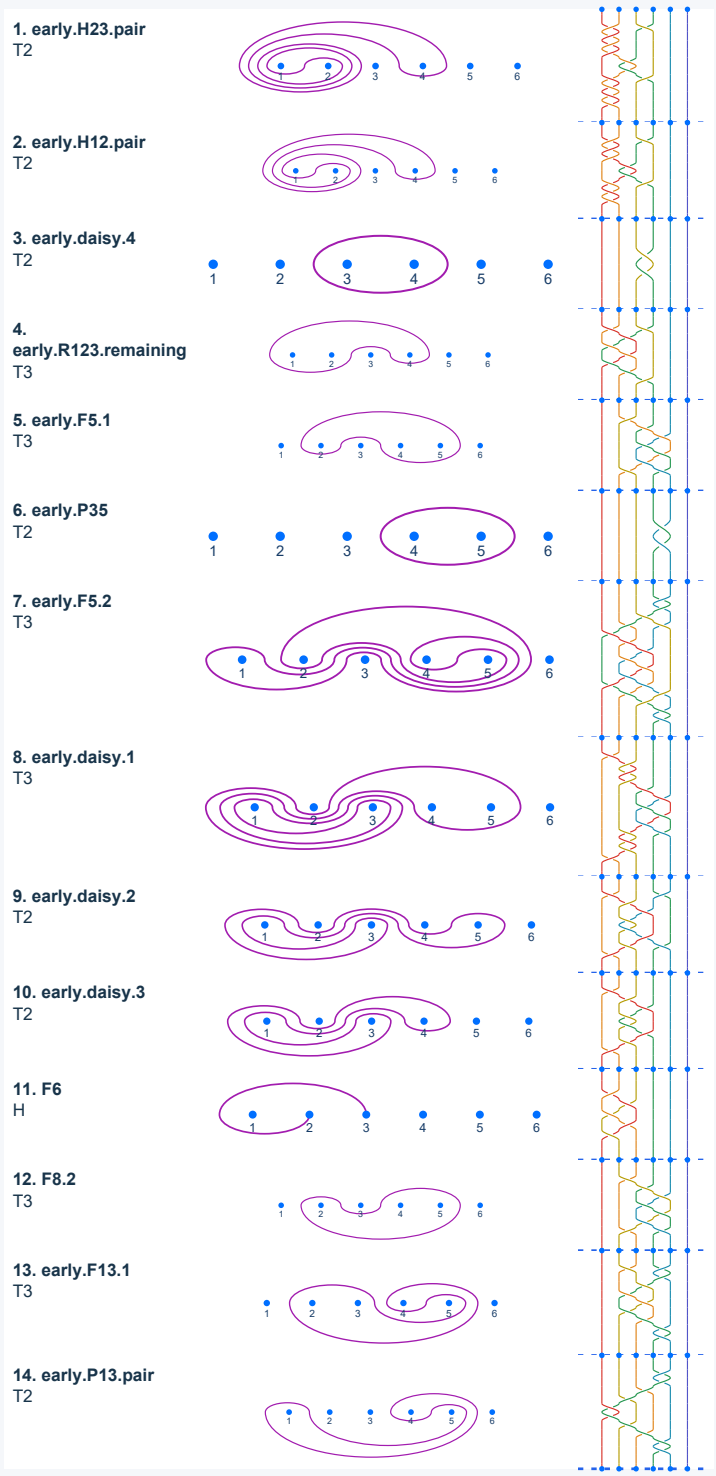
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



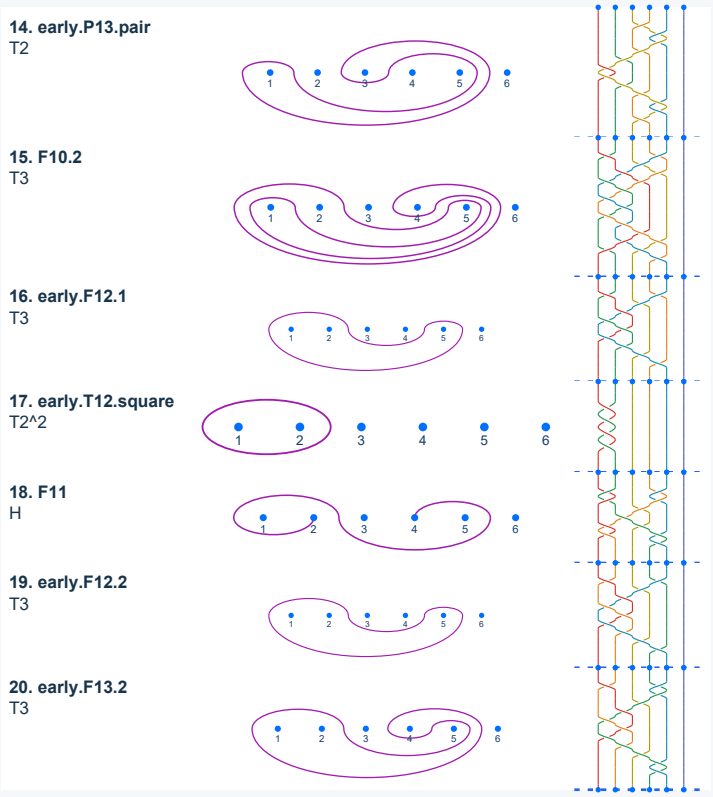
II - step 045 - continued

Common disk frame removes the shared winding before the last gather

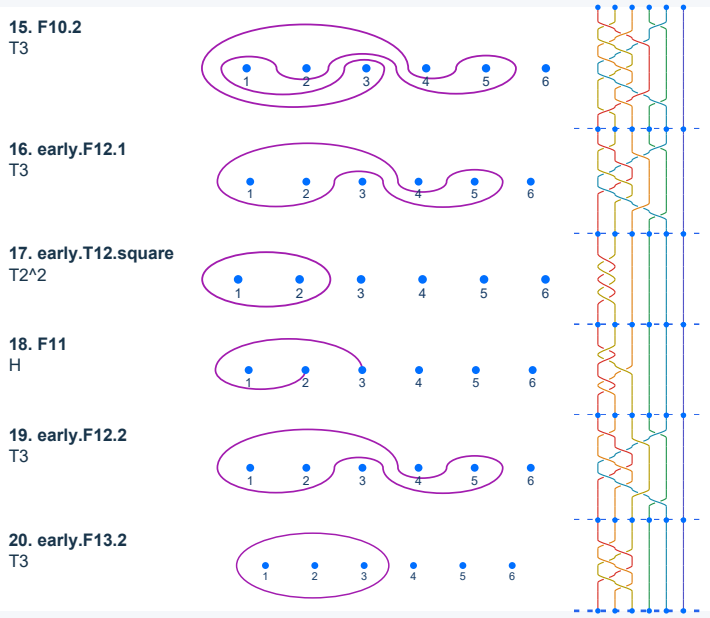
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



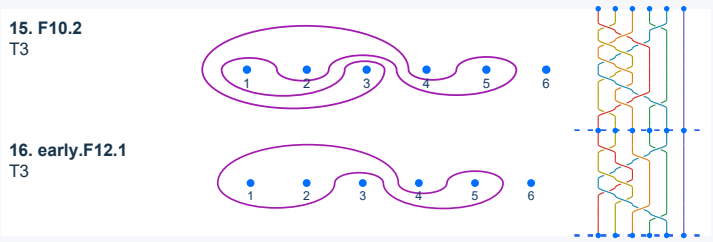
II - step 046

Reorder C12 before C10 to keep the next transport bounded

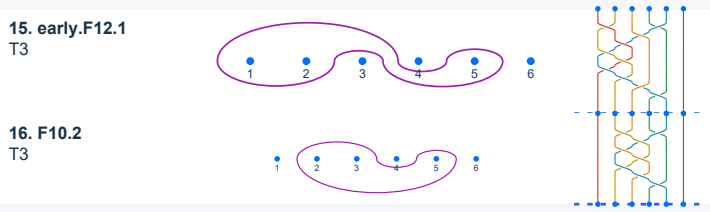
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



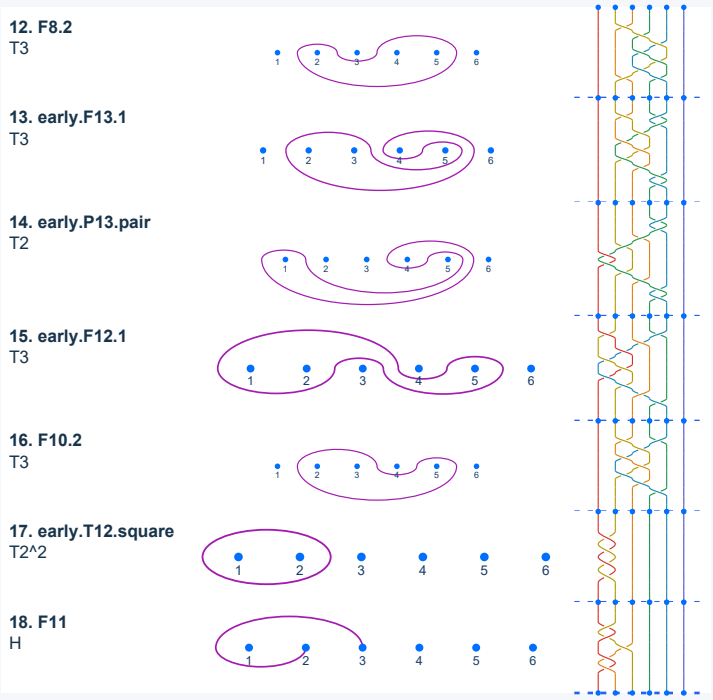
II - step 047

Bring the last matched half twists together

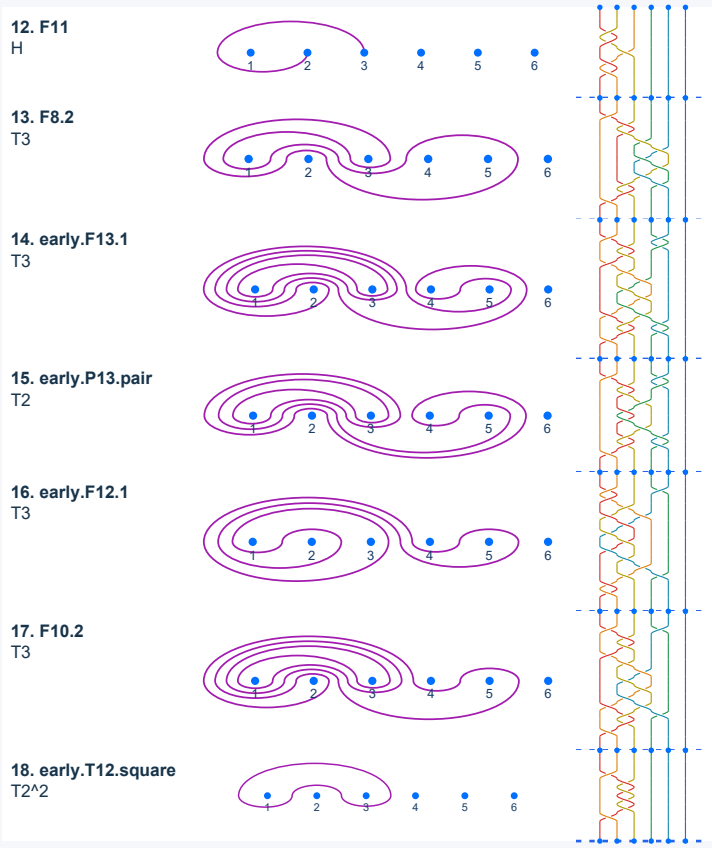
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



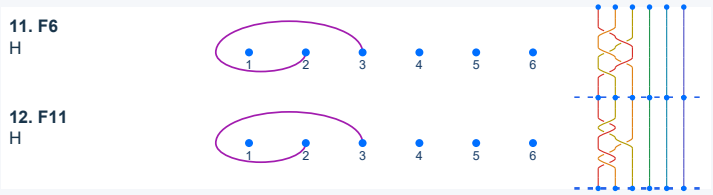
II - step 048

Combine the matching F6 and F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 8 below. Only the changed interval is shown.

Before



After



III. The first two cluster contractions

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

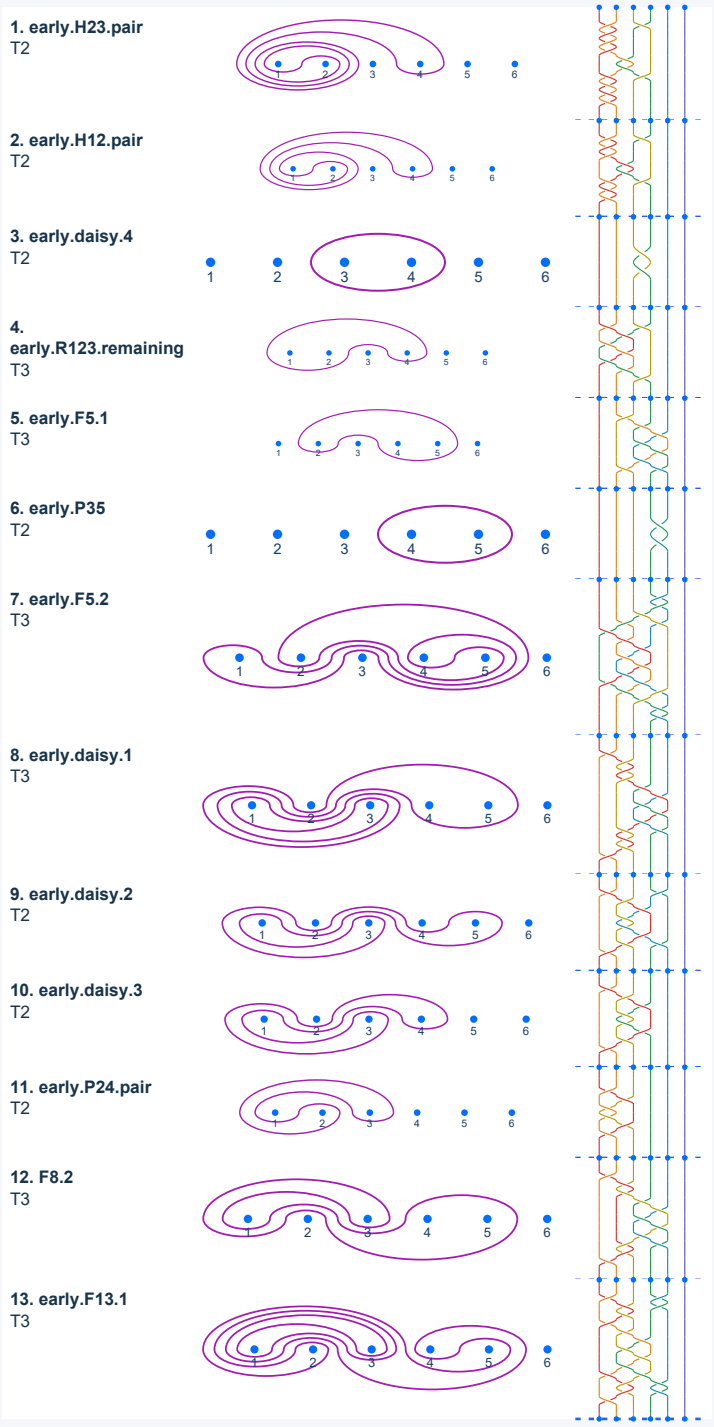
III - step 049

Put the next four-point outer curve in its round disk chart

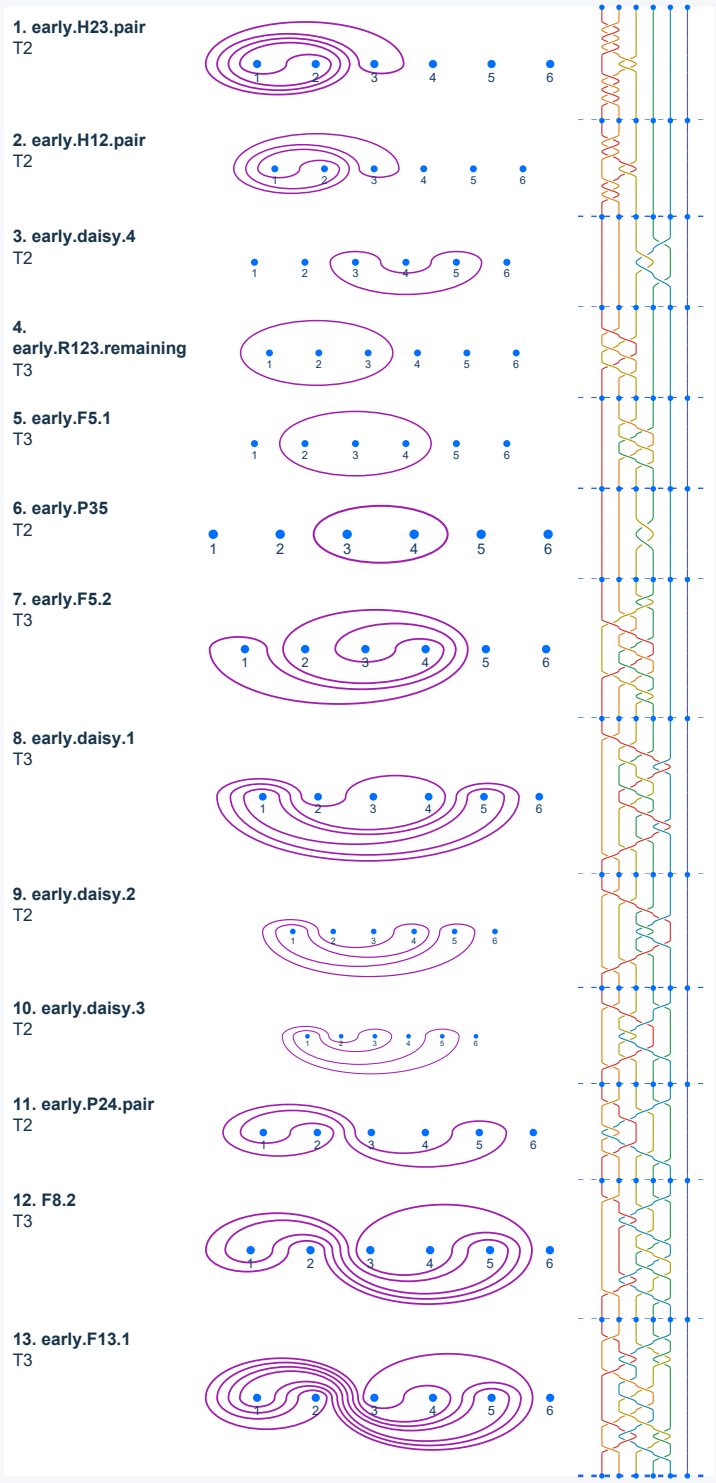
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



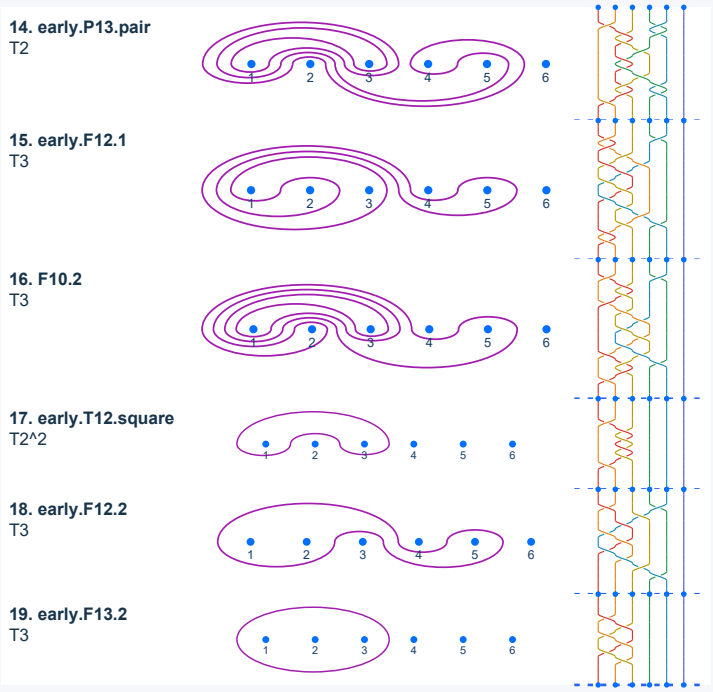
III - step 049 - continued

Put the next four-point outer curve in its round disk chart

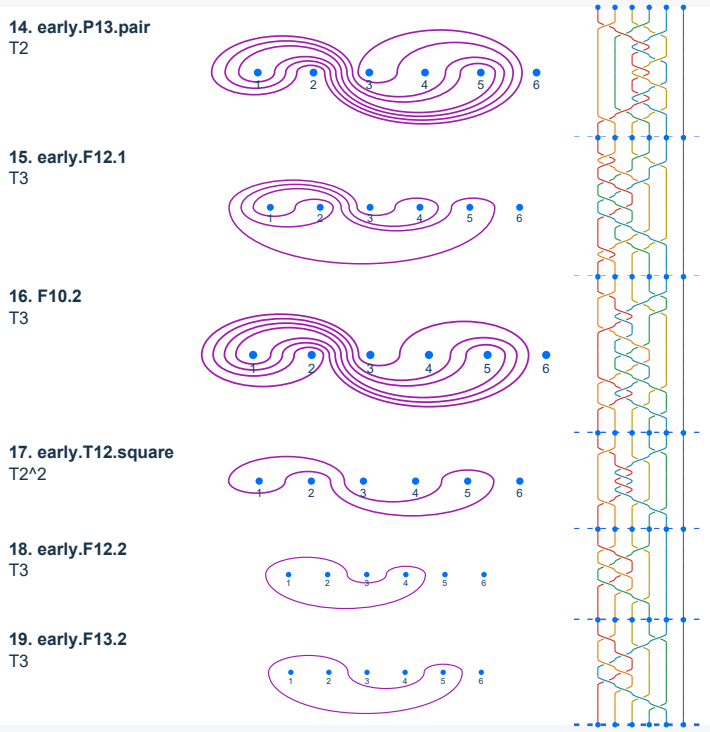
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



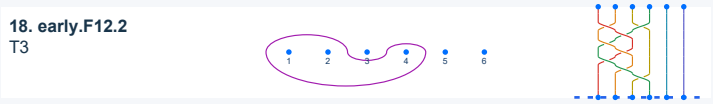
III - step 050

PC daisy of the actual C12 unit exposes Q15

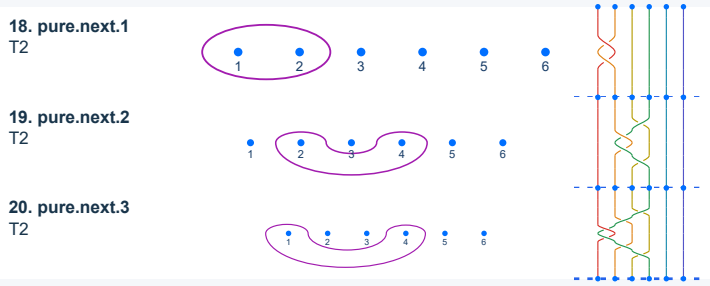
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

17 unchanged displayed factors above; 1 below. Only the changed interval is shown.

Before



After



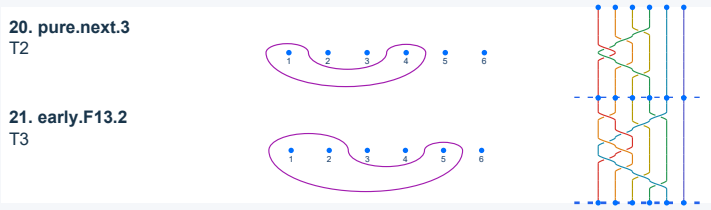
III - step 051

Carry Q15 past the final three-point root

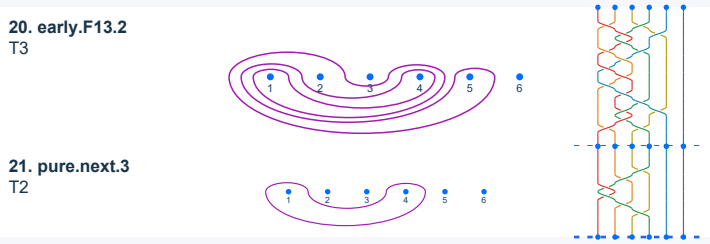
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

19 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



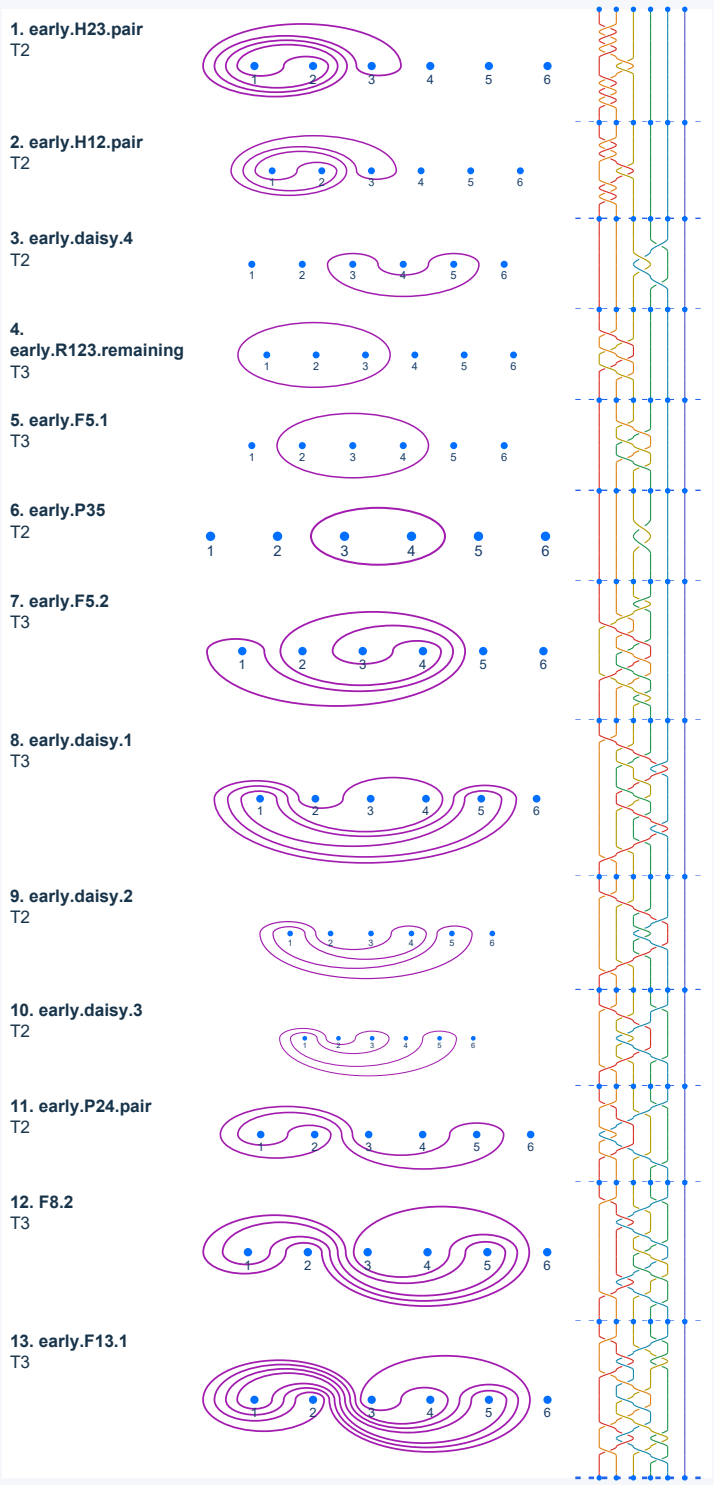
III - step 052

Rotate the last Q15 to the top of the central disk product

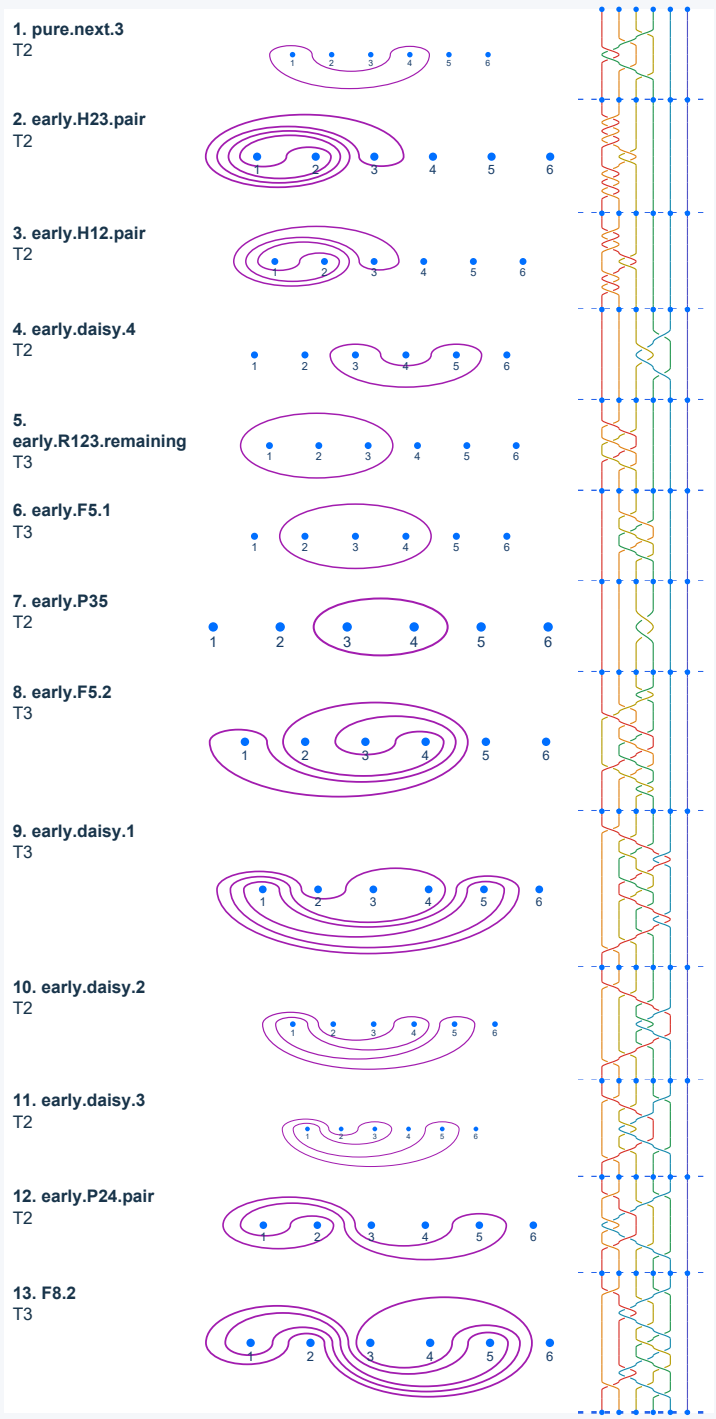
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



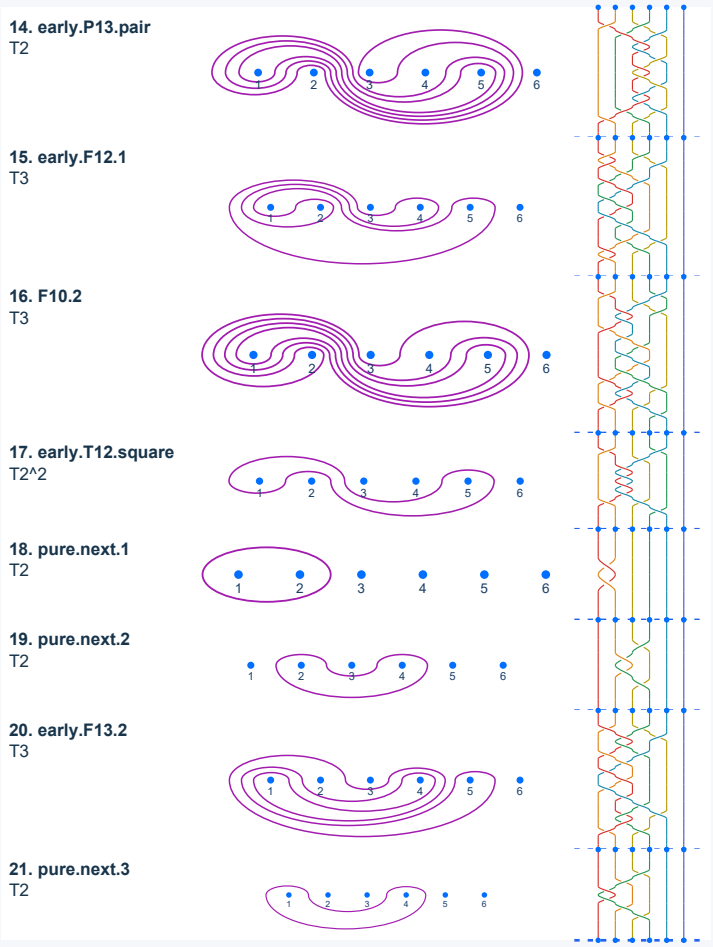
III - step 052 - continued

Rotate the last Q15 to the top of the central disk product

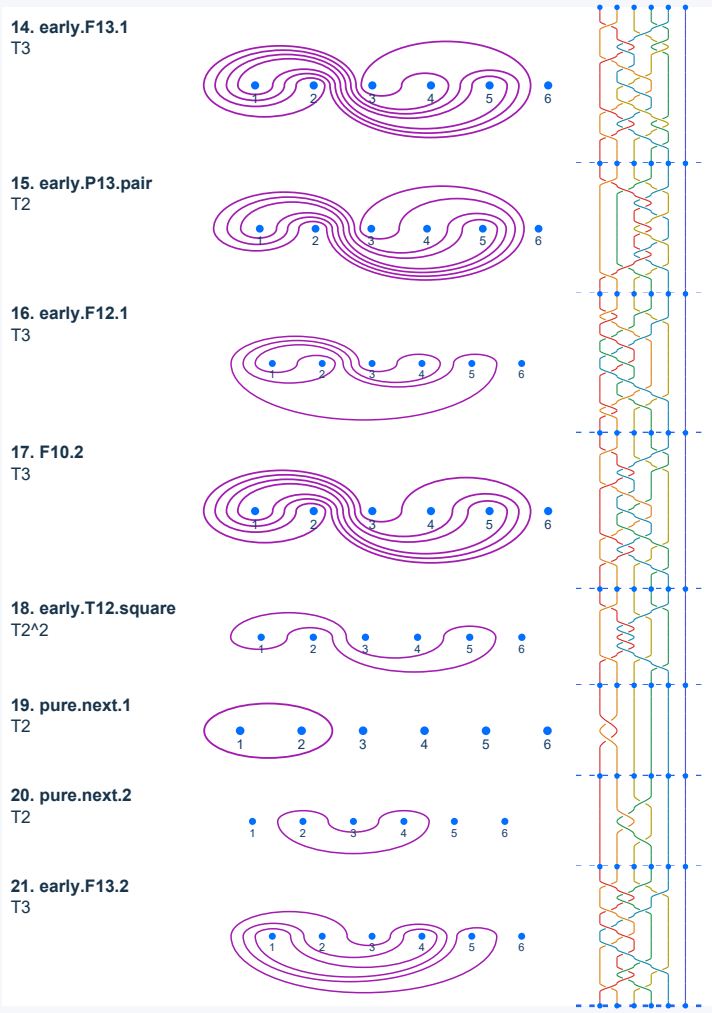
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



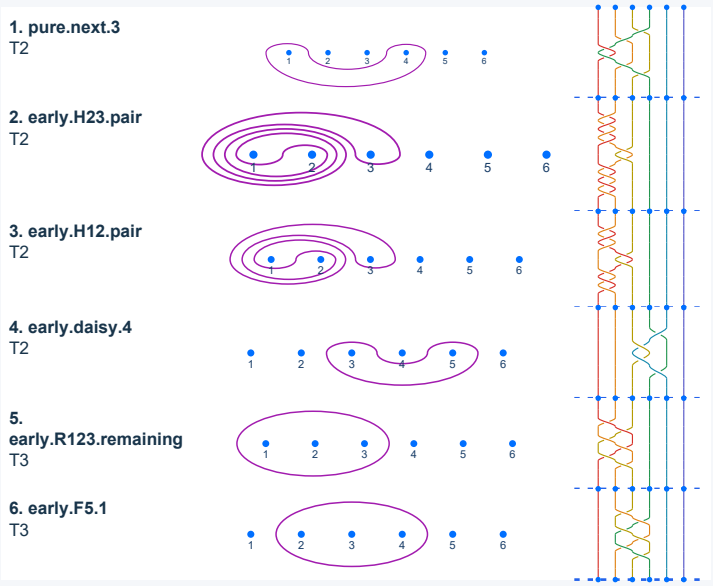
III - step 053

Bring Q15 through the leading pair curves and the two roots

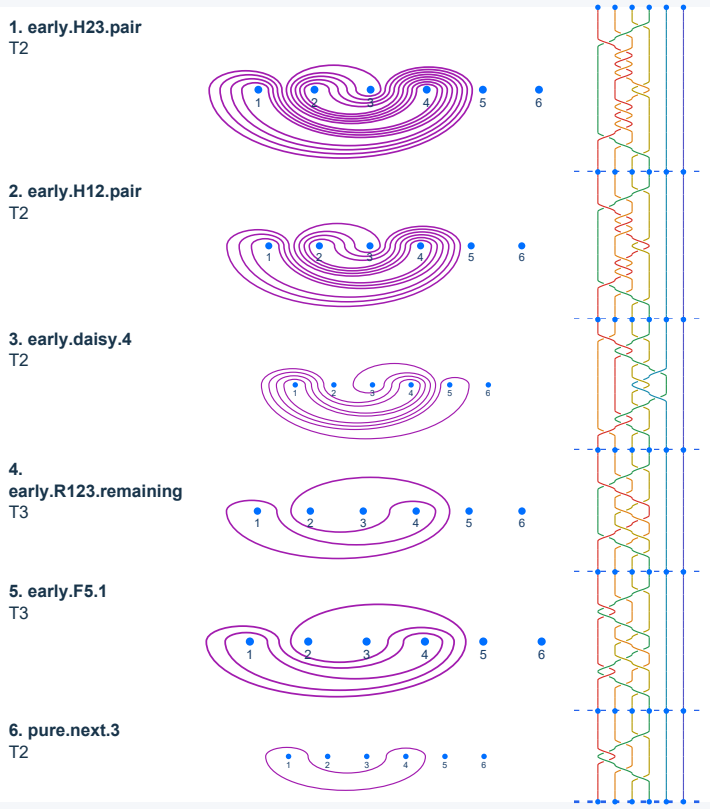
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



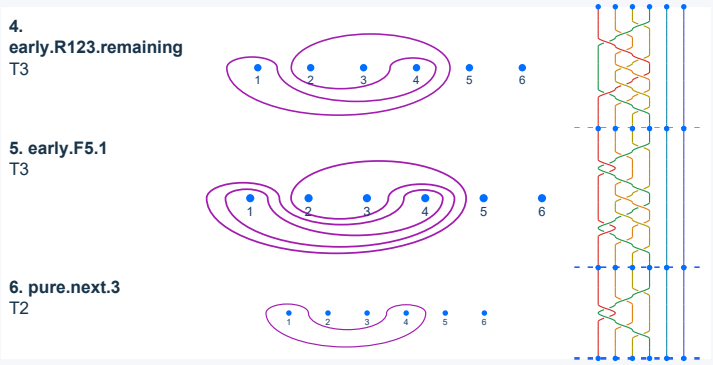
III - step 054

PC cluster lantern produces a round four-point curve and T23

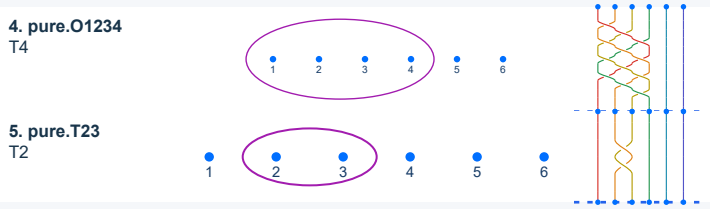
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 15 below. Only the changed interval is shown.

Before



After



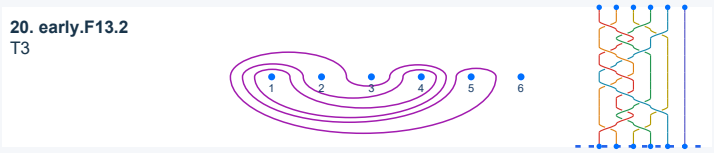
III - step 055

The final C13 daisy supplies the precise missing prefix petal

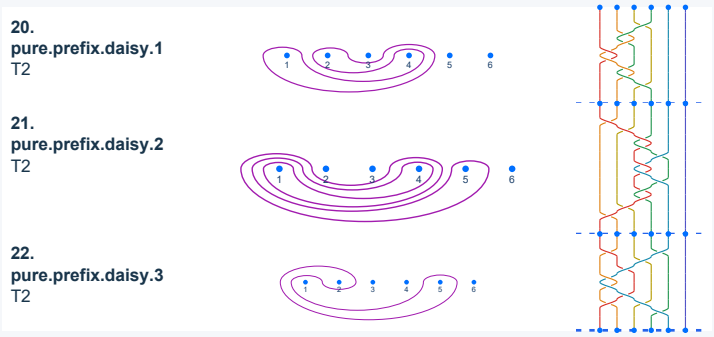
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

19 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



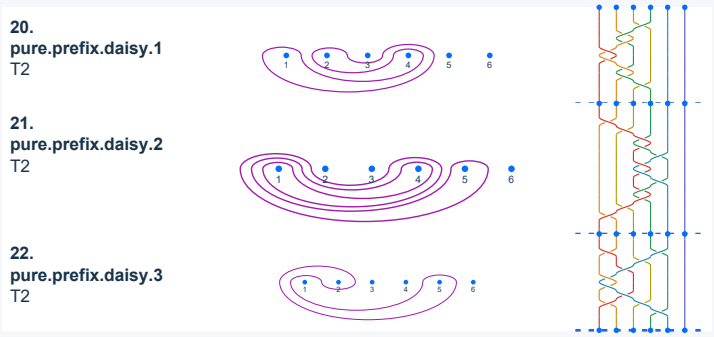
III - step 056

Carry the first petal through the other two daisy petals

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

19 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



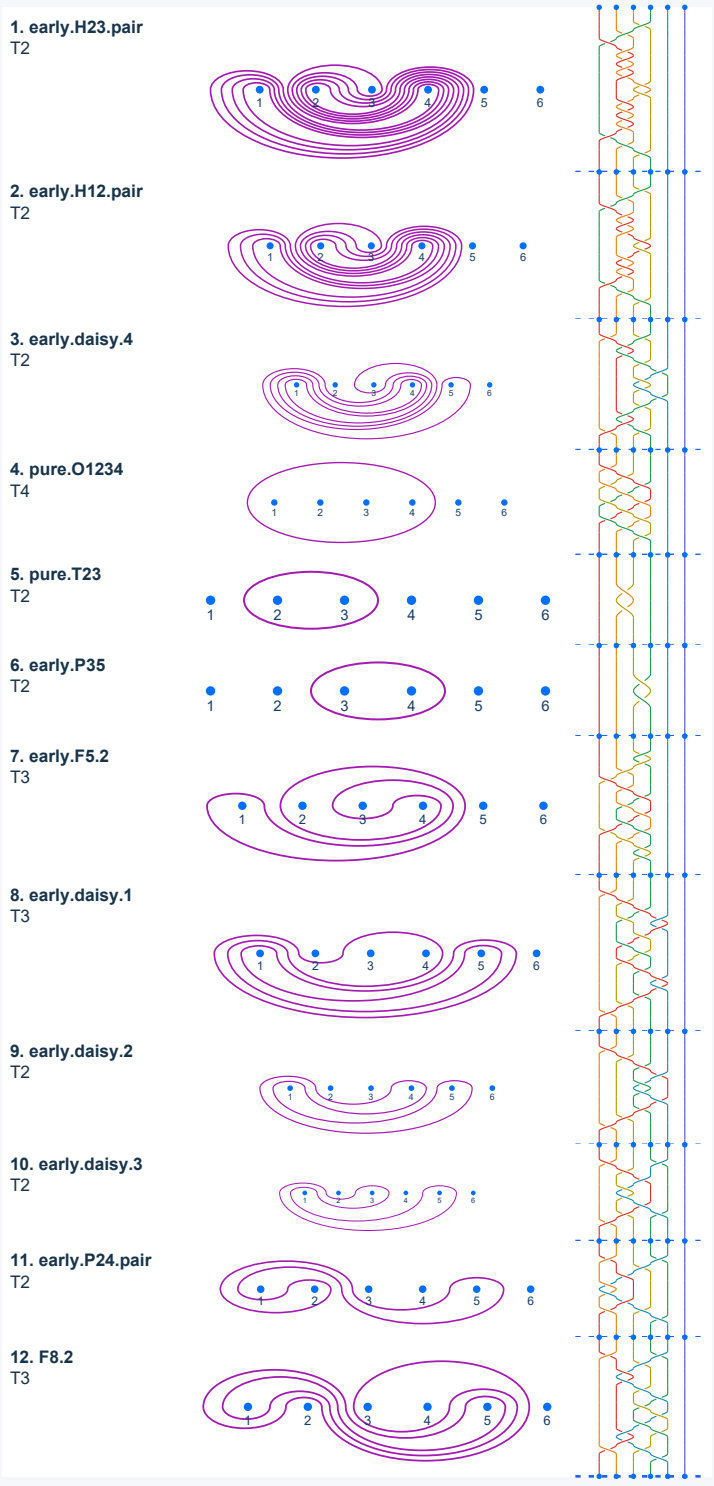
III - step 057

Rotate this last petal to the top of the central product

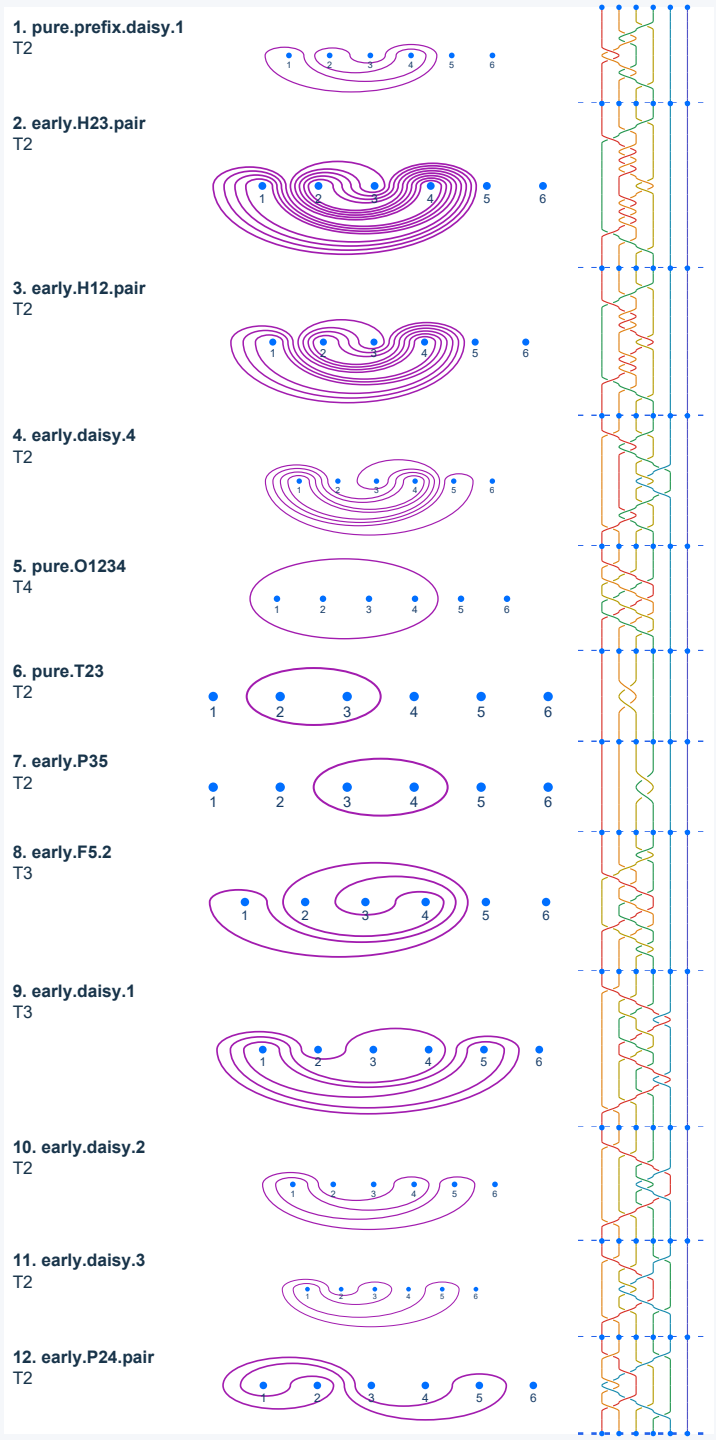
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



III - step 057 - continued

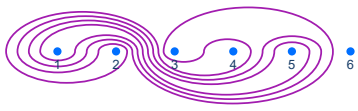
Rotate this last petal to the top of the central product

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

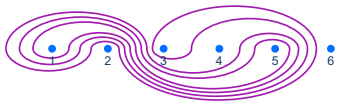
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before

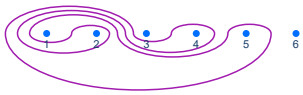
13. early.F13.1
T3



14. early.P13.pair
T2



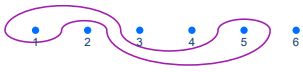
15. early.F12.1
T3



16. F10.2
T3



17. early.T12.square
T2^2



18. pure.next.1
T2



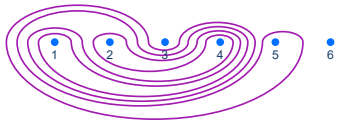
19. pure.next.2
T2



20. pure.prefix.daisy.2
T2

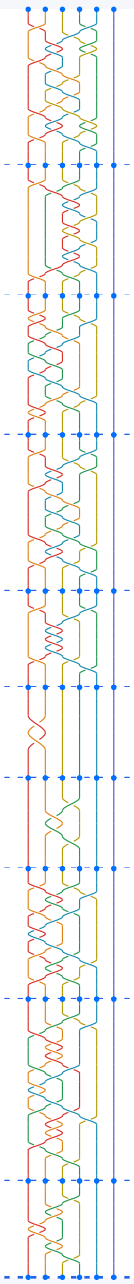


21. pure.prefix.daisy.3
T2



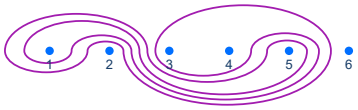
22. pure.prefix.daisy.1
T2



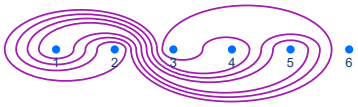


After

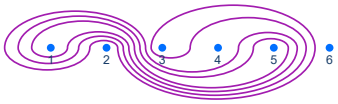
13. F8.2
T3



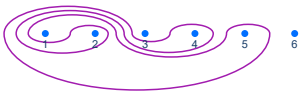
14. early.F13.1
T3



15. early.P13.pair
T2



16. early.F12.1
T3



17. F10.2
T3



18. early.T12.square
T2^2



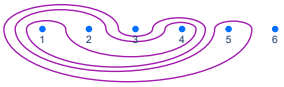
19. pure.next.1
T2



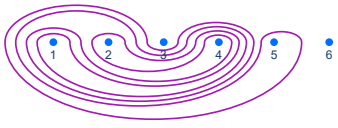
20. pure.next.2
T2

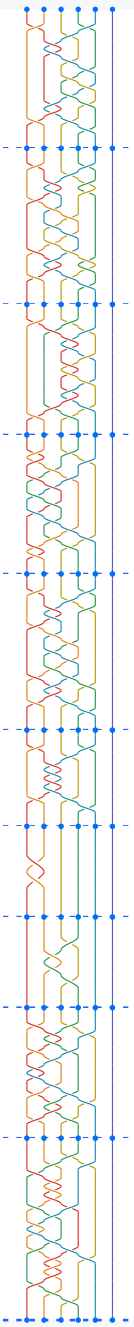


21. pure.prefix.daisy.2
T2



22. pure.prefix.daisy.3
T2





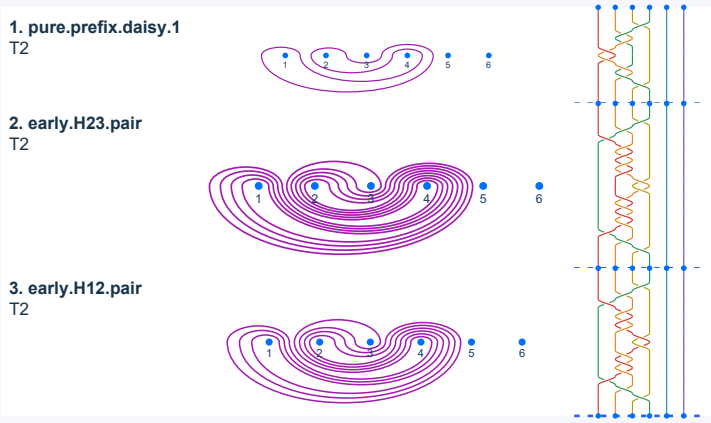
III - step 058

The supplied petal joins the two leading pair curves

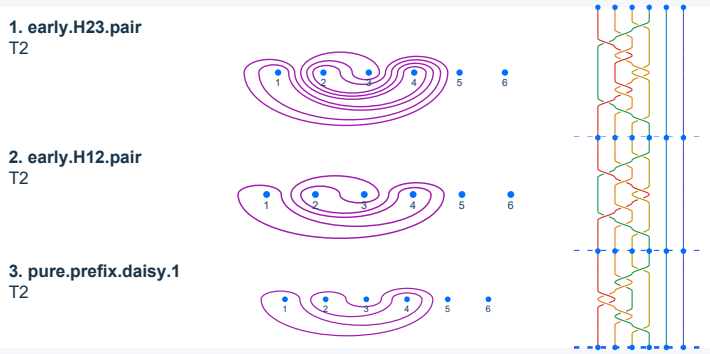
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 19 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



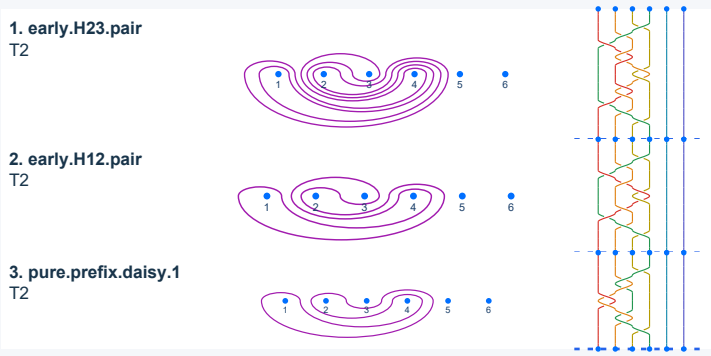
III - step 059

PC lantern contracts the two long prefix curves and their actual petal

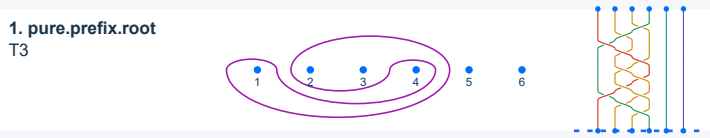
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 19 below. Only the changed interval is shown.

Before



After



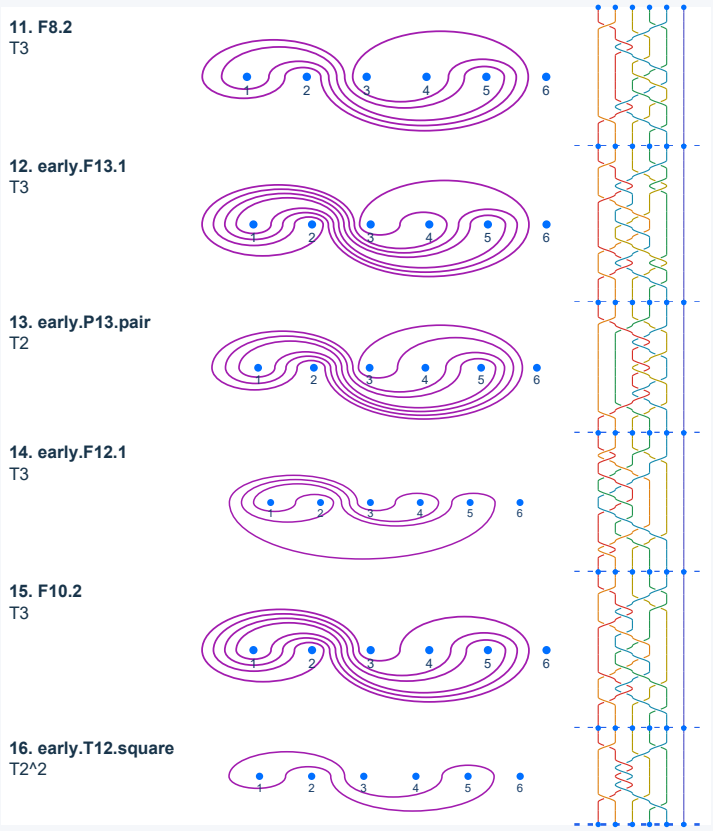
III - step 060

Carry the fixed pair square up across five neighbors; geometric crossings drop

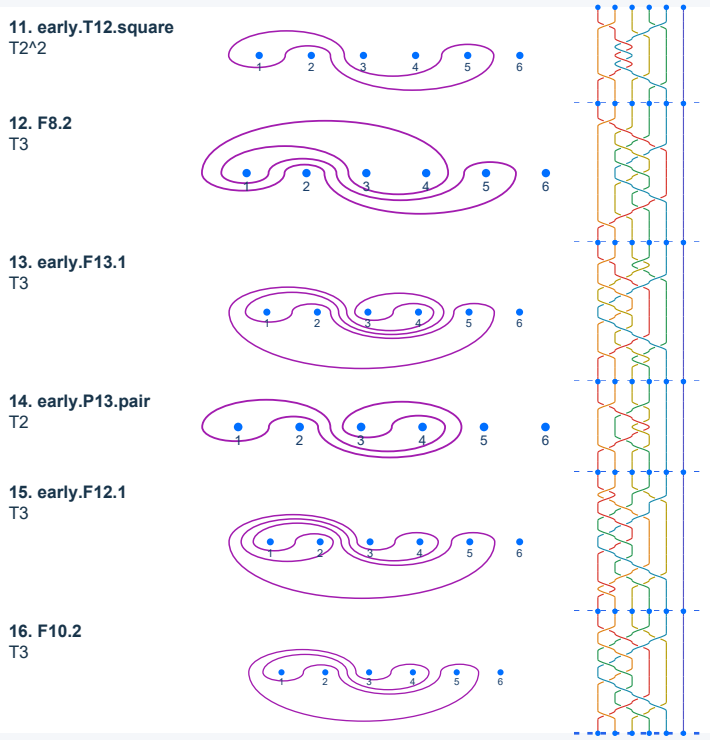
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 5.

Before



After



IV. Extract two visible exterior twists

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

IV - step 061

Split early.T12.square into its two positive units

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before

11. early.T12.square
T2^2



After

11. exterior.split.1
T2

12. exterior.split.2
T2



IV - step 062

One positive square unit crosses F8 and transports B to the exact exterior-lantern curve

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

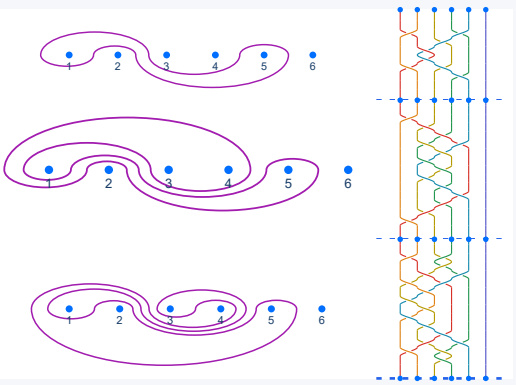
11 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before

12. exterior.split.2
T2

13. F8.2
T3

14. early.F13.1
T3

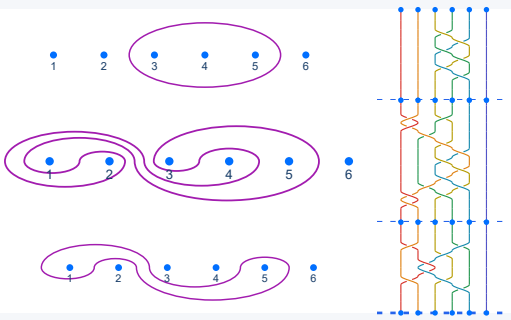


After

12. F8.2
T3

13. early.F13.1
T3

14. exterior.split.2
T2



IV - step 063

Choose a cyclic start at the matched B in the central disk product

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

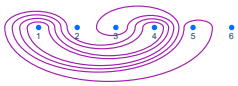
0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before

1. pure.prefix.root
T3



2. early.daisy.4
T2



3. pure.O1234
T4



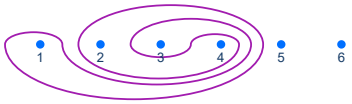
4. pure.T23
T2



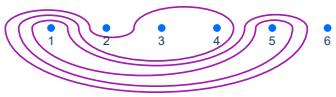
5. early.P35
T2



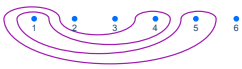
6. early.F5.2
T3



7. early.daisy.1
T3



8. early.daisy.2
T2



9. early.daisy.3
T2



10. early.P24.pair
T2



11. exterior.split.1
T2

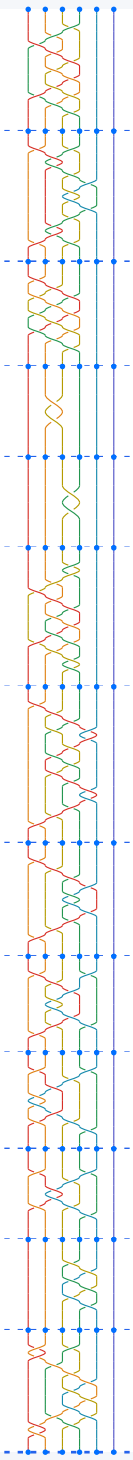


12. F8.2
T3



13. early.F13.1
T3





After

1. early.F13.1
T3



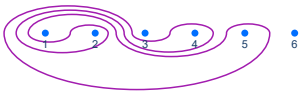
2. exterior.split.2
T2



3. early.P13.pair
T2



4. early.F12.1
T3



5. F10.2
T3



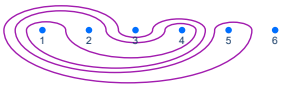
6. pure.next.1
T2



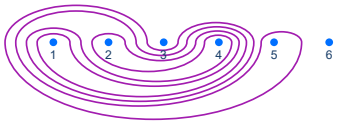
7. pure.next.2
T2



8. pure.prefix.daisy.2
T2



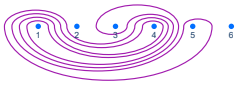
9. pure.prefix.daisy.3
T2



10. pure.prefix.root
T3



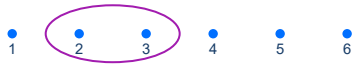
11. early.daisy.4
T2

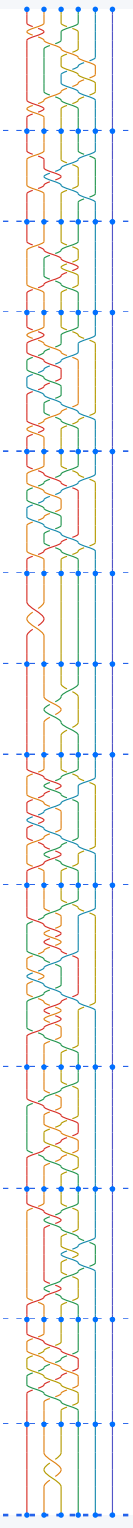


12. pure.O1234
T4



13. pure.T23
T2





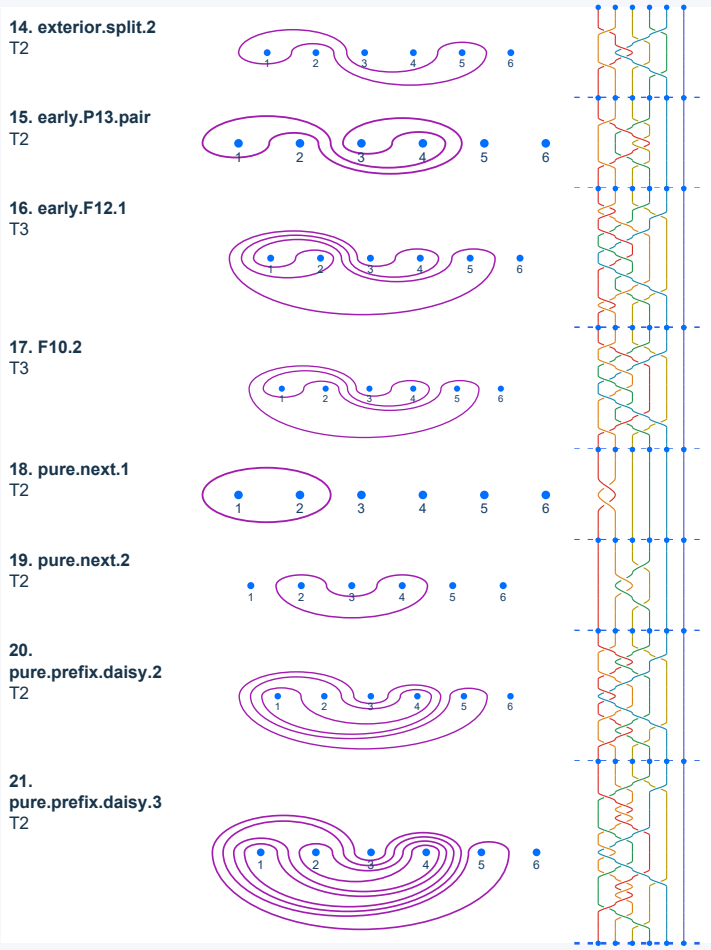
IV - step 063 - continued

Choose a cyclic start at the matched B in the central disk product

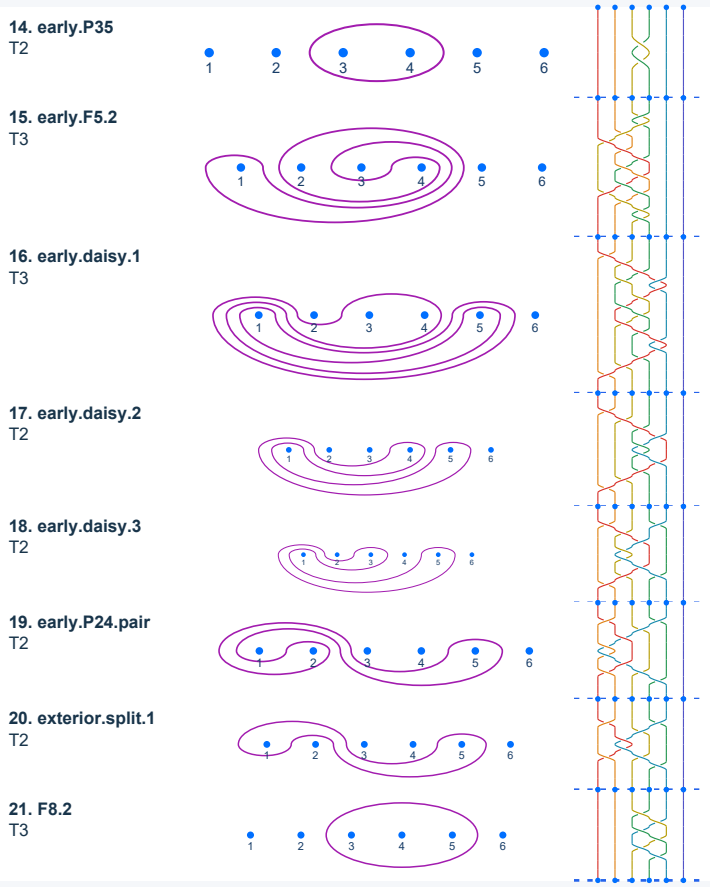
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



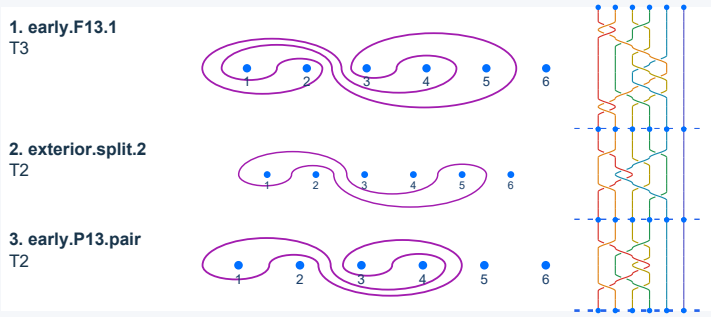
IV - step 064

Carry matched B across the square unit and P13, stopping before C

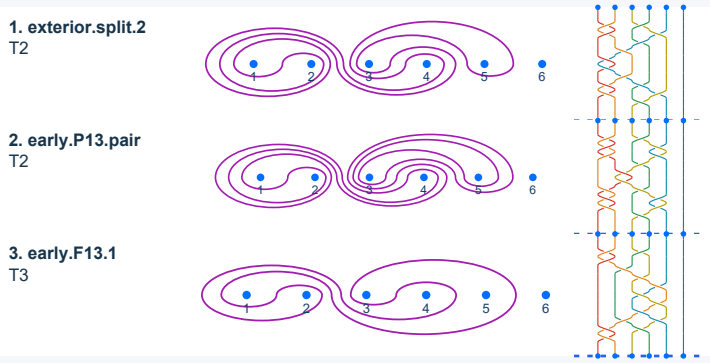
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 18 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



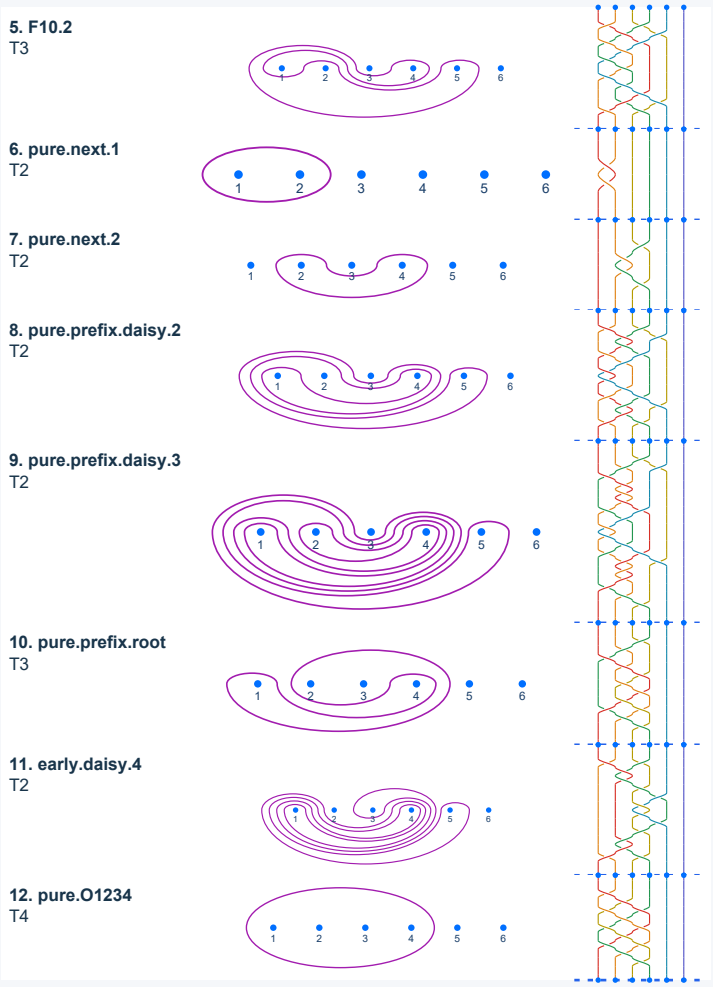
IV - step 065

Bring the round four-point curve up around the short cyclic side, immediately after C

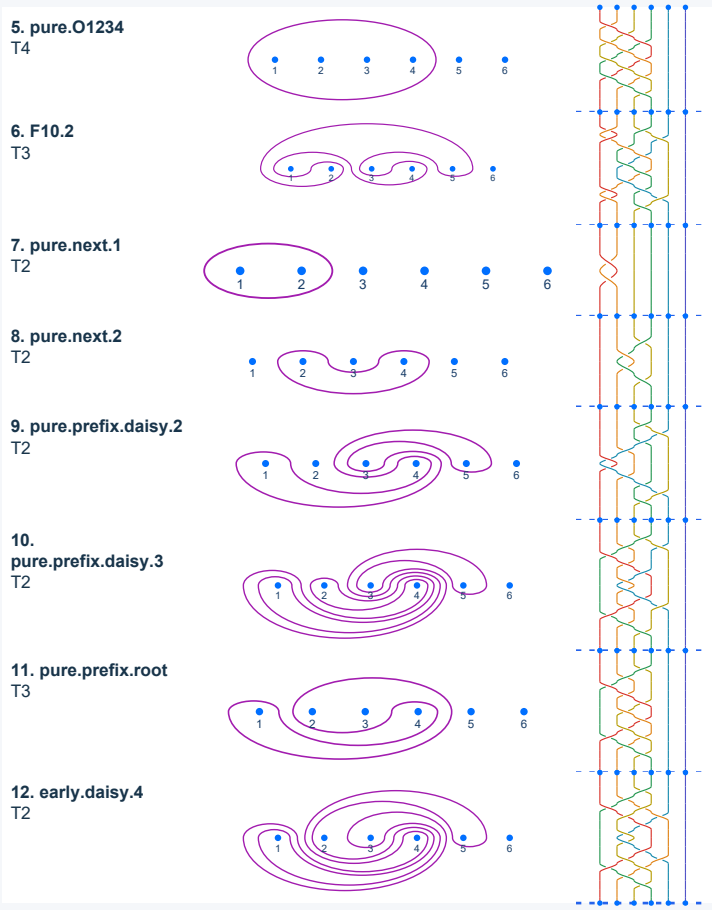
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 7.

Before



After



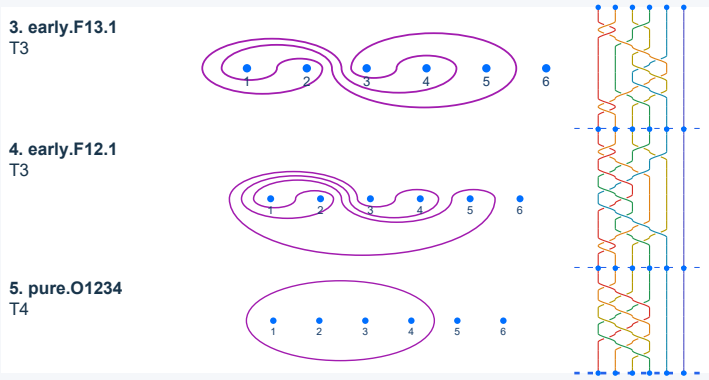
IV - step 066

Extract an actual exterior twist by the positive PC cluster lantern

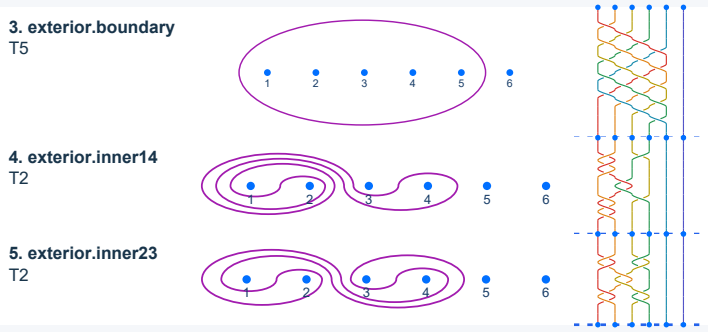
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 16 below. Only the changed interval is shown.

Before



After



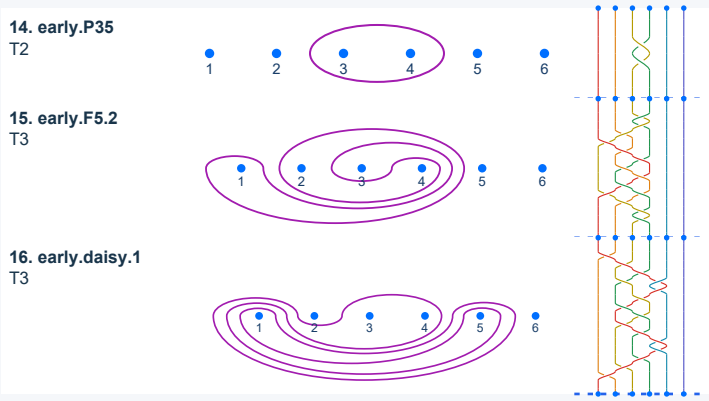
IV - step 067

Carry the actual T34 petal through two roots to its matching petals

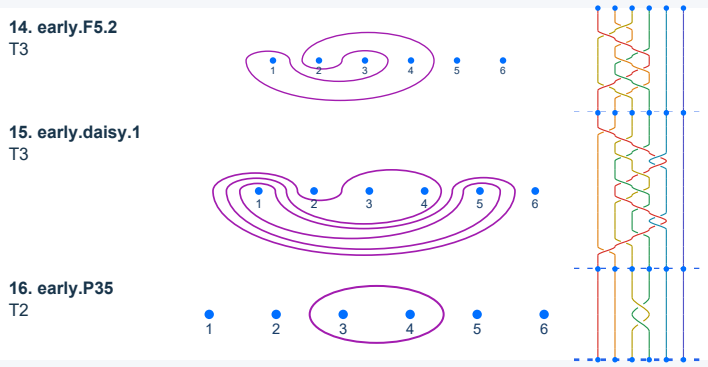
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



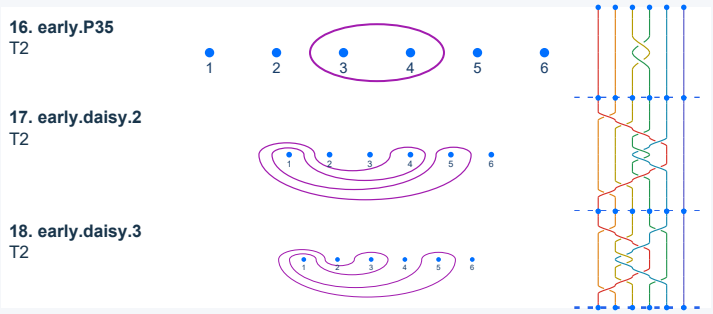
IV - step 068

Contract the three actual pair petals by the PC lantern

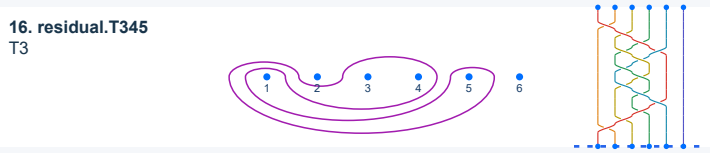
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



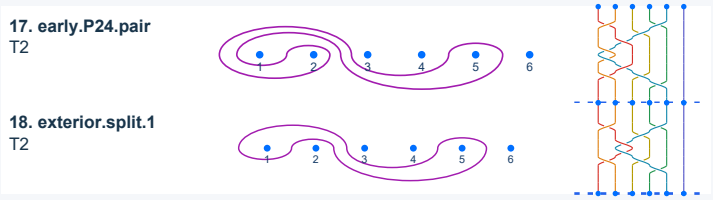
IV - step 069

First inverse Hurwitz swap of the adjacent pair twists

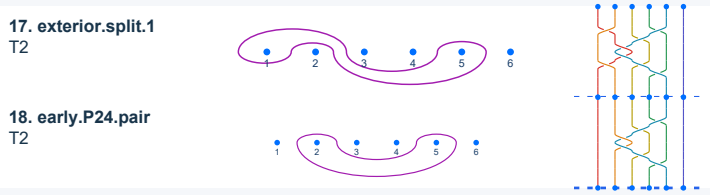
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



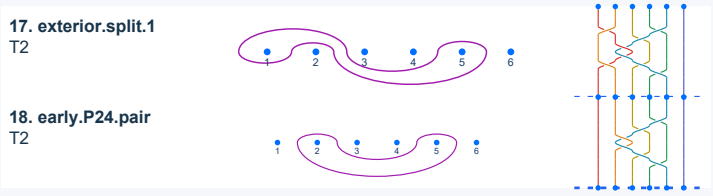
IV - step 070

Second inverse Hurwitz swap of the same two pair factors

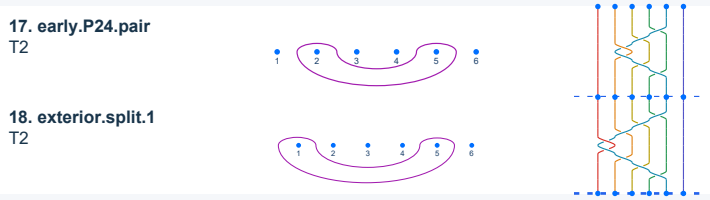
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



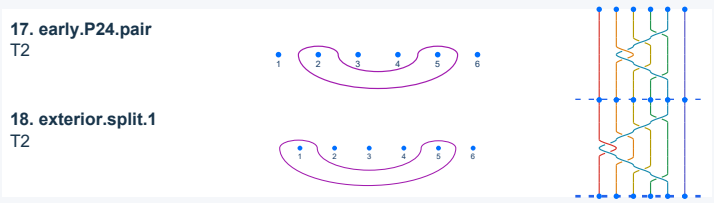
IV - step 071

Third inverse Hurwitz swap exposes the exact missing pair curve

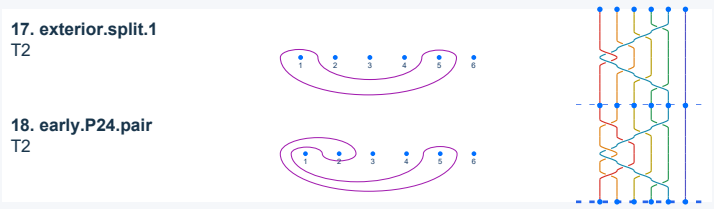
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

16 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



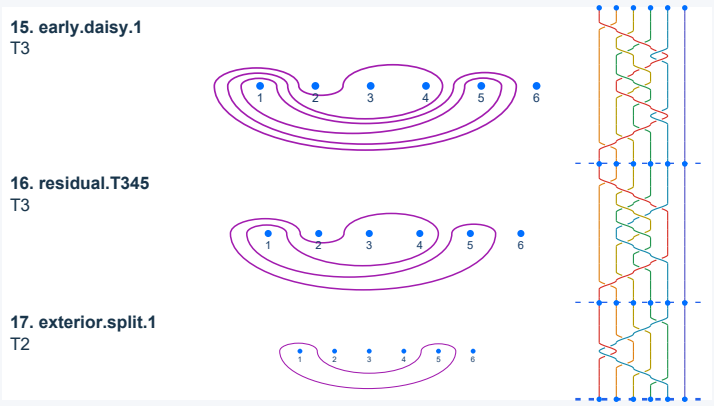
IV - step 072

PC cluster lantern contracts the two actual roots and the exposed petal

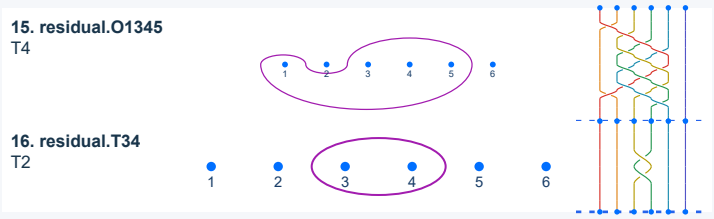
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



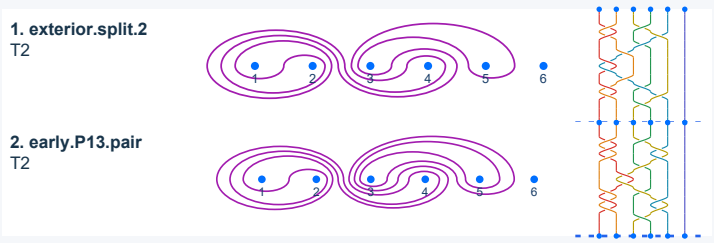
IV - step 073

Unwind the leading pair by carrying the fixed neighboring pair down once

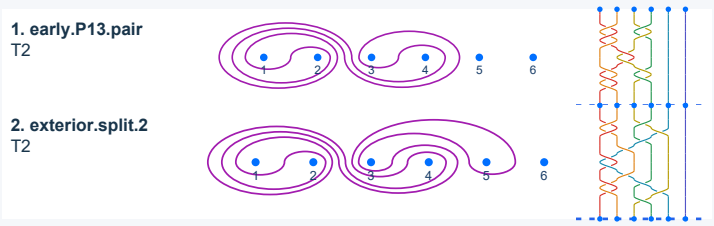
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 16 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



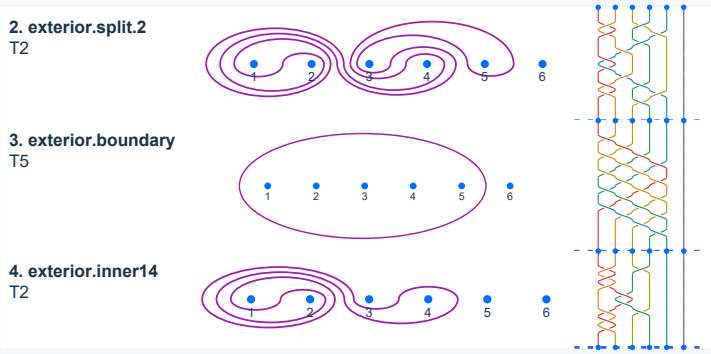
IV - step 074

The new inner curve unwinds the other leading pair across the visible exterior

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



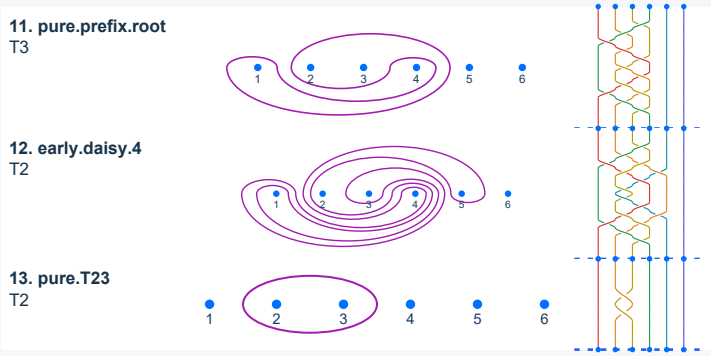
IV - step 075

Carry the 123 curve through the 35 petal and commuting T23; the petal becomes short

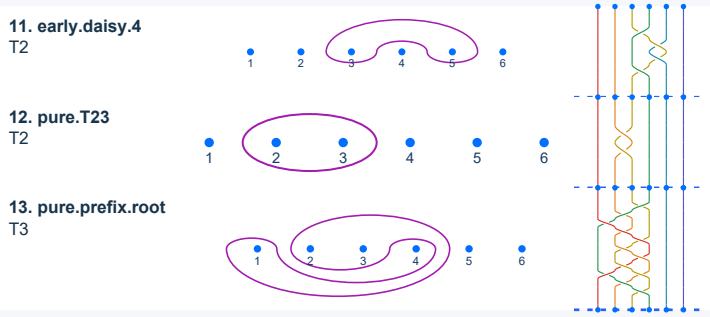
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



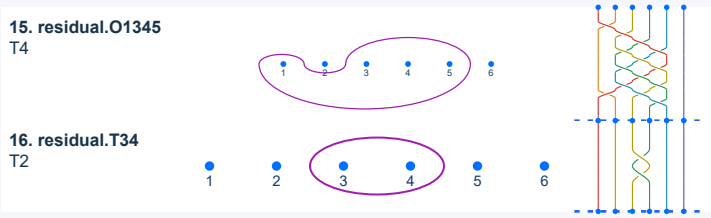
IV - step 076

Commute the round T34 past its containing O1345 curve

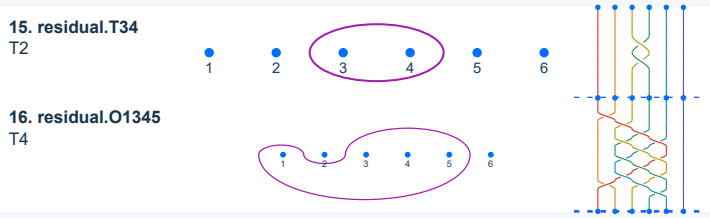
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



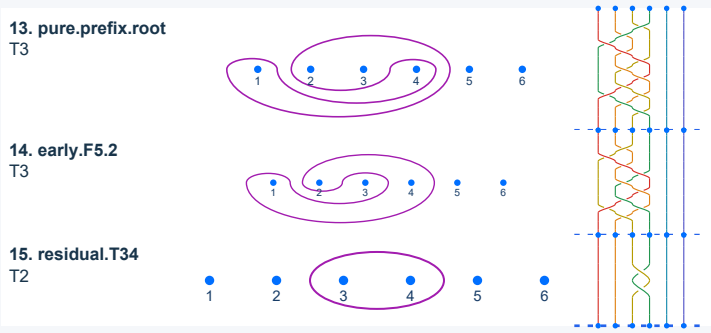
IV - step 077

PC lantern for the clusters 12, 3, 4 produces round T1234 and the common 12 curve

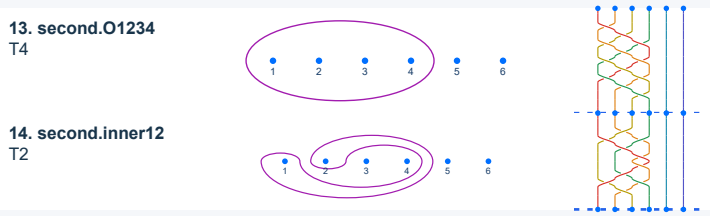
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



IV - step 078

Commute the new outer curve past its common inner curve to the actual O1345 and 25 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



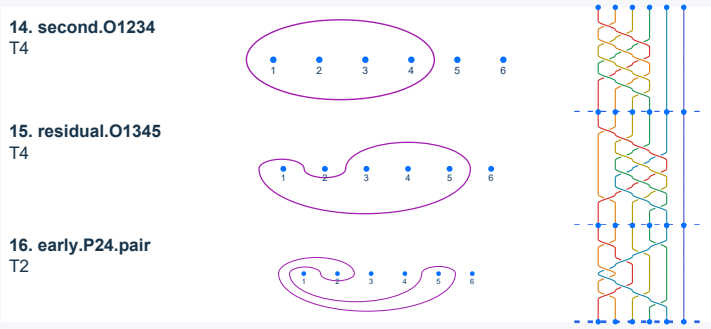
IV - step 079

PC lantern for the clusters 134, 2, 5 extracts the SECOND visible exterior twist

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 1 below. Only the changed interval is shown.

Before



After



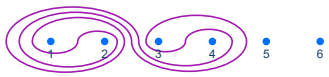
V. Rebuild round T1234

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

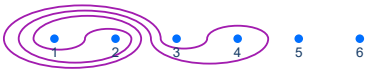
Complete factorization before Rebuild round T1234

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

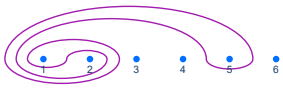
1. early.P13.pair
T2



2. exterior.inner14
T2



3. exterior.split.2
T2



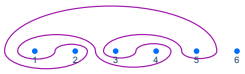
4. exterior.boundary
T5



5. exterior.inner23
T2



6. F10.2
T3



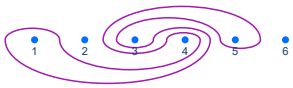
7. pure.next.1
T2



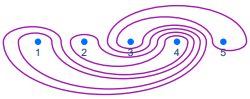
8. pure.next.2
T2



9. pure.prefix.daisy.2
T2



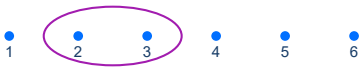
10. pure.prefix.daisy.3
T2



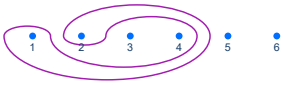
11. early.daisy.4
T2



12. pure.T23
T2



13. second.inner12
T2



14. second.exterior
T5

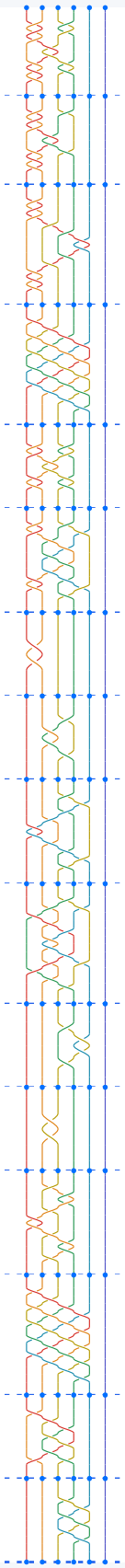


15. second.inner134
T3



16. F8.2
T3





V - step 080

PC rose of the actual round T345 supplies its round 34 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before

16. F8.2
T3

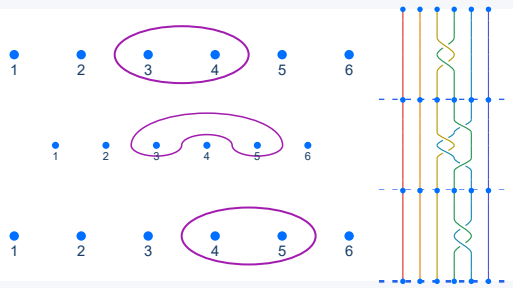


After

16. round.rose345.0
T2

17. round.rose345.1
T2

18. round.rose345.2
T2



V - step 081

Carry the under-going 24 petal through three pairs to its actual 23 and 34 companions

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 3.

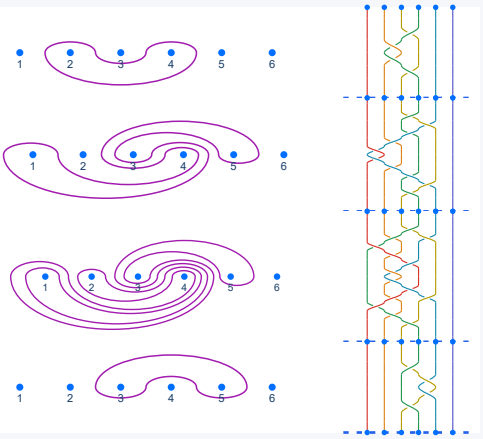
Before

8. pure.next.2
T2

9. pure.prefix.daisy.2
T2

10. pure.prefix.daisy.3
T2

11. early.daisy.4
T2



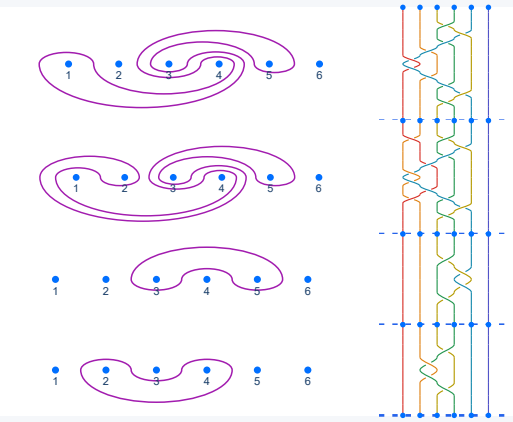
After

8. pure.prefix.daisy.2
T2

9. pure.prefix.daisy.3
T2

10. early.daisy.4
T2

11. pure.next.2
T2



V - step 082

Commute the round 34 petal up past the 134 root, exterior and routed 12 curve

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



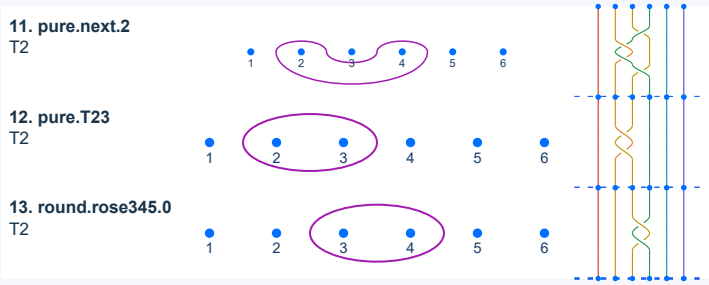
V - step 083

PC lantern contracts the actual 24, 23, 34 petals to ROUND T234

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



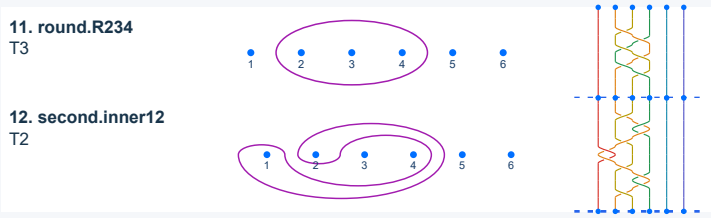
V - step 084

The actual T234 removes exactly the winding of the common 12 curve

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



V - step 085

Commute the 134 root past the visible central exterior to the matched 12 and 234 curves

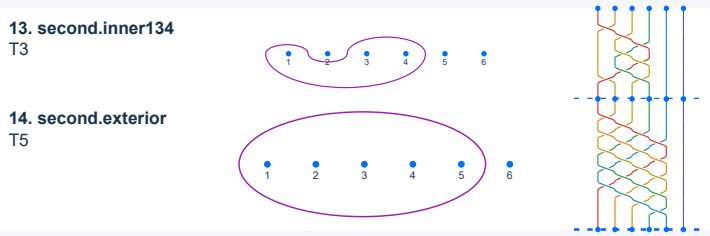
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



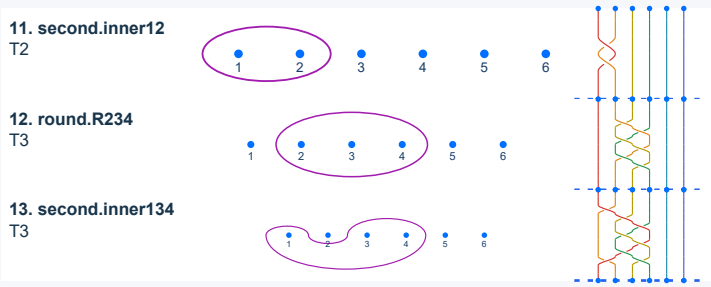
V - step 086

PC cluster lantern for 1, 2, 34 produces ROUND T1234 and T34

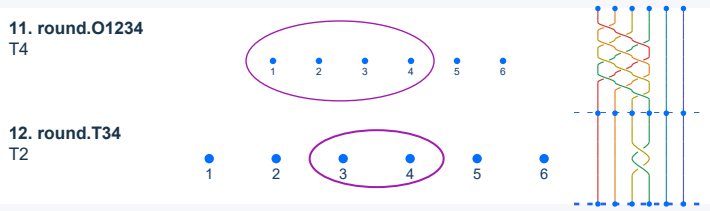
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



VI. Collect and close the four-point rose

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

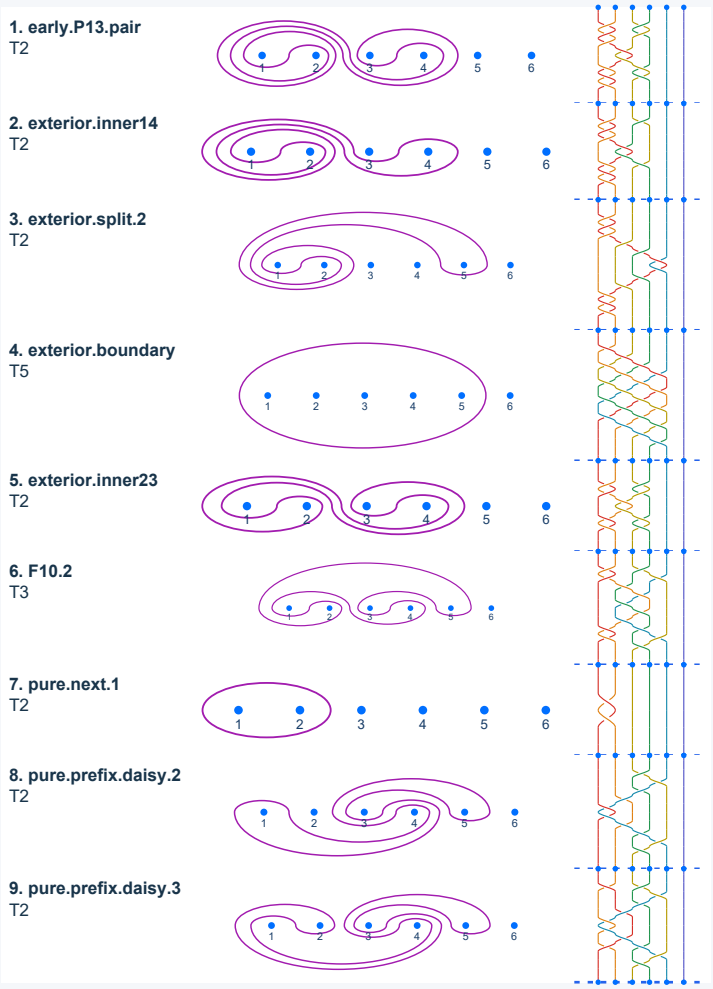
VI - step 087

Explicit common T12 frame removes the shared winding of the leading curves

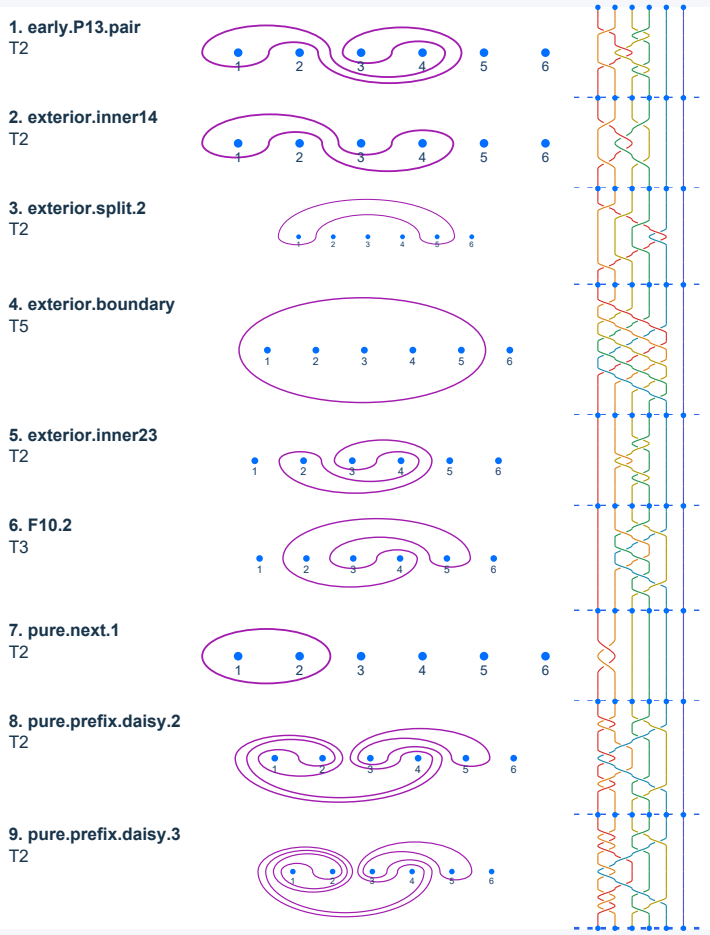
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



VI - step 088

PC rose of the actual 245 curve separates its inside-four petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 9 below. Only the changed interval is shown.

Before

6. F10.2
T3

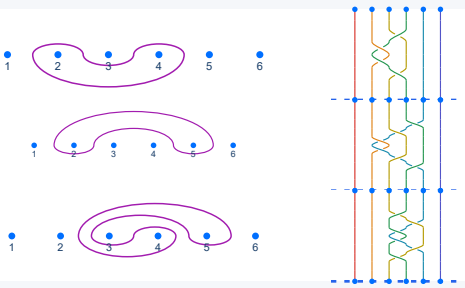


After

6. close.rose245.0
T2

7. close.rose245.1
T2

8. close.rose245.2
T2



VI - step 089

Collect exterior.inner23 with the preceding inside-four-point curves

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

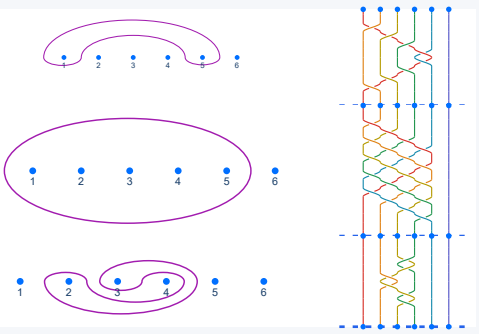
2 unchanged displayed factors above; 12 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before

3. exterior.split.2
T2

4. exterior.boundary
T5

5. exterior.inner23
T2

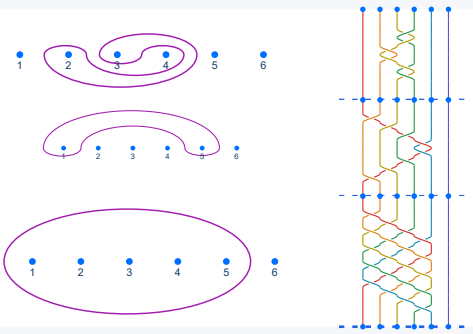


After

3. exterior.inner23
T2

4. exterior.split.2
T2

5. exterior.boundary
T5



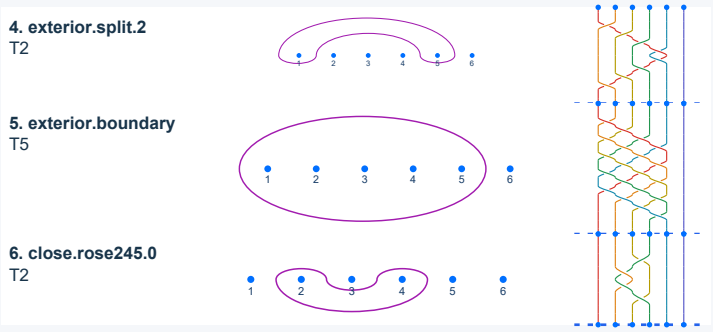
VI - step 090

Collect close.rose245.0 with the preceding inside-four-point curves

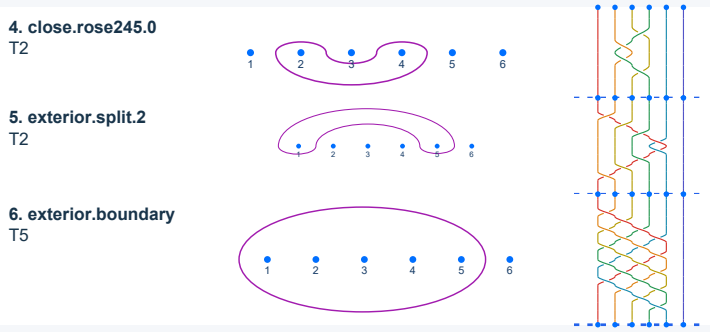
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 11 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



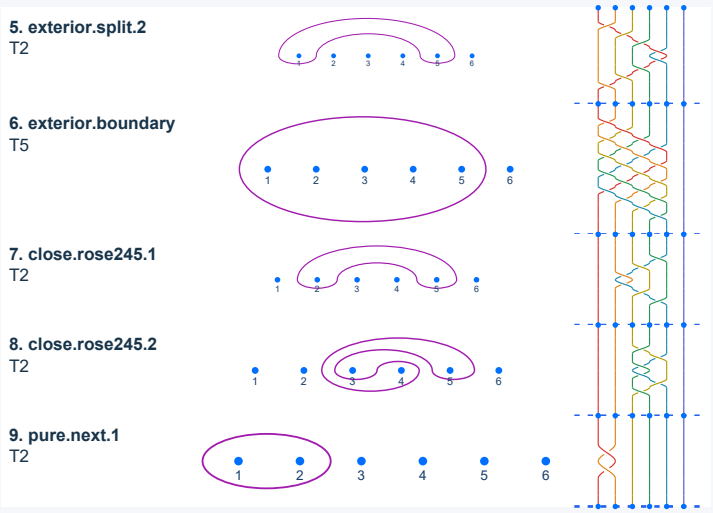
VI - step 091

Collect pure.next.1 with the preceding inside-four-point curves

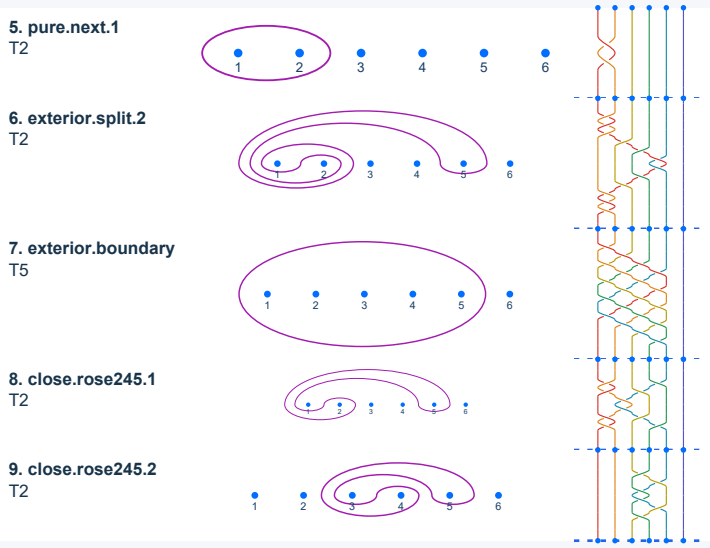
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



VI - step 092

Carry round T34 down the short side across the two end petals and the exterior

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



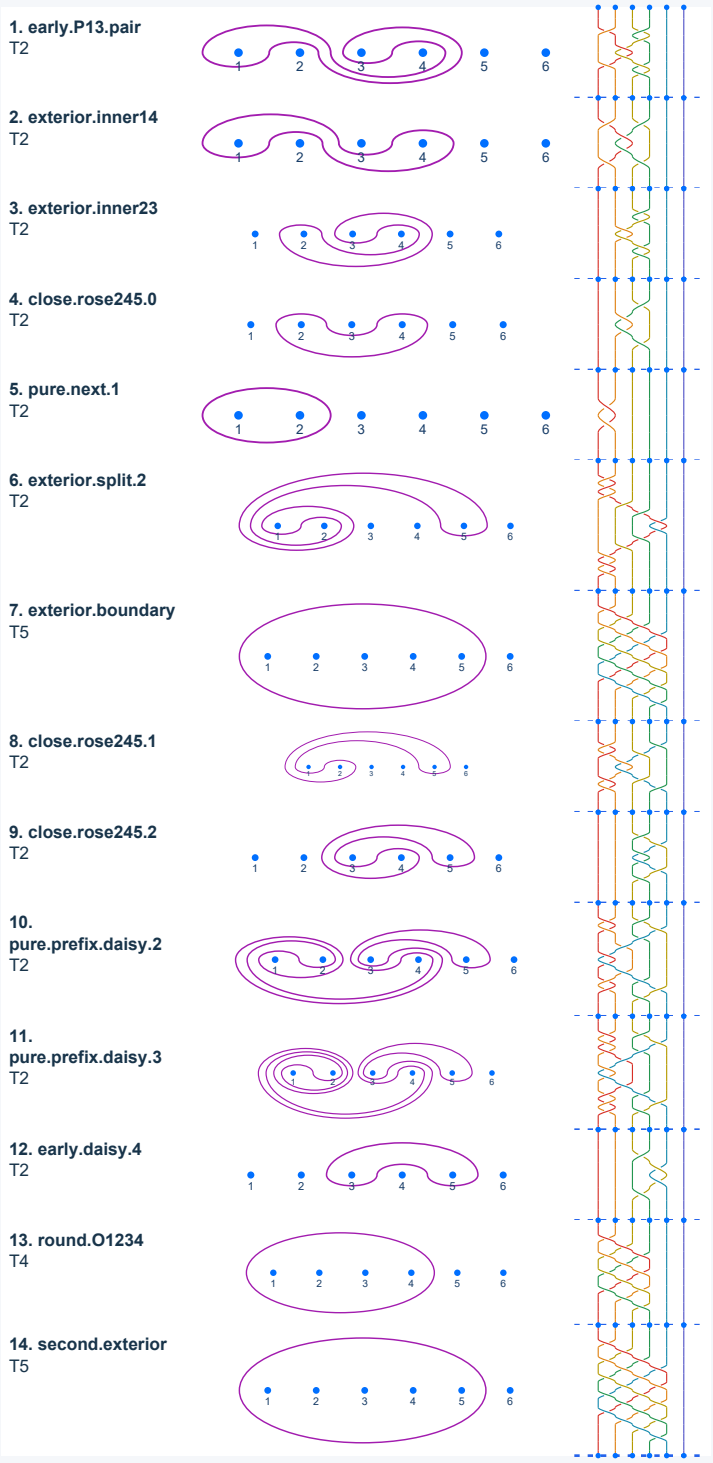
VI - step 093

Choose the cyclic start at round T34; its full common frame is retained

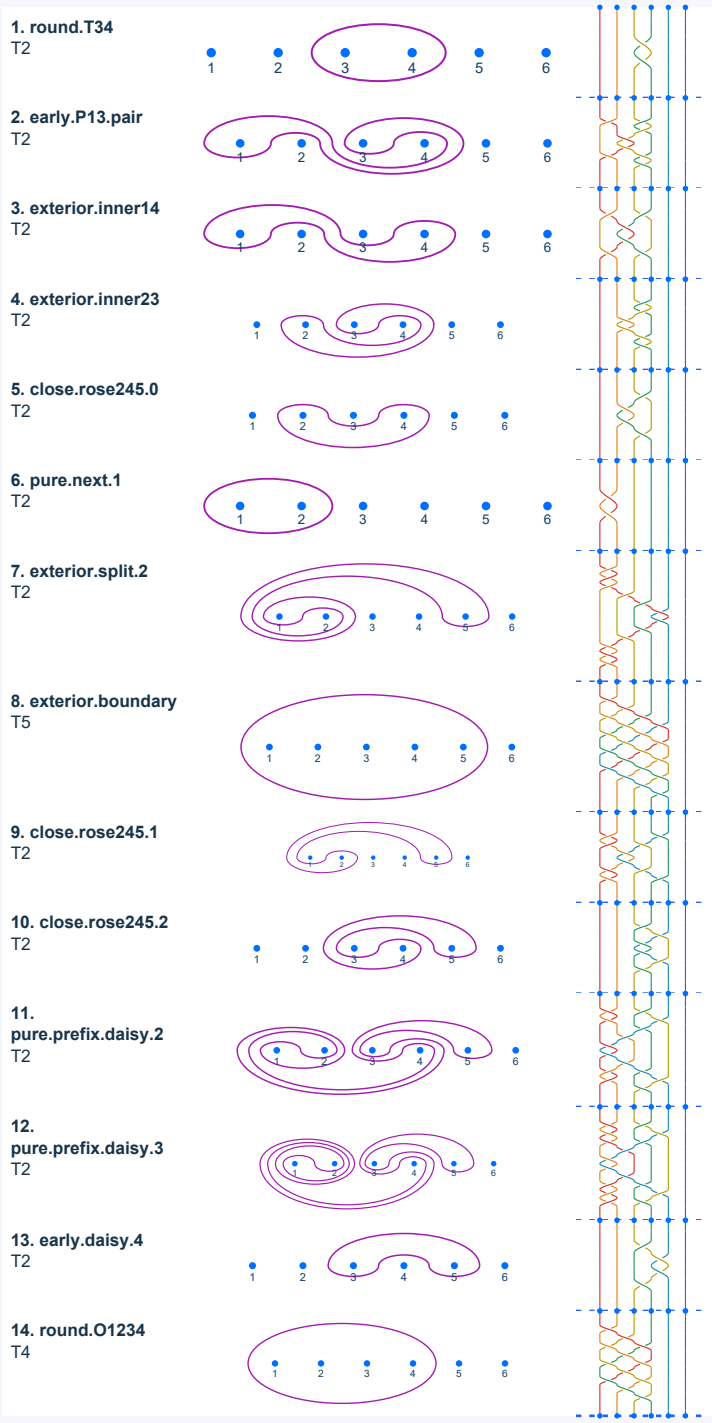
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



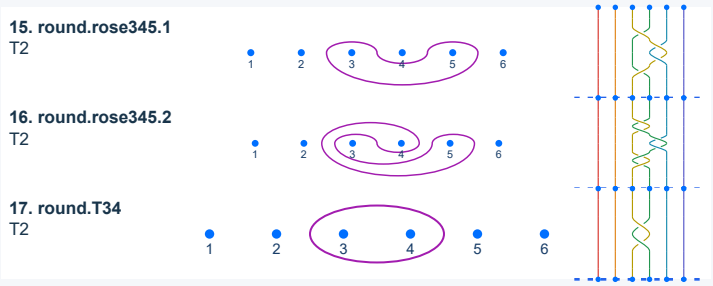
VI - step 093 - continued

Choose the cyclic start at round T34; its full common frame is retained

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



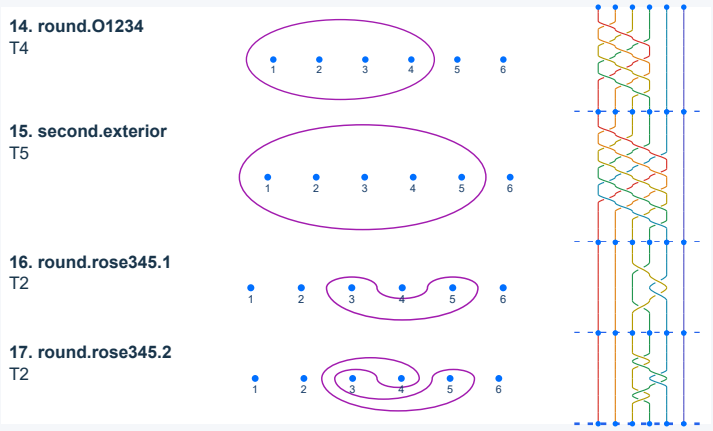
VI - step 094

Carry round T1234 down the short side instead of through the five long petals

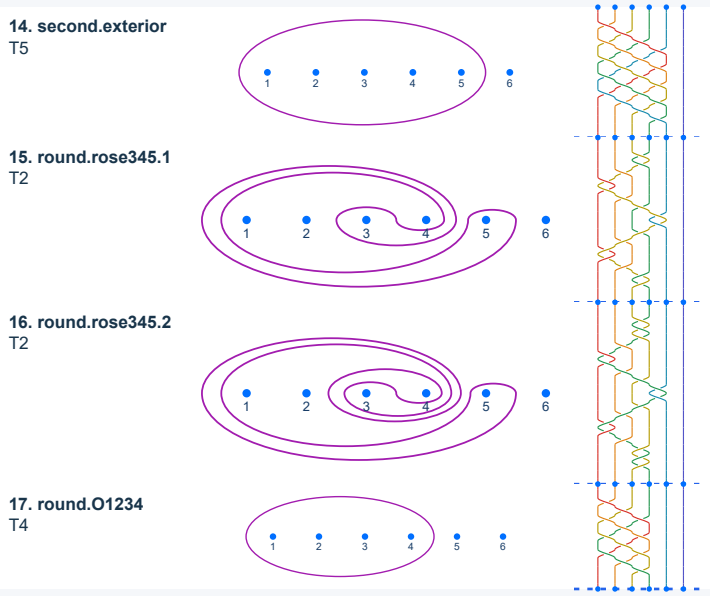
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

13 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



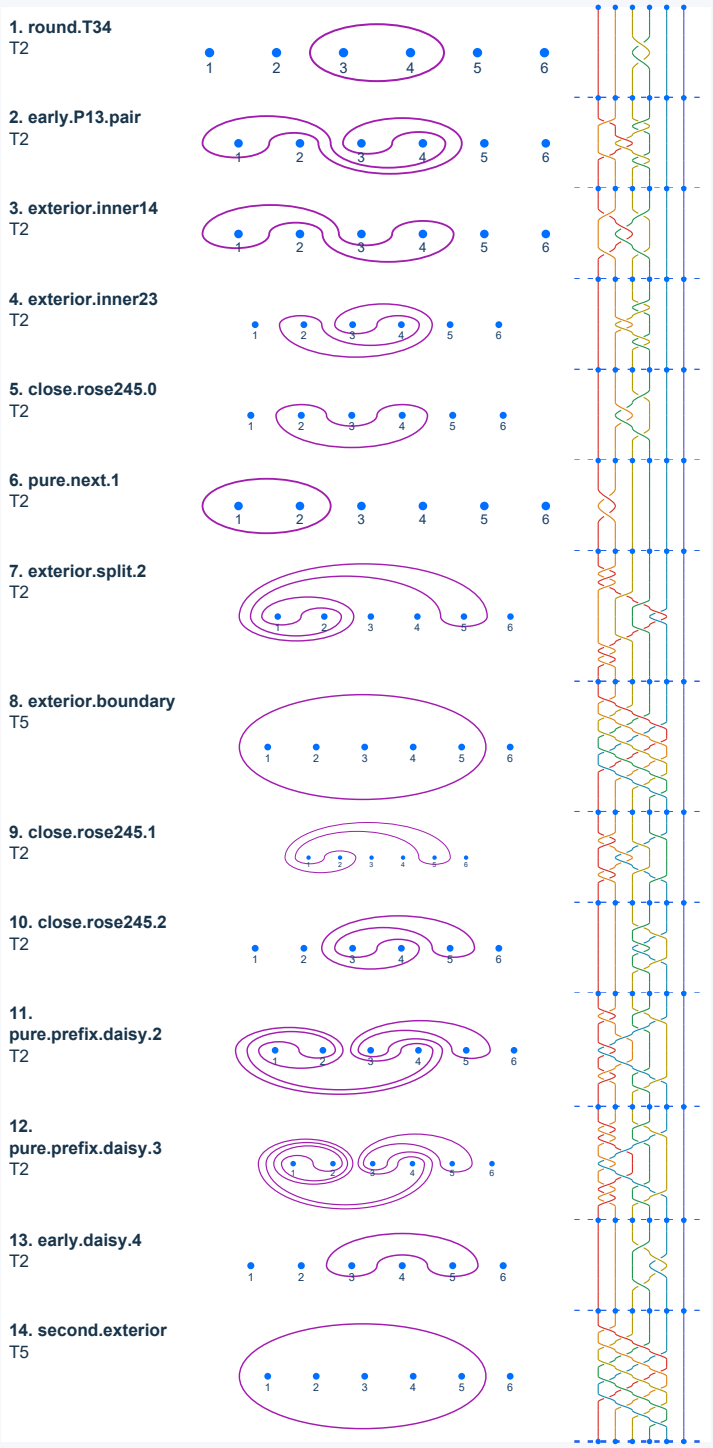
VI - step 095

Choose the cyclic start at round T1234; all inside-four factors are now consecutive

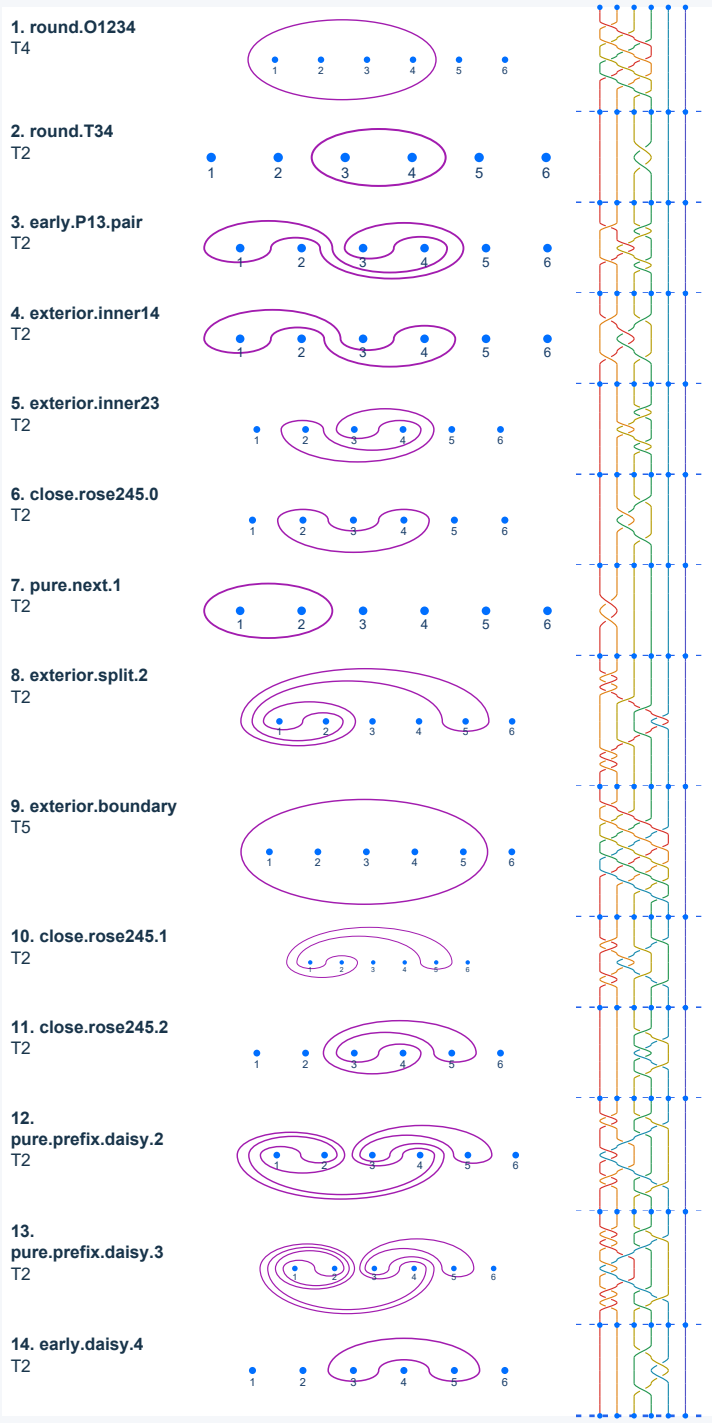
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



VI - step 095 - continued

Choose the cyclic start at round T1234; all inside-four factors are now consecutive

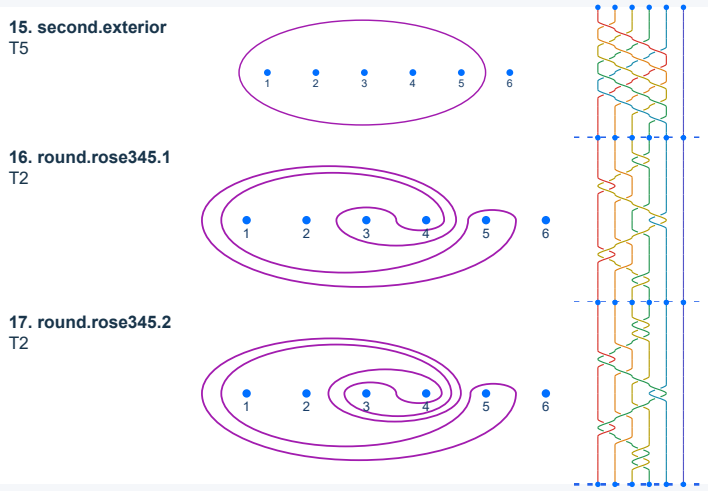
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

Before



After



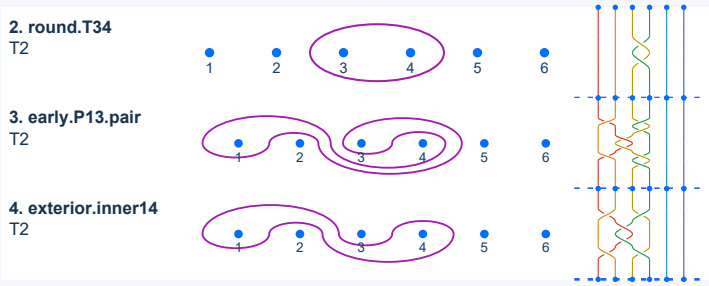
VI - step 096

PC lantern contracts the actual 34, 13, 14 petals to the 134 root

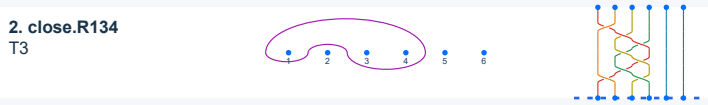
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

Before



After



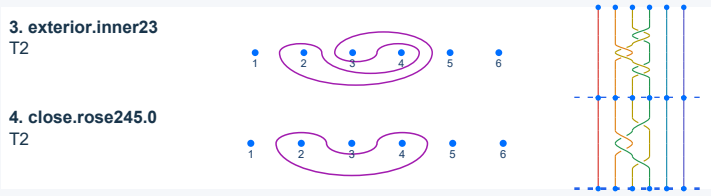
VI - step 097

One inverse Hurwitz swap makes the remaining 23 petal round

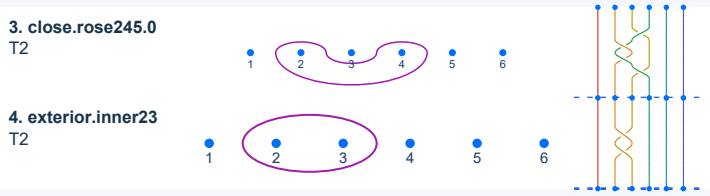
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



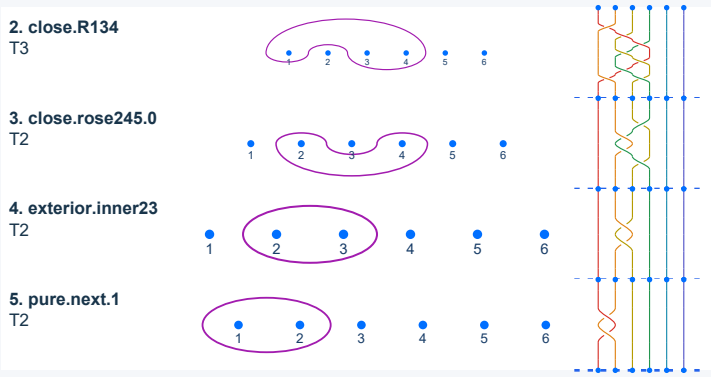
VI - step 098

The four actual factors contract by the PC daisy in common frame sigma2 sigma3

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 10 below. Only the changed interval is shown.

Before



After



VII. Untwist the point-pushing petals

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

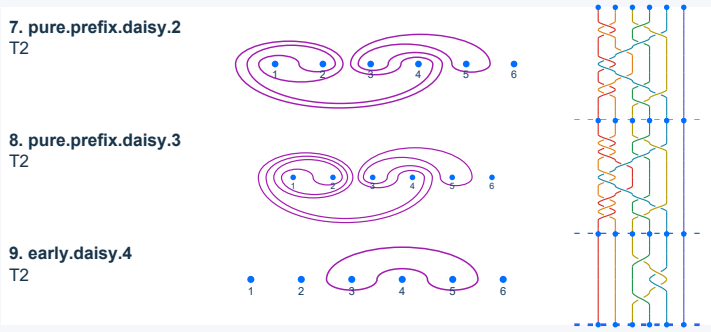
VII - step 099

Carry the SAME point-5 petal early.daisy.4 across its successive neighbors

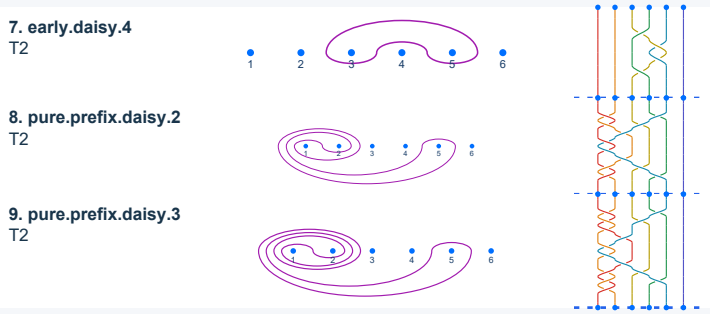
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



VII - step 100

Untwist the point-5 petal close.rose245.1 by a disk Hurwitz move

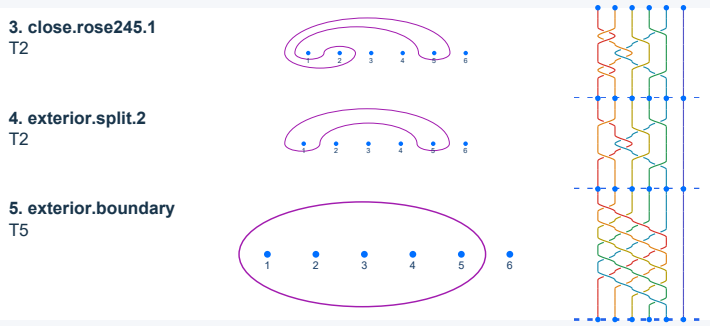
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



VII - step 101

Untwist the point-5 petal exterior.split.2 by a disk Hurwitz move

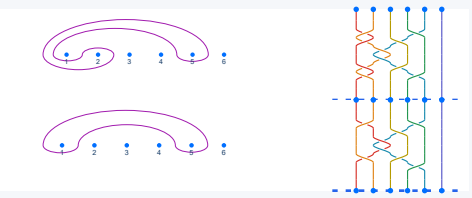
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before

3. close.rose245.1
T2

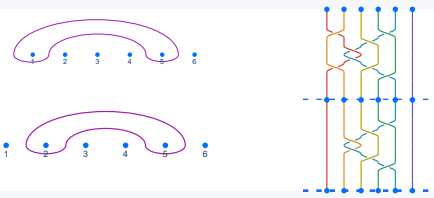
4. exterior.split.2
T2



After

3. exterior.split.2
T2

4. close.rose245.1
T2



VII - step 102

Untwist the point-5 petal early.daisy.4 by a disk Hurwitz move

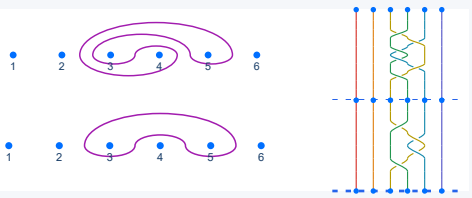
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before

6. close.rose245.2
T2

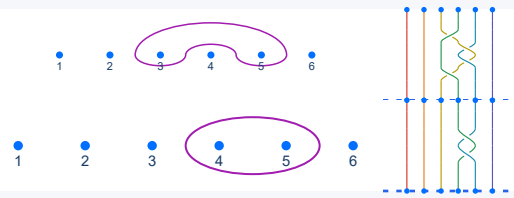
7. early.daisy.4
T2



After

6. early.daisy.4
T2

7. close.rose245.2
T2



VII - step 103

Untwist the point-5 petal pure.prefix.daisy.2 by a disk Hurwitz move

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before

8. pure.prefix.daisy.2
T2

9. pure.prefix.daisy.3
T2



After

8. pure.prefix.daisy.3
T2

9. pure.prefix.daisy.2
T2



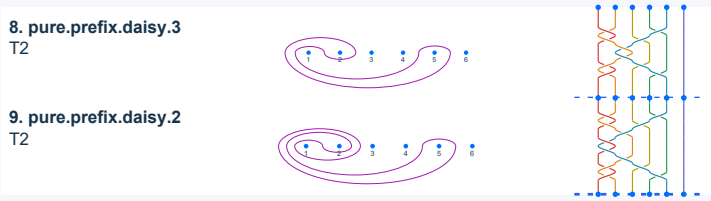
VII - step 104

Untwist the point-5 petal pure.prefix.daisy.3 by a disk Hurwitz move

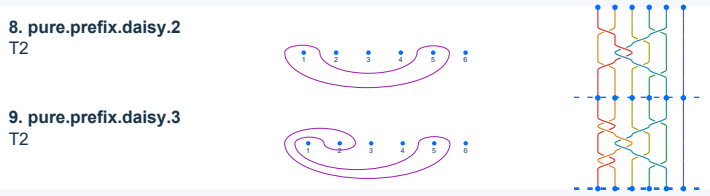
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



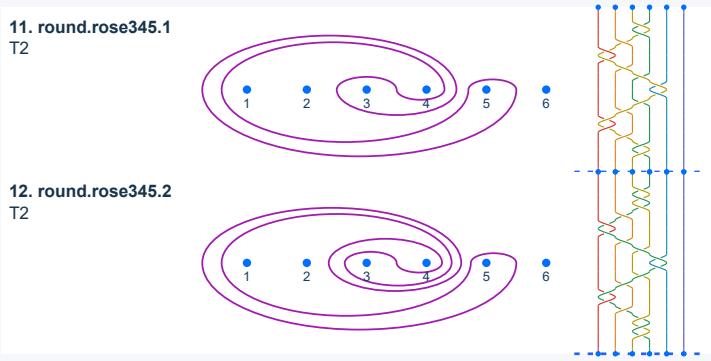
VII - step 105

Untwist the point-5 petal round.rose345.1 by a disk Hurwitz move

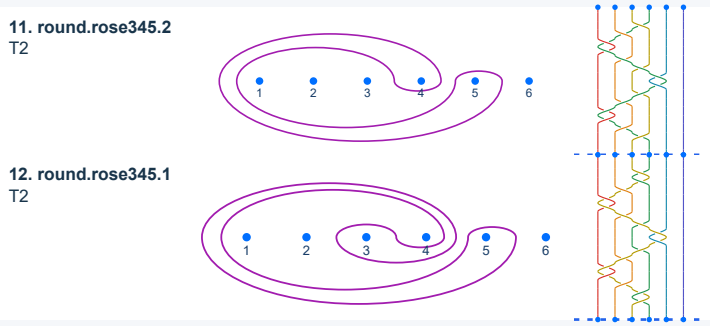
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



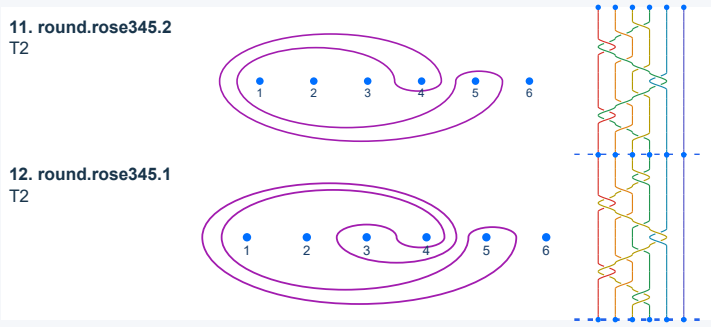
VII - step 106

Untwist the point-5 petal round.rose345.2 by a disk Hurwitz move

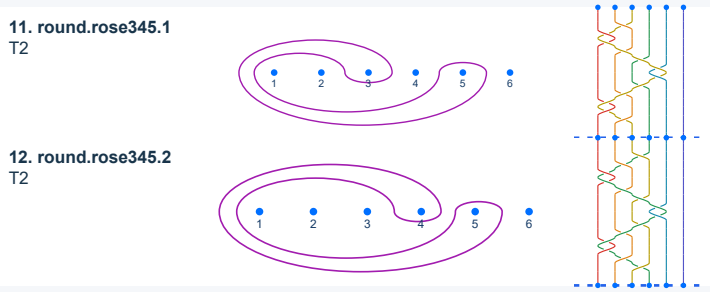
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



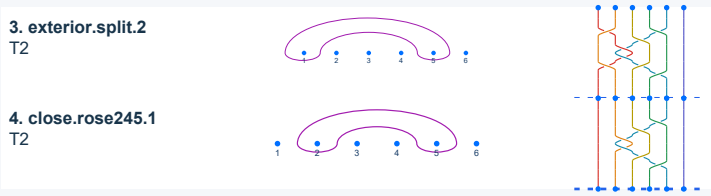
VII - step 107

Untwist the point-5 petal close.rose245.1 by a disk Hurwitz move

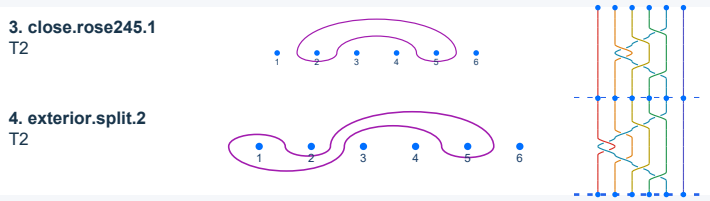
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



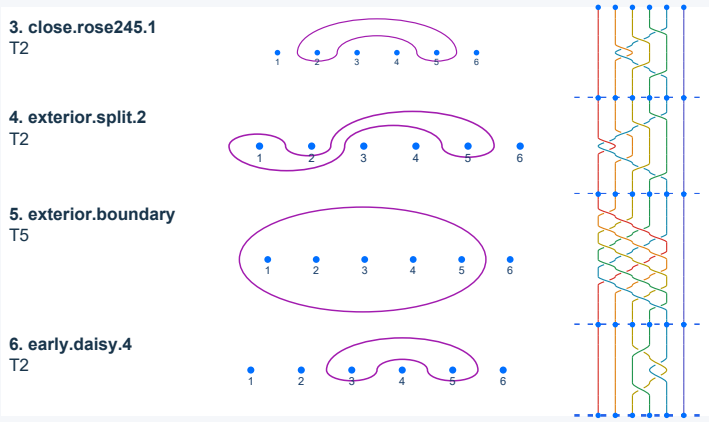
VII - step 108

Carry the SAME point-5 petal early.daisy.4 across its successive neighbors

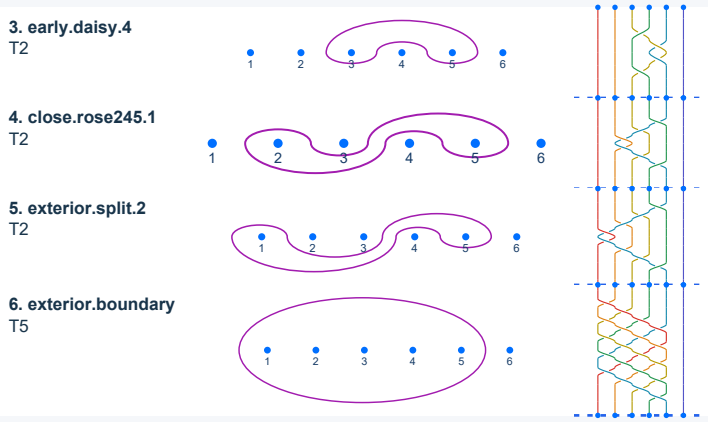
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



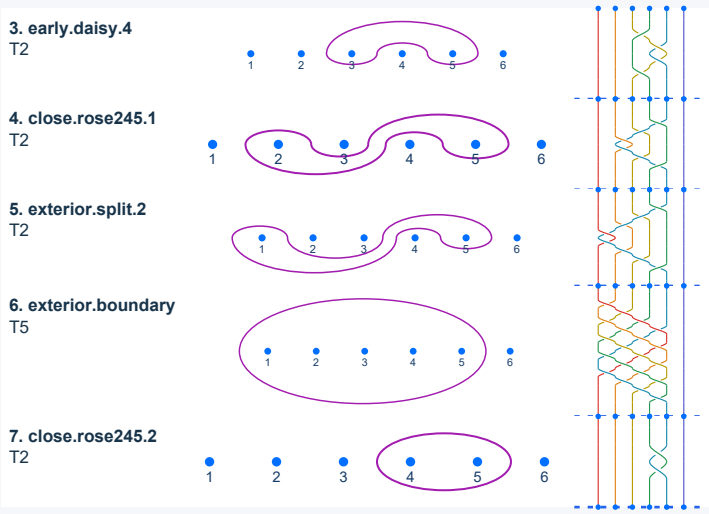
VII - step 109

Carry the SAME point-5 petal close.rose245.2 across its successive neighbors

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 4.

Before



After



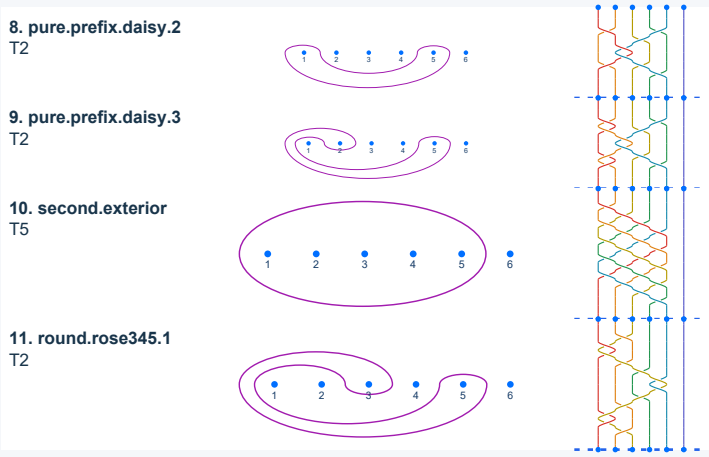
VII - step 110

Carry the SAME point-5 petal pure.prefix.daisy.2 across its successive neighbors

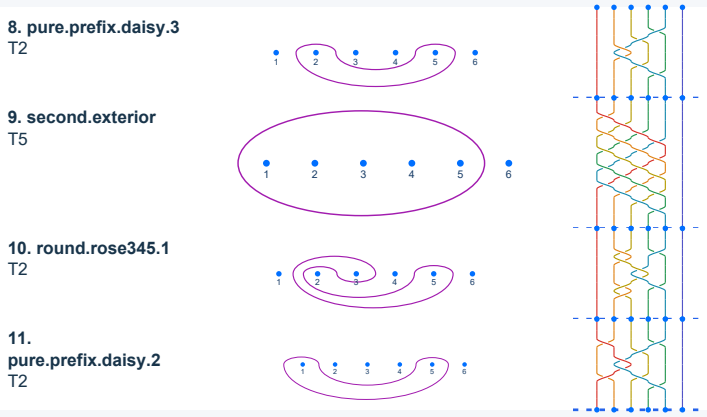
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 3.

Before



After



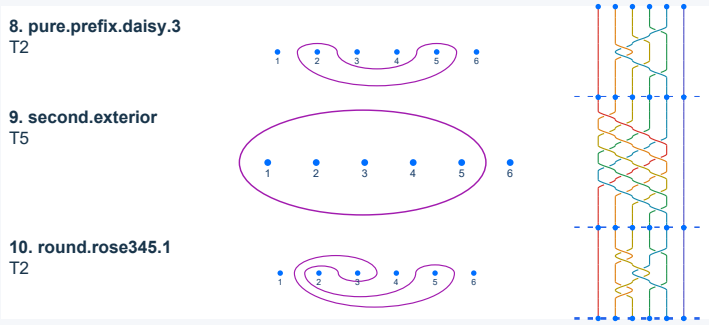
VII - step 111

Untwist the point-5 petal pure.prefix.daisy.3 by a disk Hurwitz move

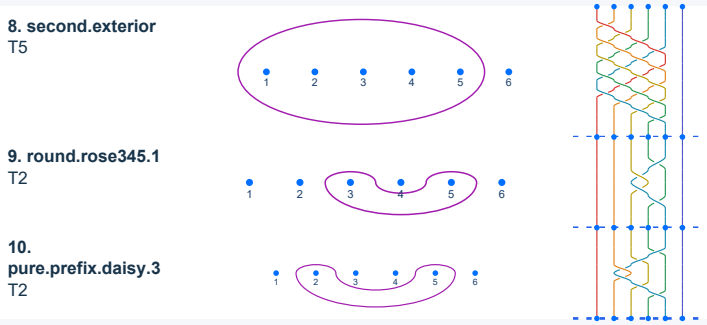
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 2.

Before



After



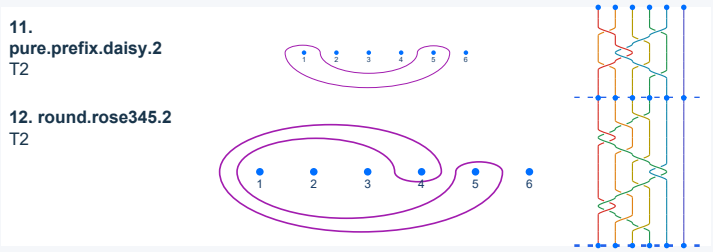
VII - step 112

Untwist the point-5 petal pure.prefix.daisy.2 by a disk Hurwitz move

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



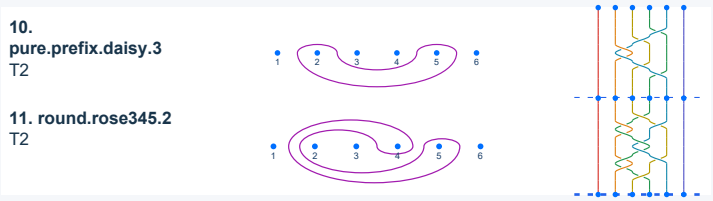
VII - step 113

Untwist the point-5 petal pure.prefix.daisy.3 by a disk Hurwitz move

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



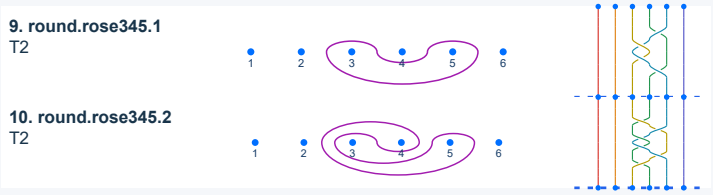
VII - step 114

Untwist the point-5 petal round.rose345.1 by a disk Hurwitz move

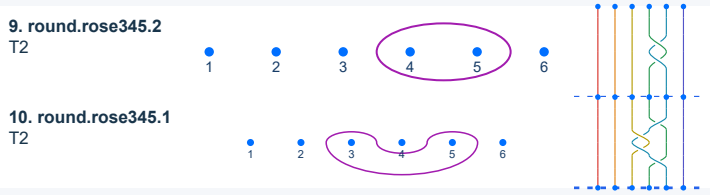
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



VIII. Two daisies, four exteriors and the sphere cap

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

Complete factorization before Two daisies, four exteriors and the sphere cap

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. round.O1234
T4

2. close.O1234
T4

3. close.rose245.2
T2

4. early.daisy.4
T2

5. close.rose245.1
T2

6. exterior.split.2
T2

7. exterior.boundary
T5

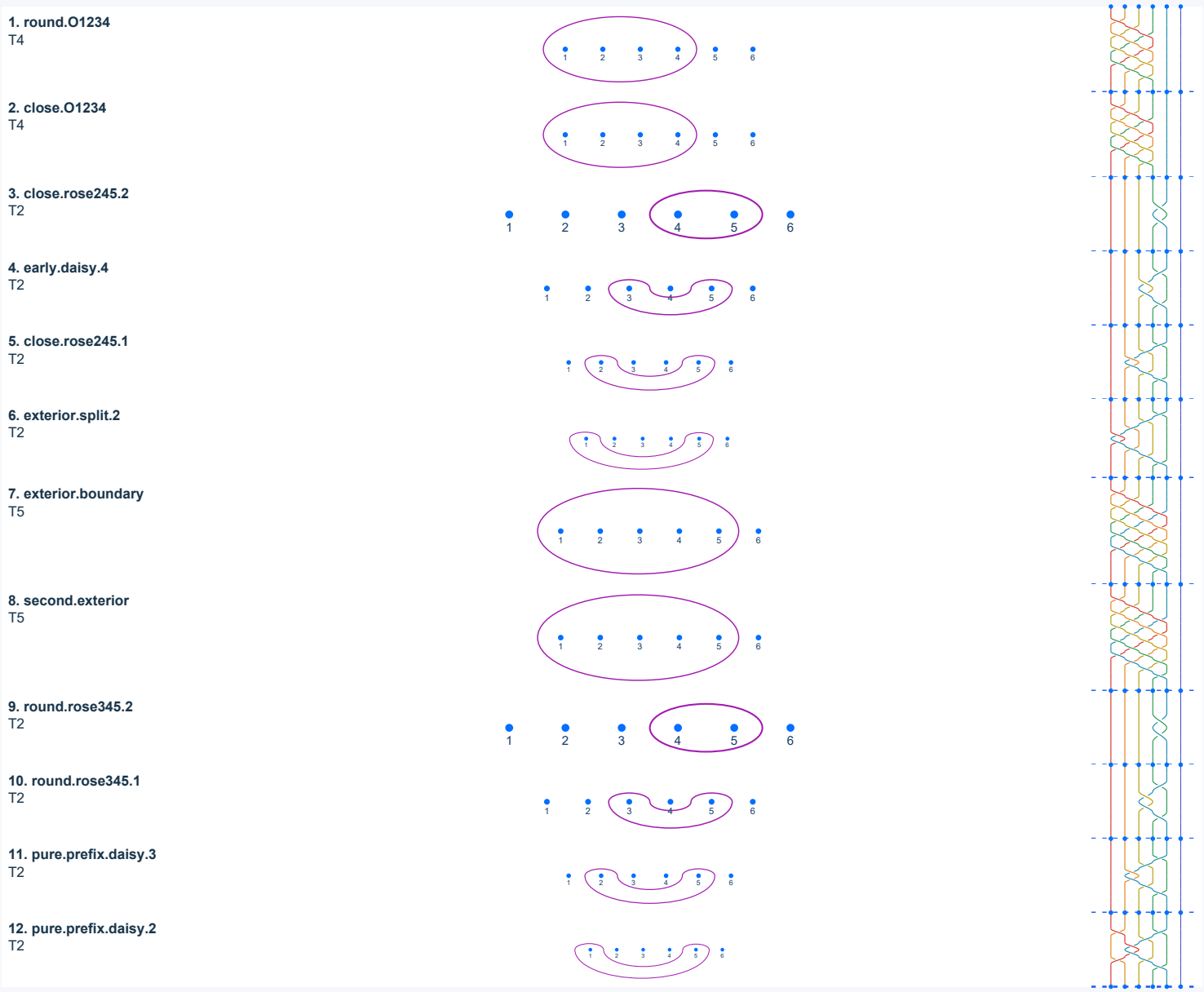
8. second.exterior
T5

9. round.rose345.2
T2

10. round.rose345.1
T2

11. pure.prefix.daisy.3
T2

12. pure.prefix.daisy.2
T2



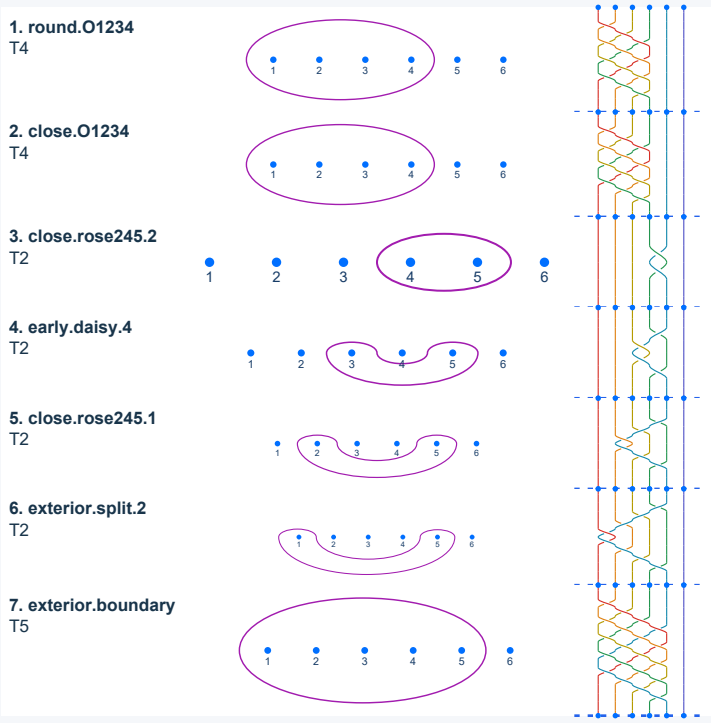
VIII - step 115

Commute the already visible exterior twist out of the daisy petals

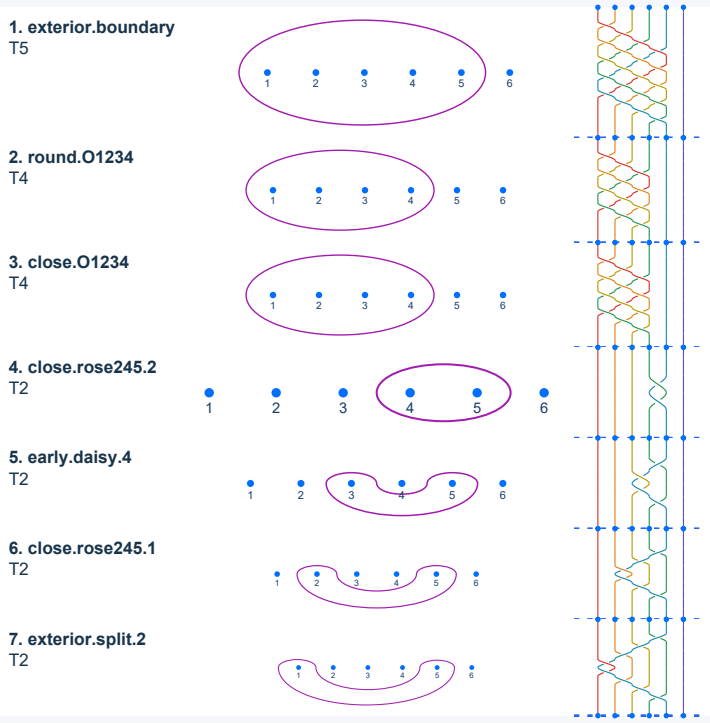
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 6.

Before



After



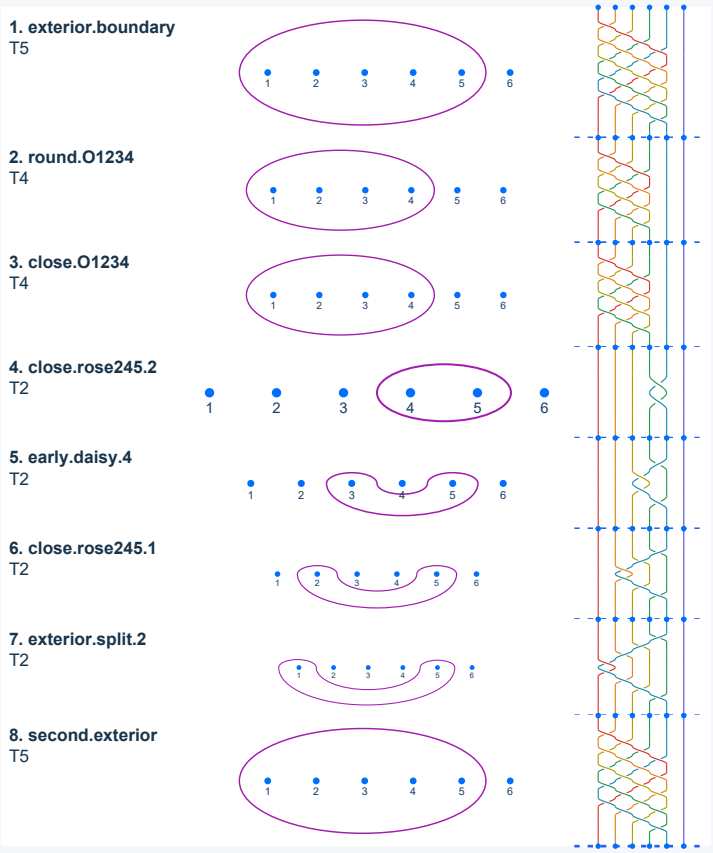
VIII - step 116

Commute the already visible exterior twist out of the daisy petals

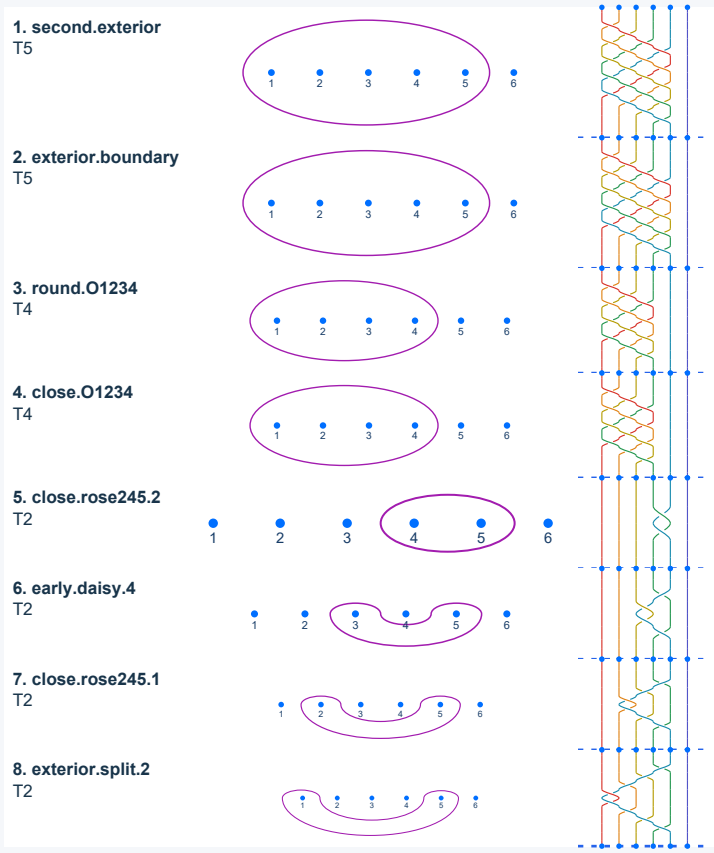
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 7.

Before



After



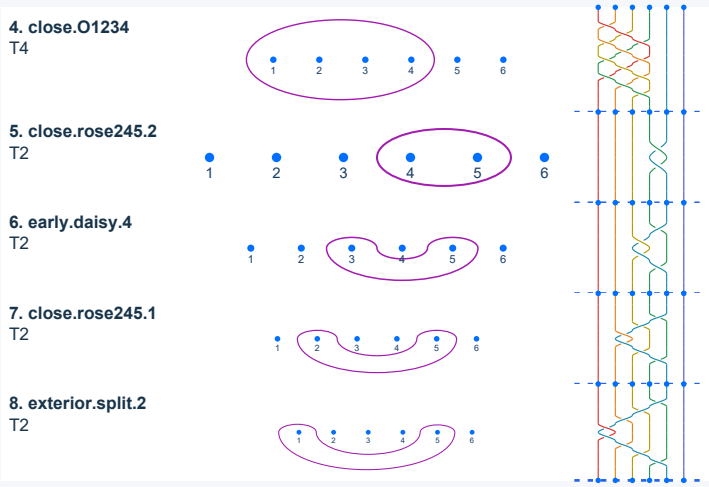
VIII - step 117

PC daisy contracts T1234 and the FIRST four actual canonical petals

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



VIII - step 118

Commute the third visible exterior past the remaining T1234

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 1.

Before



After



VIII - step 119

PC daisy contracts T1234 and the SECOND four actual canonical petals

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



VIII - step 120

Combine the FOUR identical visible exterior twists

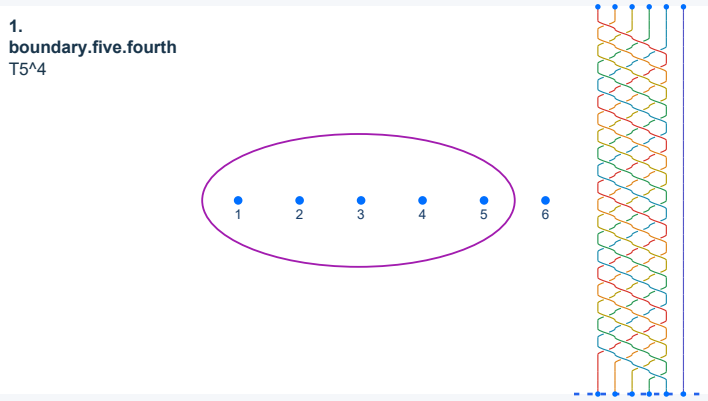
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



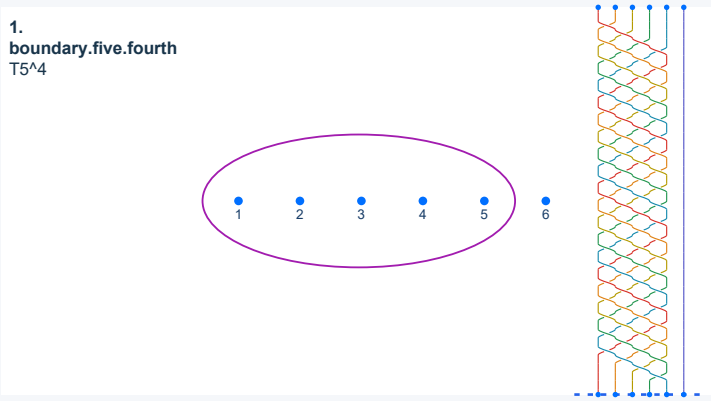
VIII - step 121

SPHERE MOVE: cap the FOURTH boundary power

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

Before



After

Identity / no factors in this interval.

IX. Sphere discrepancies and certificates

The following are ALL FOUR sphere operations in this named route. They are not disk braid equalities. Each discrepancy is written in signed generator indices; the exact sphere certificate and +I lift are retained online.

SPHERE MOVE at step 14

SPHERE LANTERN: absorb the matched pair and the two prefix roots

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (66 signed braid letters):

5 -4 2 3 2 3 2 3 4 -5 -4 -2 3 3 2 -5 -2 -3 4 2 3 2 3 2
3 -4 3 2 5 4 2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1
-1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1 -4 3 3 -5 -2 -3 -2

SPHERE MOVE at step 16

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (482 signed braid letters):

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 -1 -2 3 -4 3 4 -3 2
1 5 -1 -2 3 4 3 4 3 4 3 4 3 4 3 4 3 4 2 1 -4 2 3 2 3 2
3 2 3 2 3 2 3 4 2 3 2 5 -3 -3 -4 -5 -4 -4 -3 -2 -1 -2 -1 -2
-1 -2 -1 -2 -1 -2 -1 3 -4 -3 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 3 1
1 1 -4 -3 -2 -1 2 3 4 -1 -1 -1 4 4 4 1 1 2 -4 -3 -4 -3 -4 -3
-2 -1 -1 -4 -3 -4 -3 1 2 2 1 -4 -3 -2 -1 2 3 4 -1 -2 -2 -1 3 4
3 -4 -4 -3 -2 -1 -2 -1 -2 -1 3 4 4 1 -3 -2 -3 -2 -4 -3 -2 -3 -2 -3
-2 4 -3 -2 -1 4 -2 3 -4 -3 -3 2 -4 -3 -2 3 4 -2 3 3 4 -3 2 -4
-3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -1 -2 -4 -3 -3 -2 3 3 4 2 1
-2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -1 -1 -4 2 1 1 2 3 -4 -4 -3 -2
-1 -1 -2 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1

SPHERE MOVE at step 17

SPHERE MOVE: cap the SIX framing corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (120 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4

SPHERE MOVE at step 121

SPHERE MOVE: cap the FOURTH boundary power

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

FULL DISK DISCREPANCY (80 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4

Certificates and exact scope

Complete named graphical route:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/index.html?phase=namedfull

Full raw certificates:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/certificate.json

The independent six-point sphere cut comparison has six marked points and five consecutive-point arcs. It is separately jointly certified; the individual factor supports in this book are not joint cut complexes.

The new 33-operation named boundary closure has 55 retained adjacent checks, stays positive, and never exceeds support-word maximum 14. Its endpoint is four actual exterior twists. The goal of an under-twenty complete derivation remains open.

The archived 125-page/177-step PDF is preserved separately. This edition contains the new 121-step named proof, with no action-only boundary-recognition step.

Independent jointly certified six-point sphere cut comparison

